PART 2 · AWS, AZURE AND GOOGLE CLOUD

Every database, every monitoring tool, every scope

A recorded walkthrough of RunShore deploying PostgreSQL, MySQL, MongoDB and SAP ASE, registering them with PMM, New Relic and Splunk, and running failover, replication, workload and point-in-time-recovery tests: first on local Kubernetes, then on AWS, Azure and Google Cloud. Each chapter is shown in the order it was captured, with the log of every action.

VERSION	DATE	EVIDENCE	ORDER
1.0	November 2026	129 screenshots · 129 logs	Capture time, oldest first

CHAPTERS

CHAPTER	SCOPE	DATABASES	OBSERVED WITH	STEPS	SCREENS
07 · AWS EKS	AWS · EKS	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	46	58
08 · Azure AKS	Microsoft Azure · AKS	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	44	34
09 · Google GKE	Google Cloud · GKE	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	44	37

EACH STEP IS ONE OF

1. Deploy database
2. Deploy tool
3. Register
4. Failover
5. Replication
6. Workload
7. Assessment
8. Unregister
9. Destroy

Account numbers, subscription IDs, cluster addresses, private IP addresses and usernames have been removed from logs and blurred in screenshots. Warnings in a log are shown as they happened.

07 AWS EKS

AWS · EKS

The same workflow on Amazon Web Services: RunShore builds an EKS cluster with Terraform, runs every database and monitoring tool on it, and destroys it at the end, proving nothing billable is left.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
AWS · EKS	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	58 screens · 42 logs	27 Oct–3 Nov 2026

7.1 Screenshots

What the console and dashboards showed at this point.

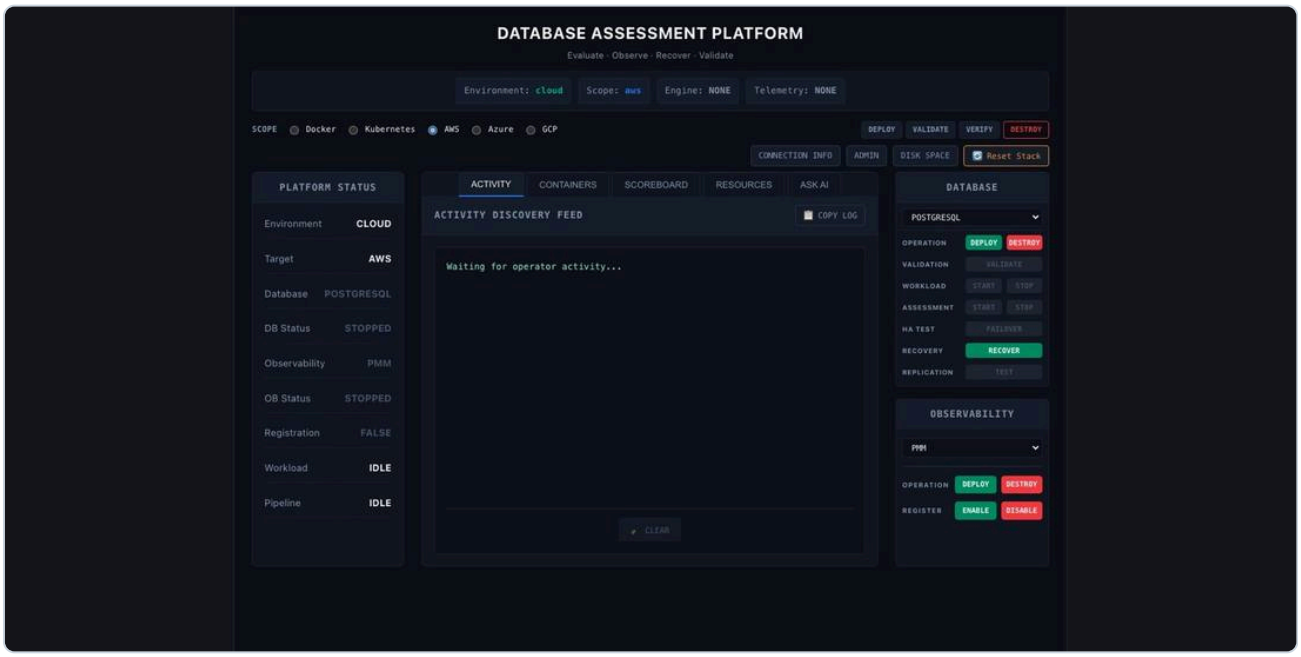


Figure 7.1 27 Oct 2026 · 02:07:36

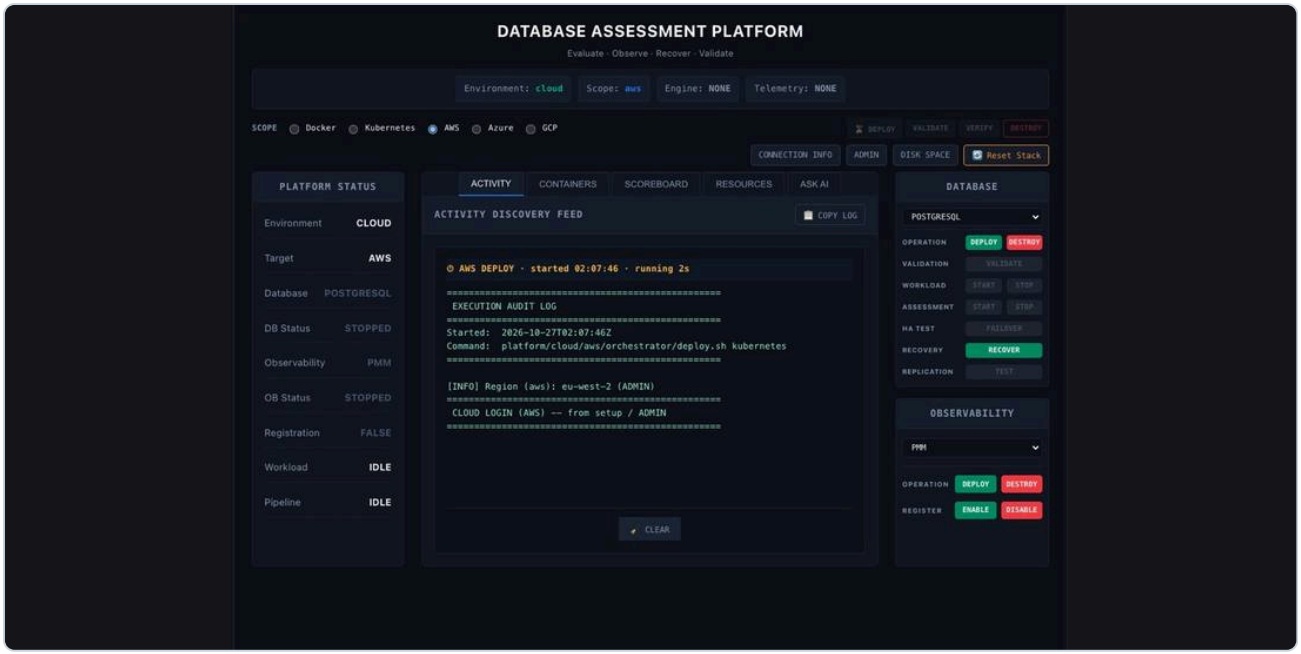


Figure 7.2 27 Oct 2026 · 02:07:49

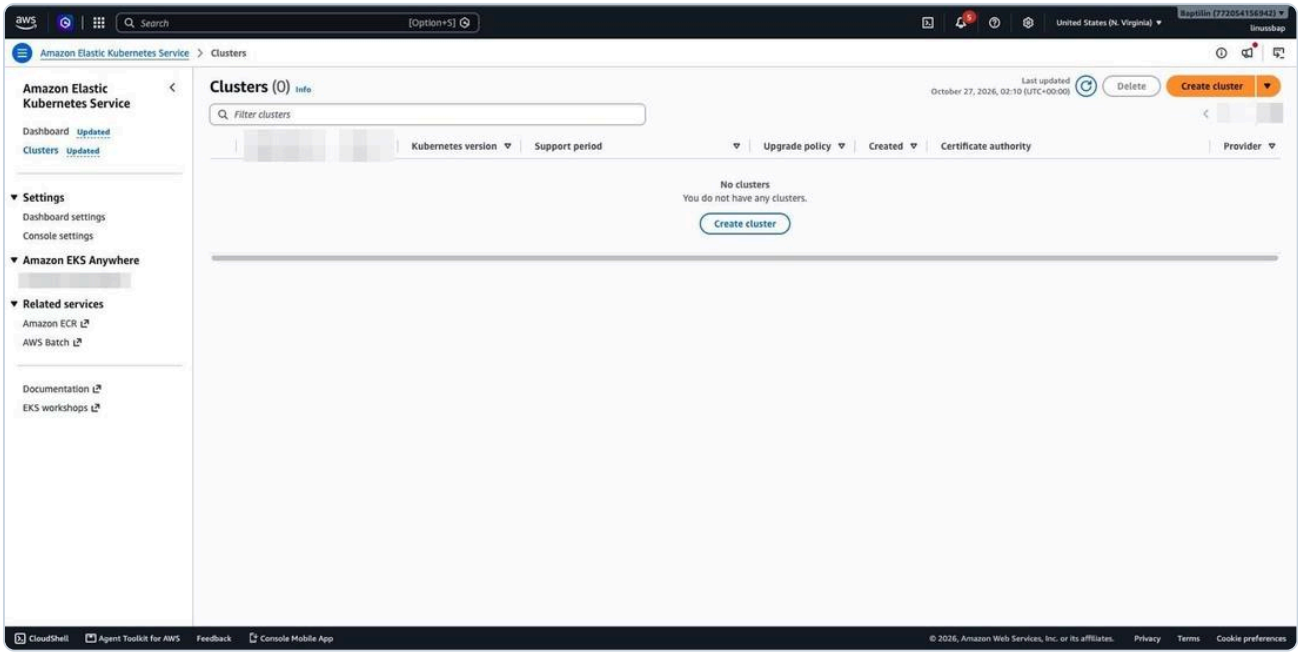


Figure 7.3 27 Oct 2026 · 02:10:25

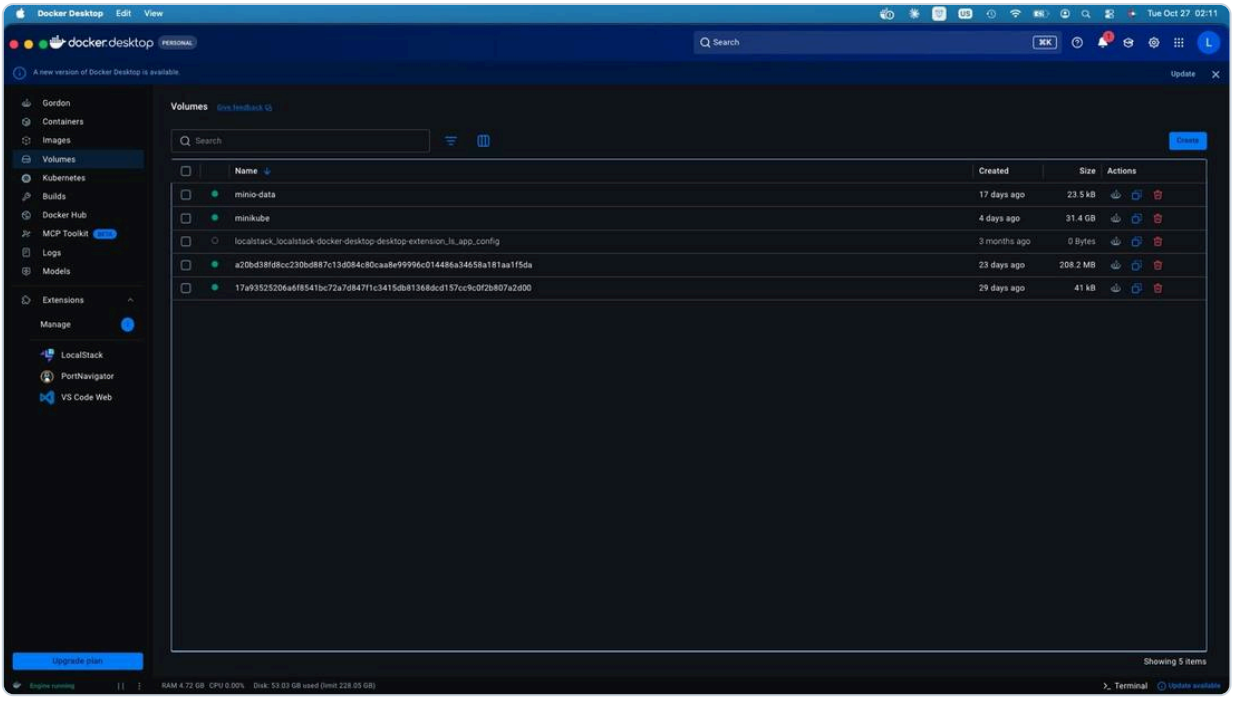


Figure 7.4 27 Oct 2026 · 02:11:58

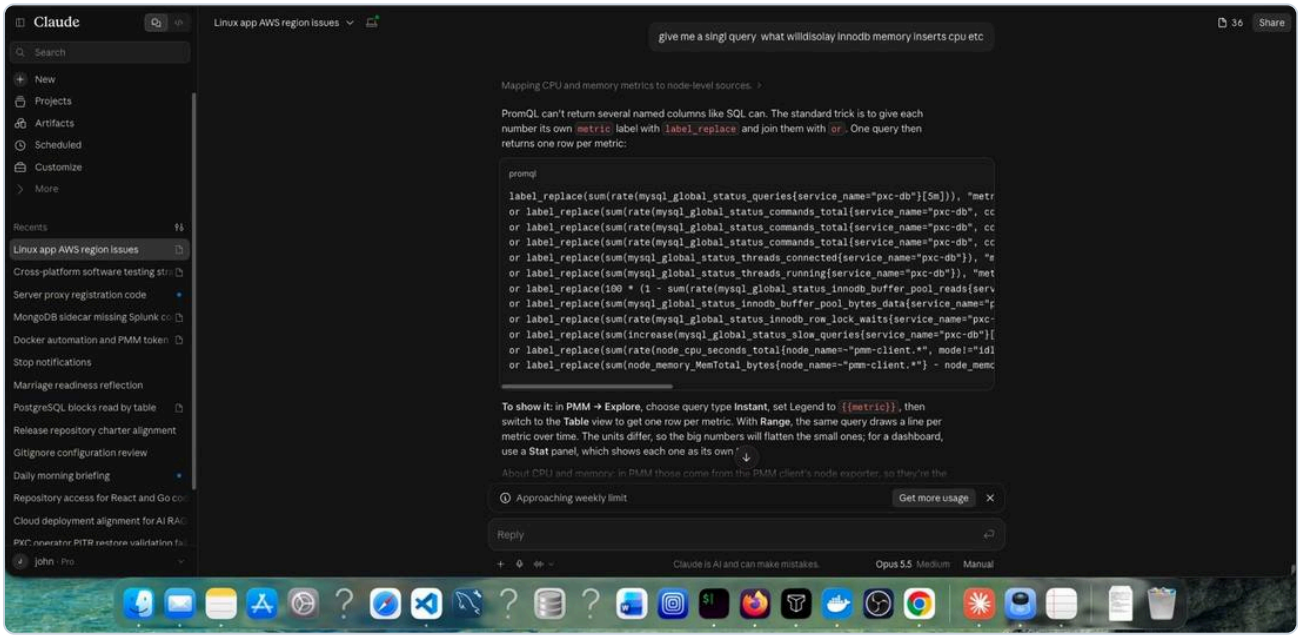


Figure 7.5 27 Oct 2026 · 02:11:58

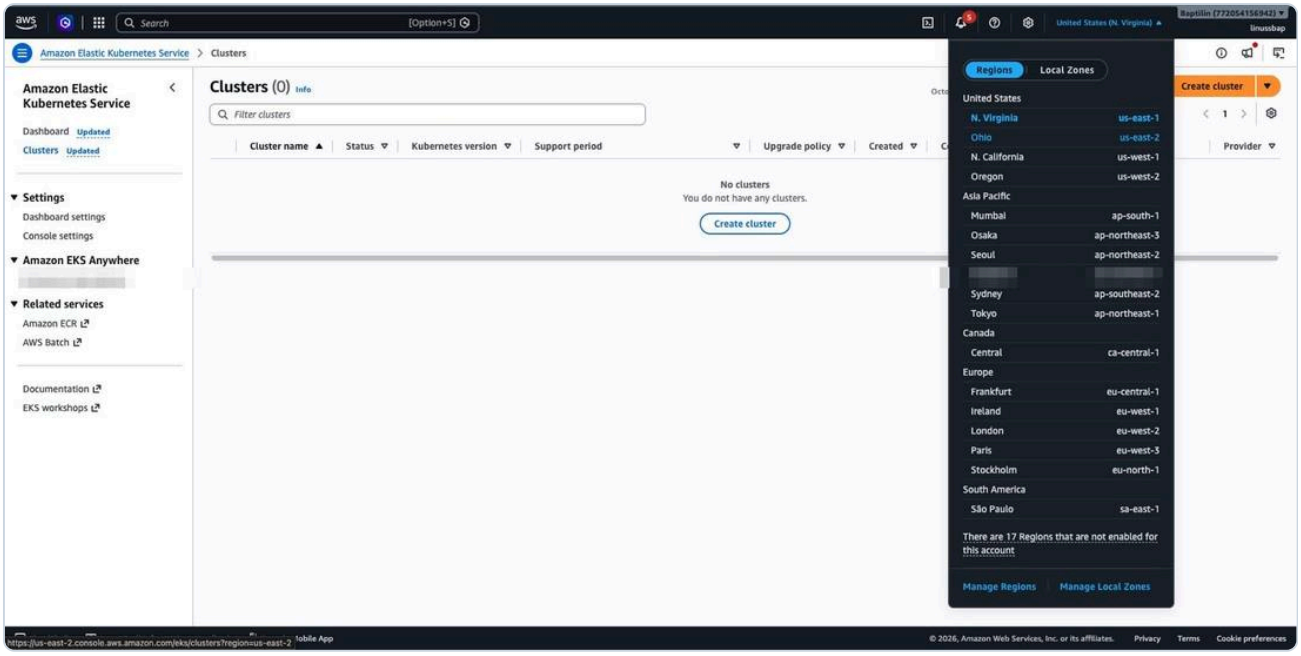


Figure 7.6 27 Oct 2026 · 02:16:56

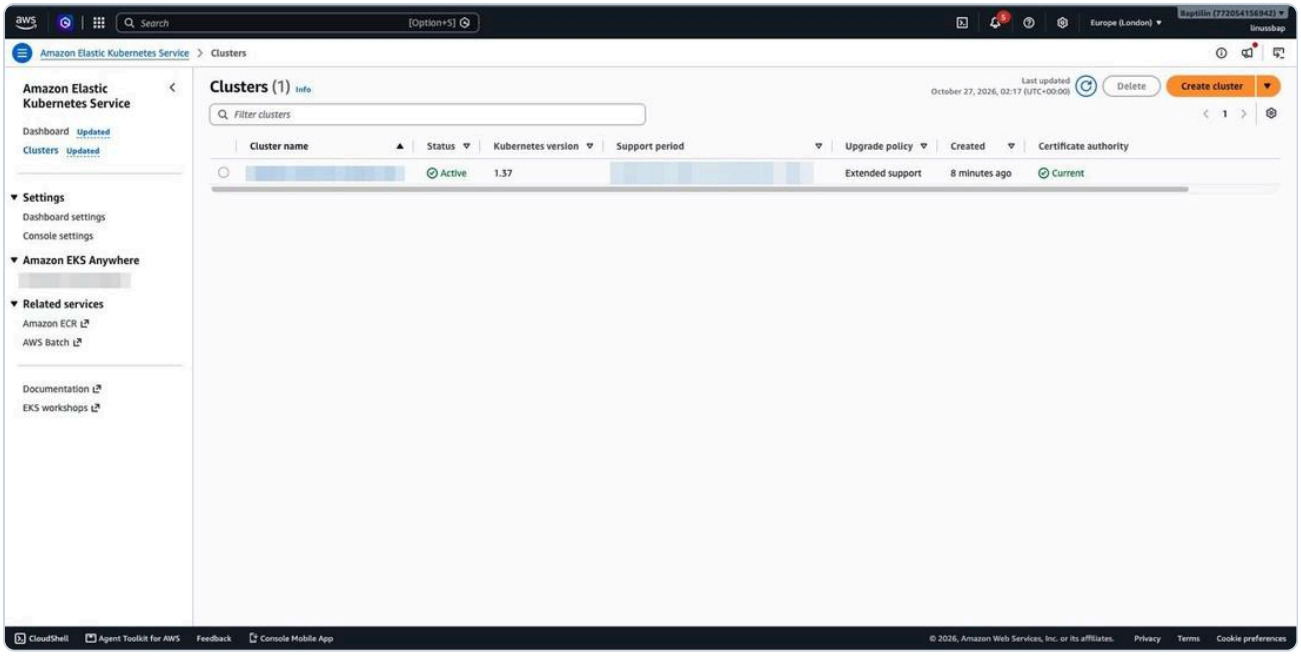


Figure 7.7 27 Oct 2026 · 02:17:13

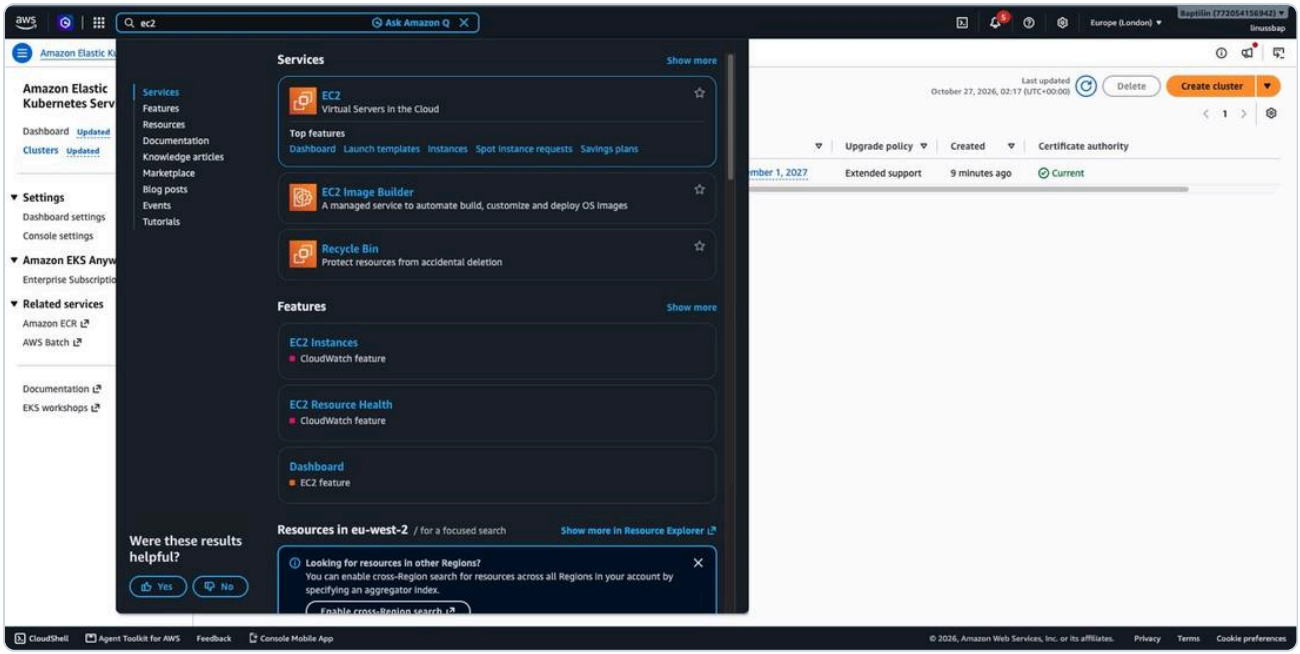


Figure 7.8 27 Oct 2026 · 02:18:16

7.2 Screenshots

What the console and dashboards showed at this point.

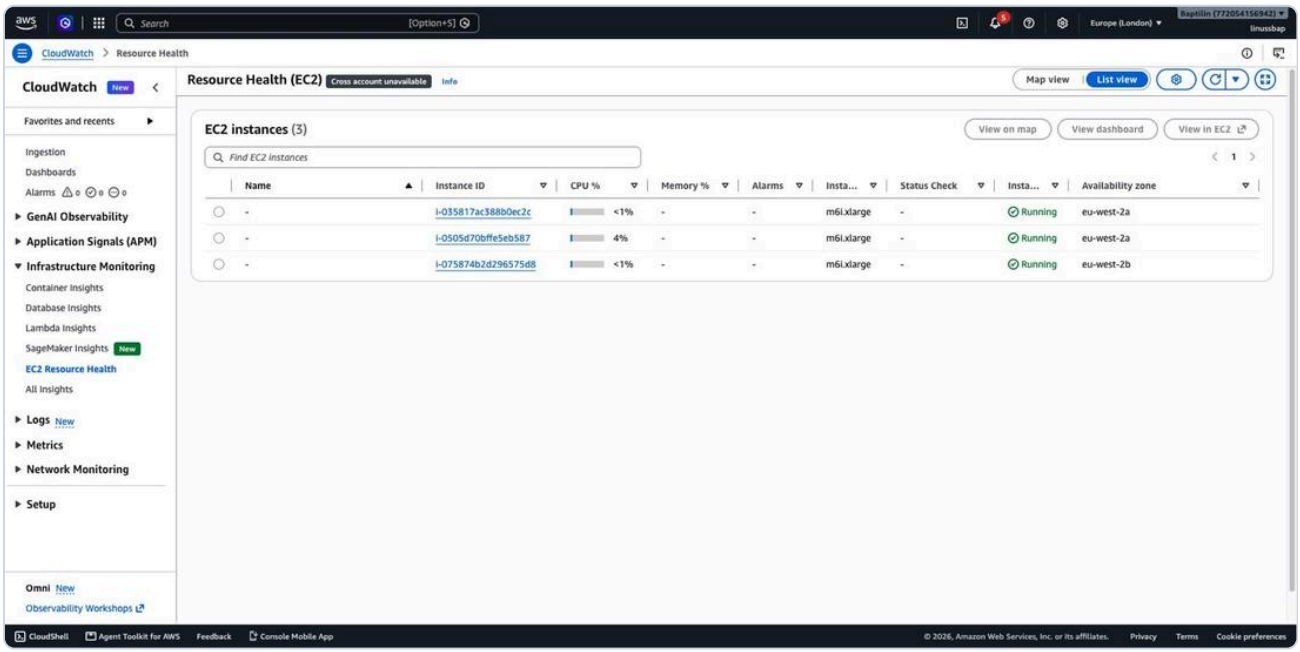


Figure 7.9 27 Oct 2026 · 02:18:37

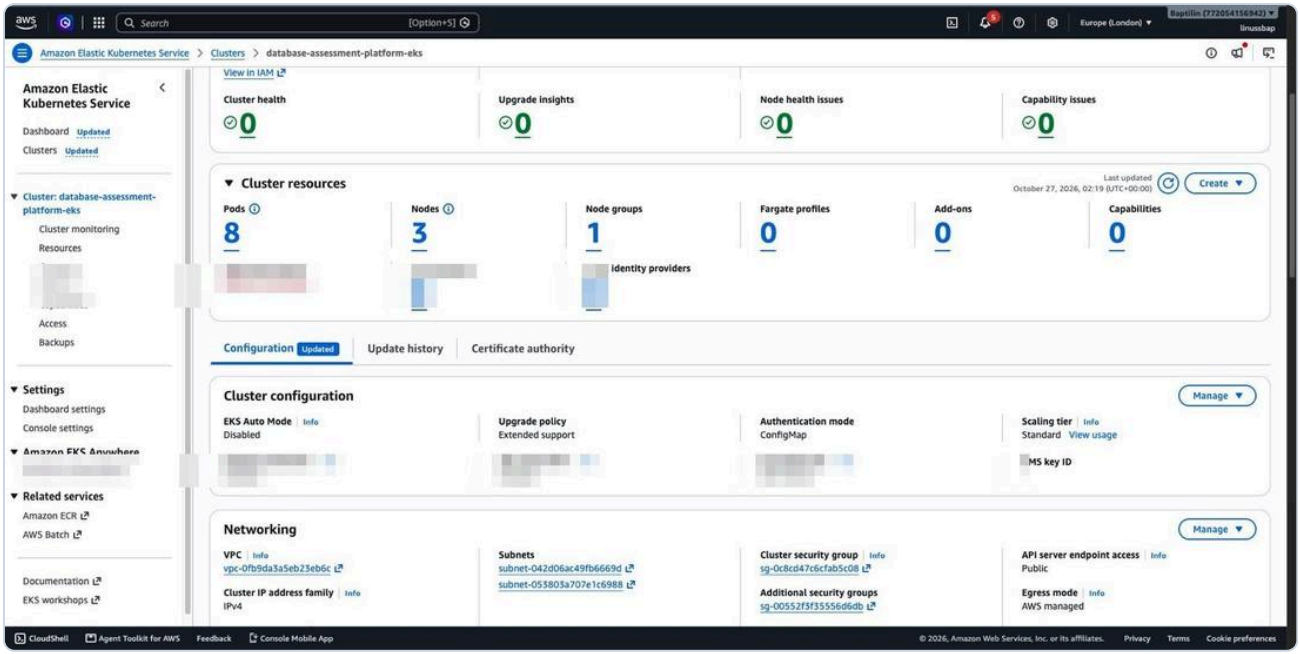


Figure 7.10 27 Oct 2026 · 02:19:49

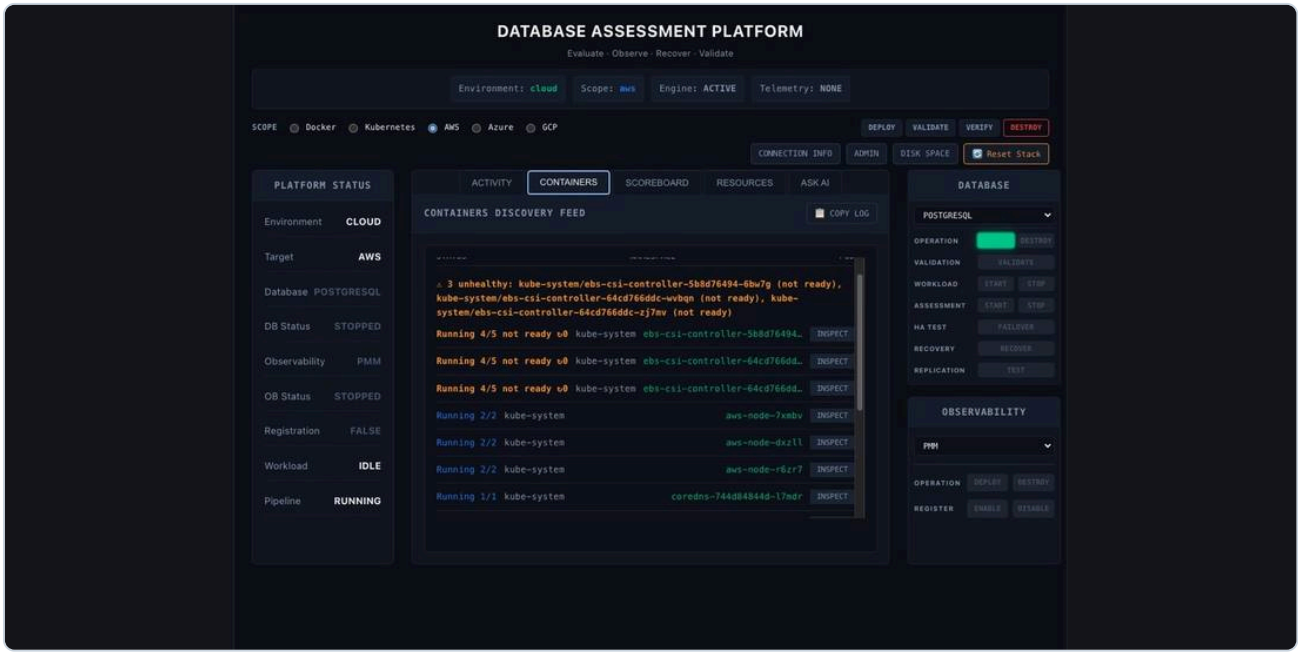


Figure 7.11 27 Oct 2026 · 02:21:28

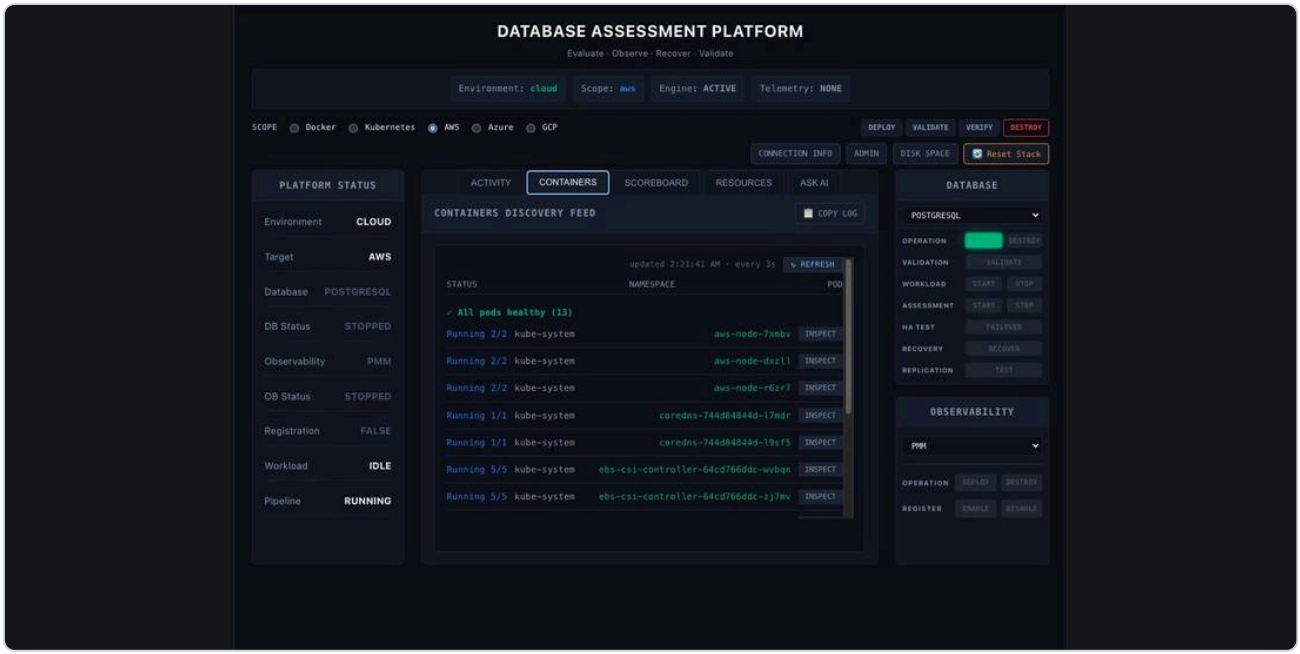


Figure 7.12 27 Oct 2026 · 02:21:42

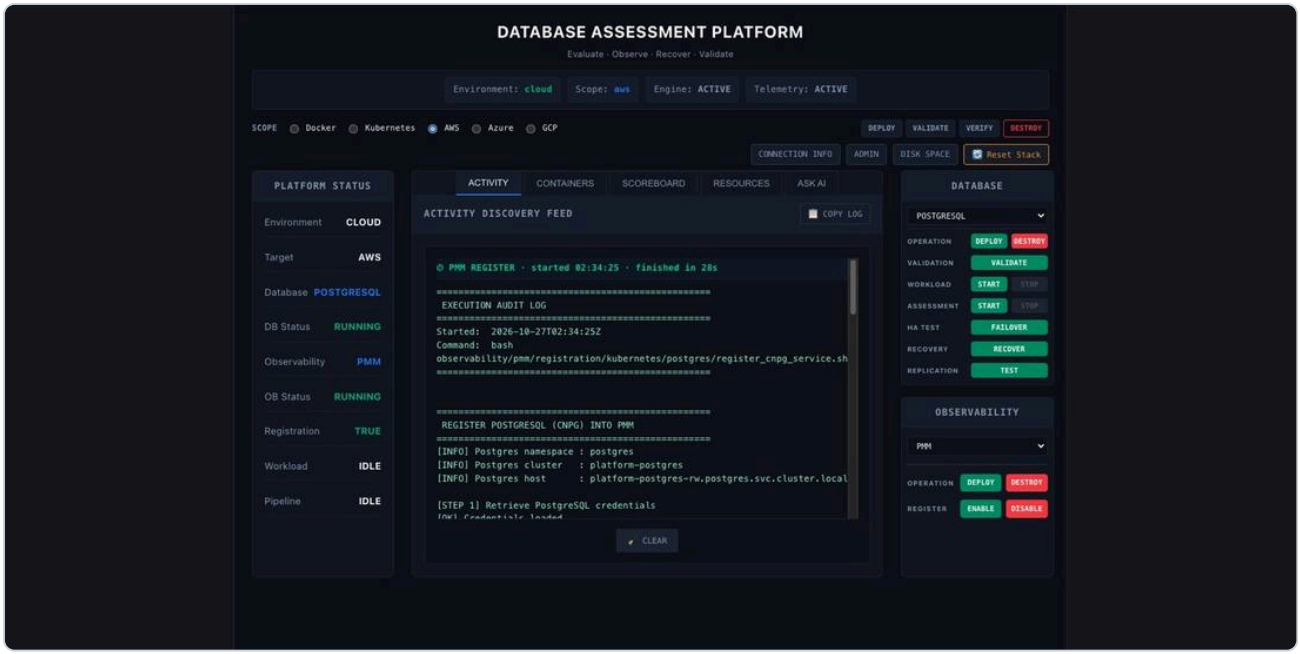


Figure 7.13 27 Oct 2026 · 02:36:42

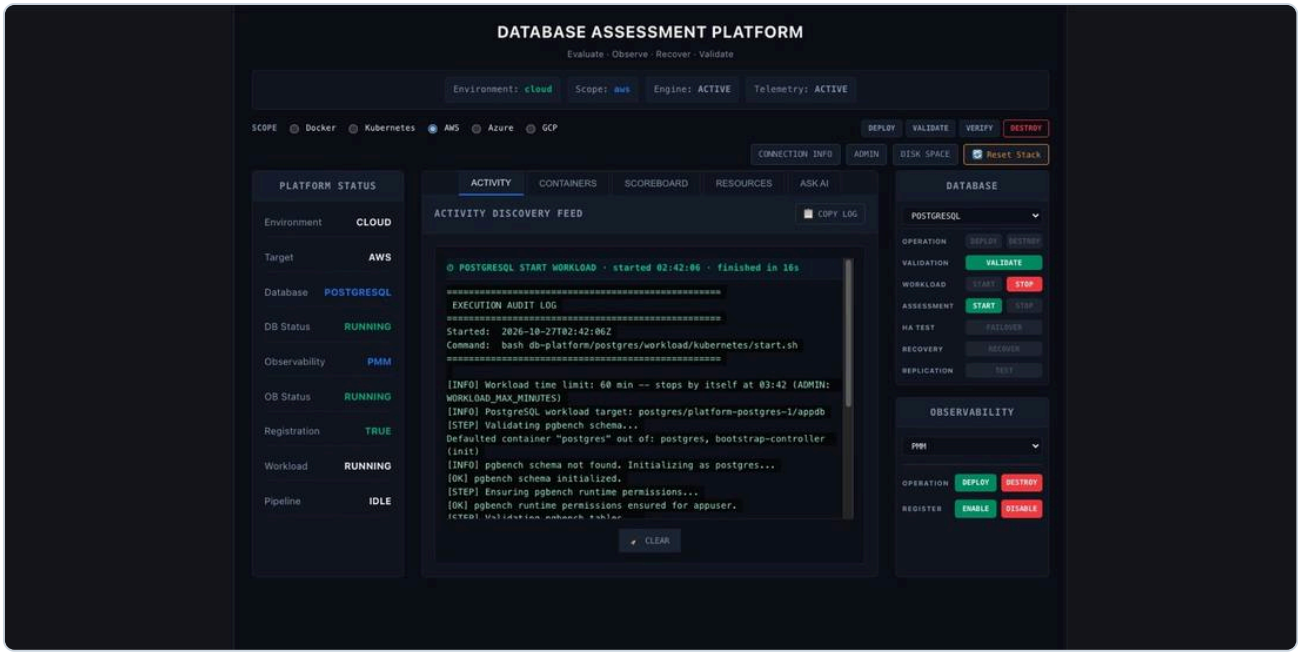


Figure 7.14 27 Oct 2026 · 02:43:11

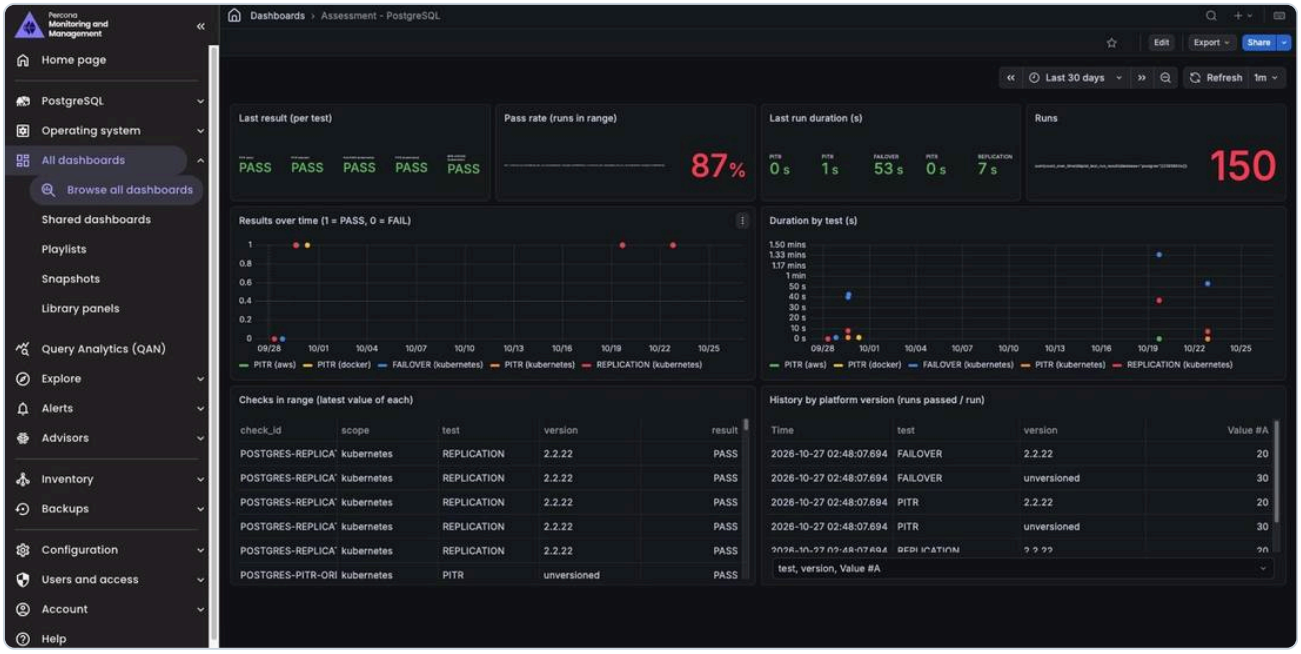


Figure 7.15 27 Oct 2026 · 02:48:47

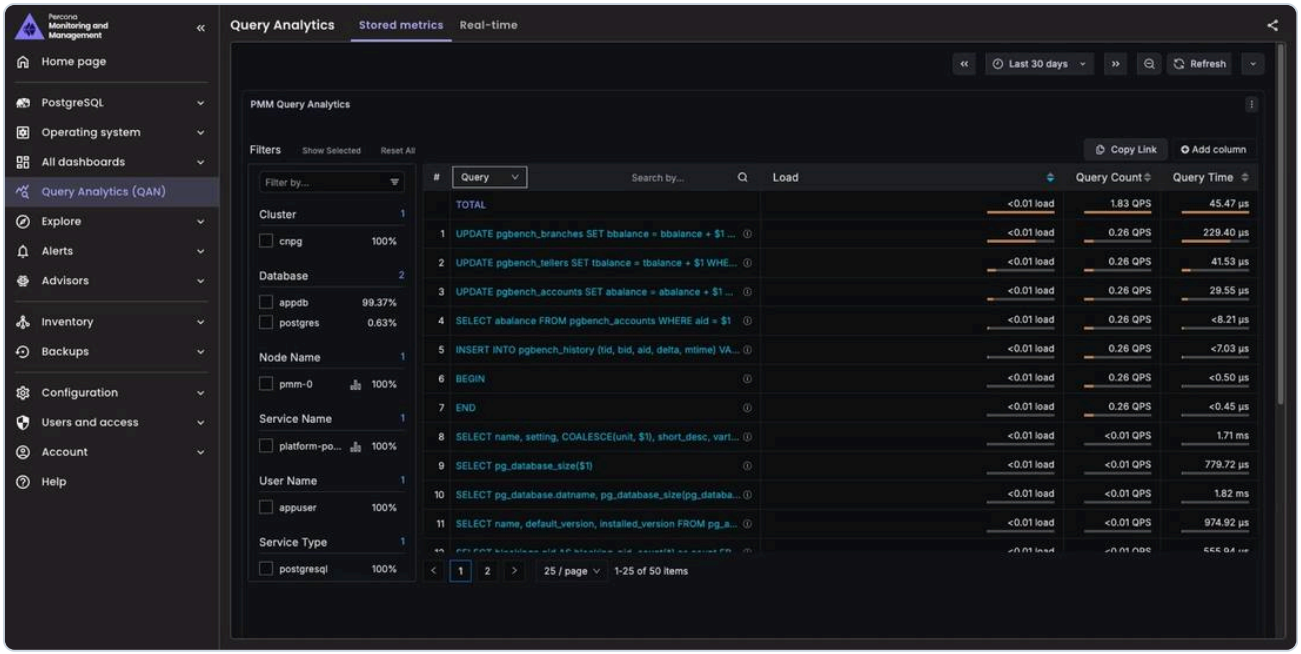


Figure 7.16 27 Oct 2026 · 02:49:30

7.3 Screenshots

What the console and dashboards showed at this point.

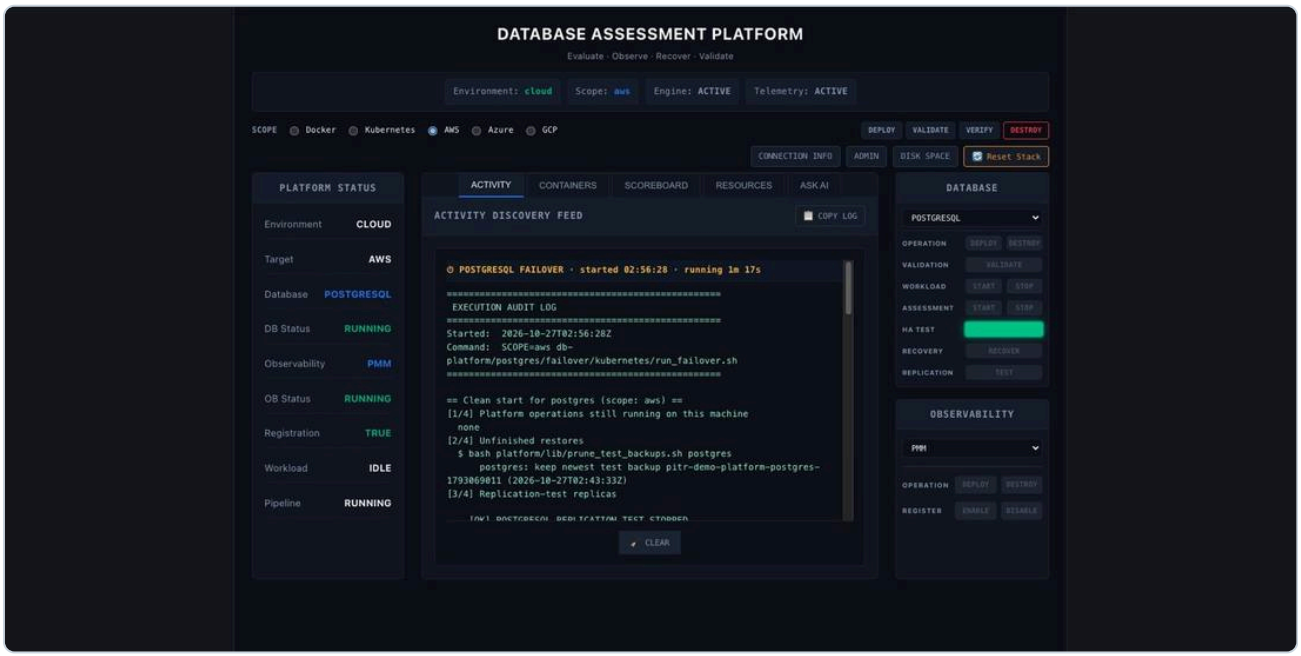


Figure 7.17 27 Oct 2026 · 02:57:45

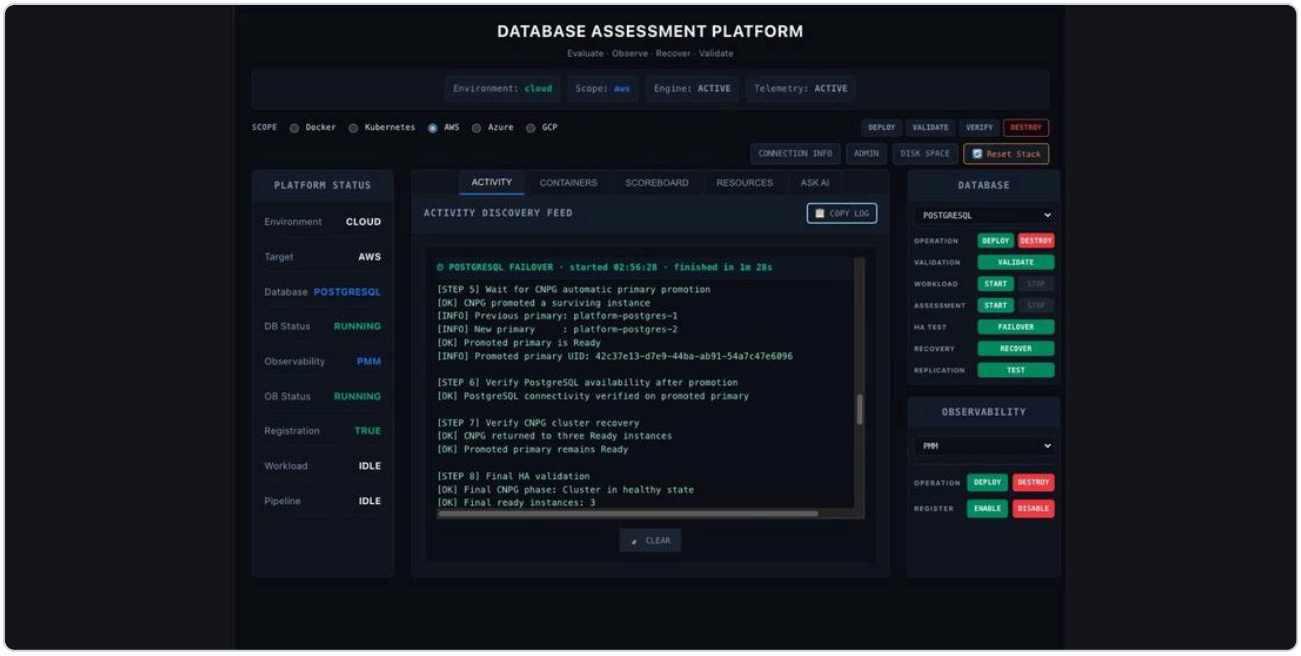


Figure 7.18 27 Oct 2026 · 02:58:10

7.4 Failover test

Failover | **Completed**

The primary database node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 27 Oct 2026, 02:56 · took 1m 28s

COMMAND

```
SCOPE=aws db-platform/postgres/failover/kubernetes/run_failover.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ Final CNPG phase: Cluster in healthy state
- ✔ Final ready instances: 3

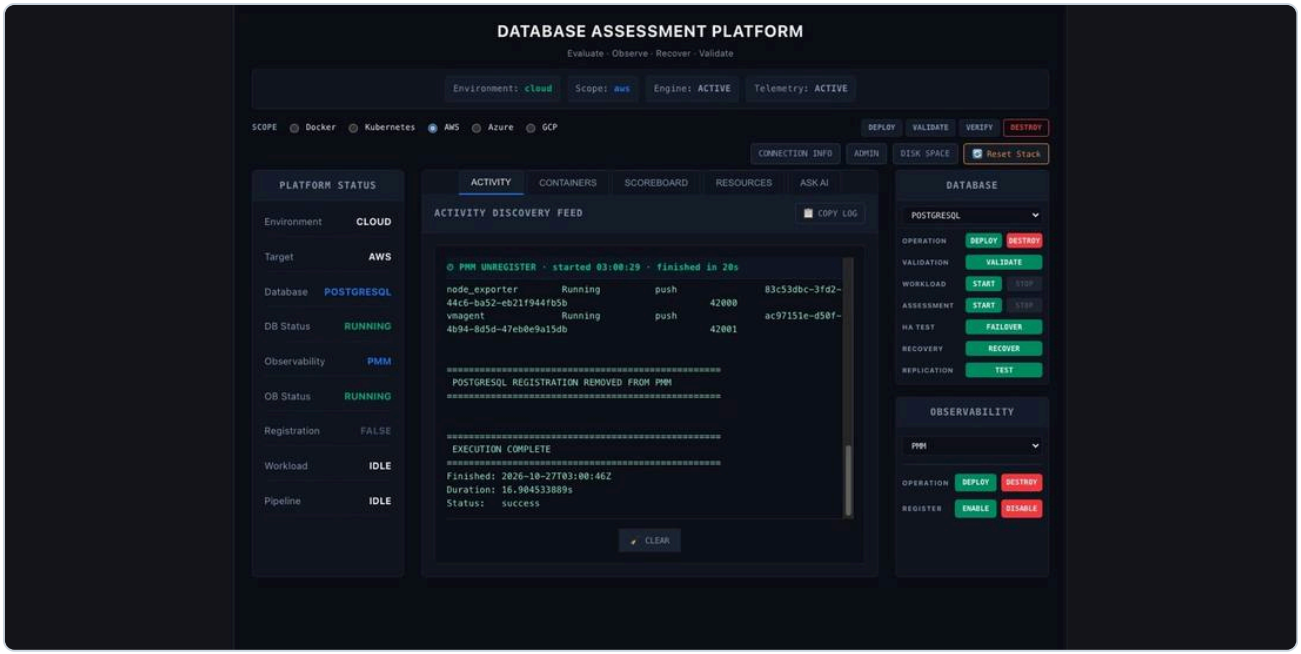


Figure 7.19 27 Oct 2026 · 03:01:03

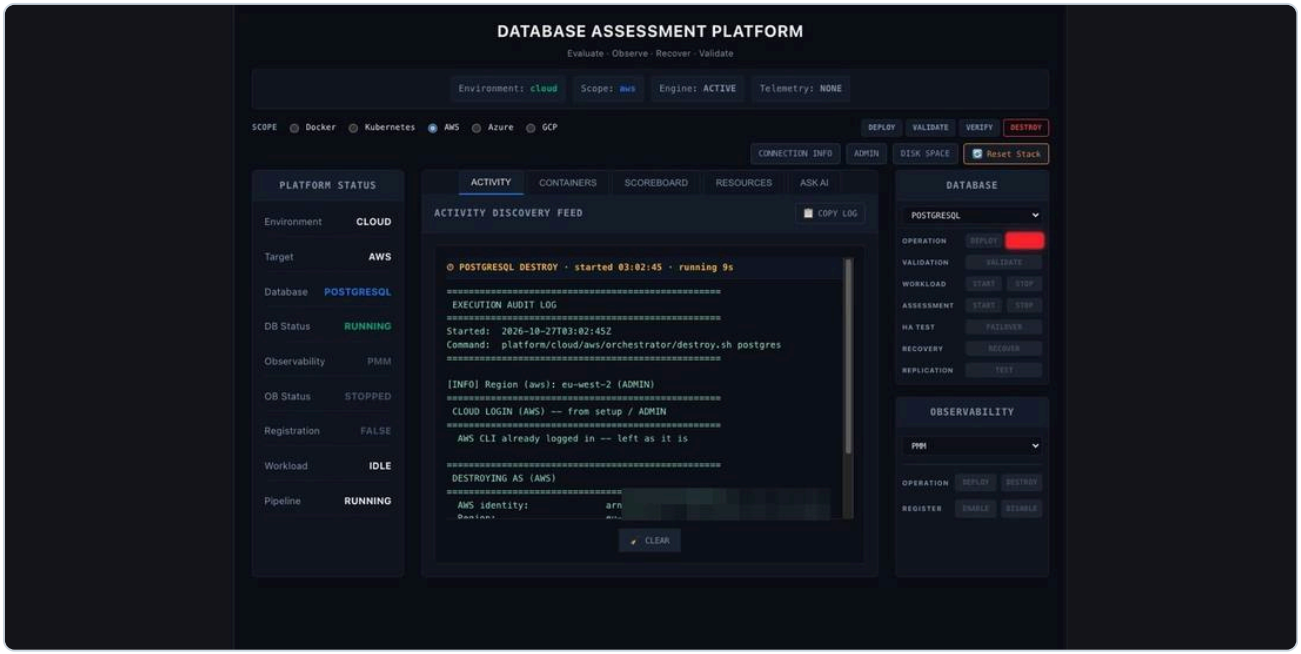


Figure 7.20 27 Oct 2026 · 03:02:54

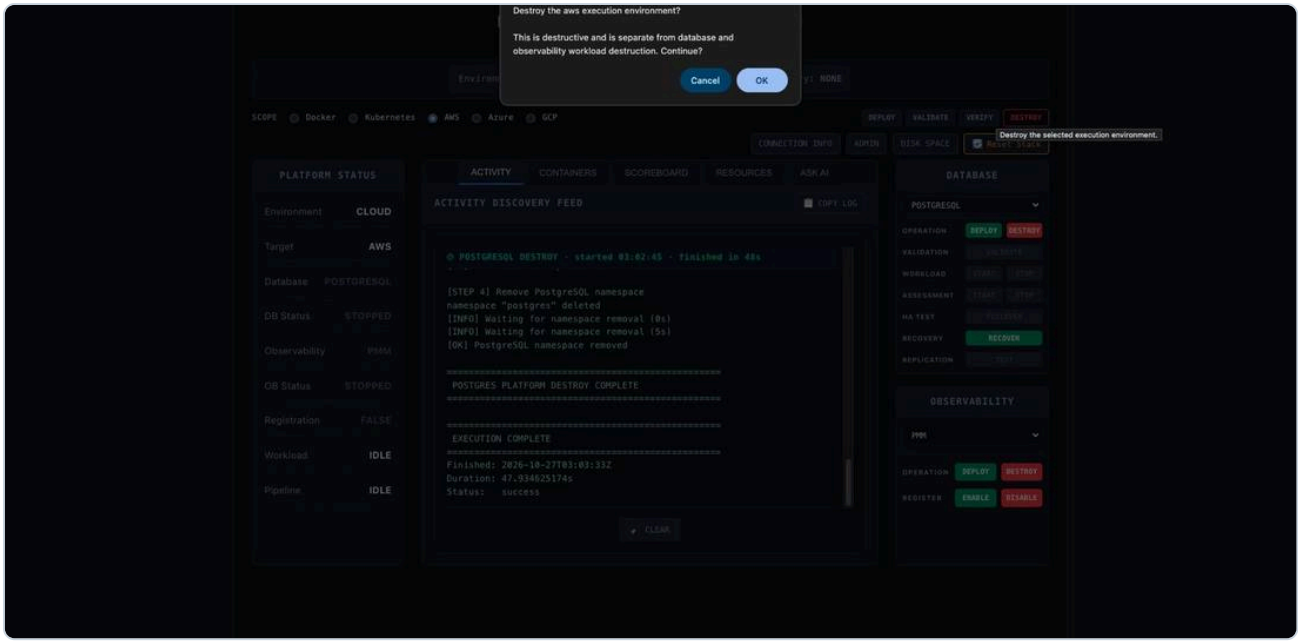


Figure 7.21 27 Oct 2026 · 03:04:22

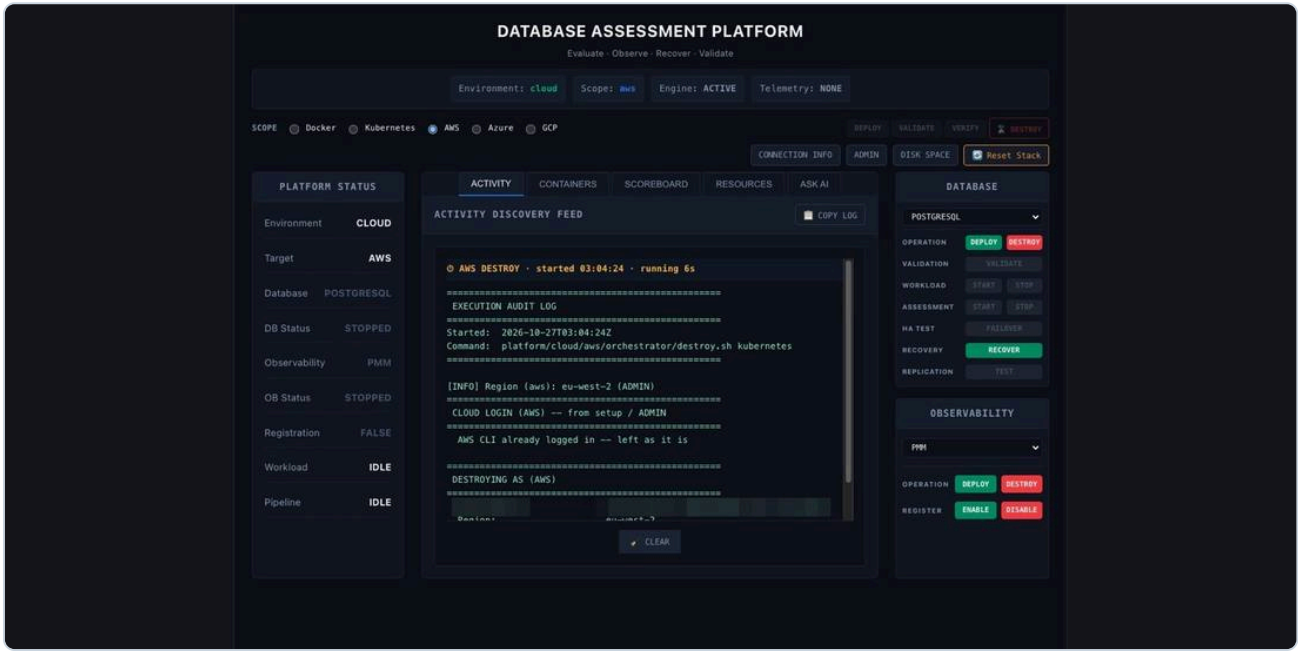


Figure 7.22 27 Oct 2026 · 03:04:30

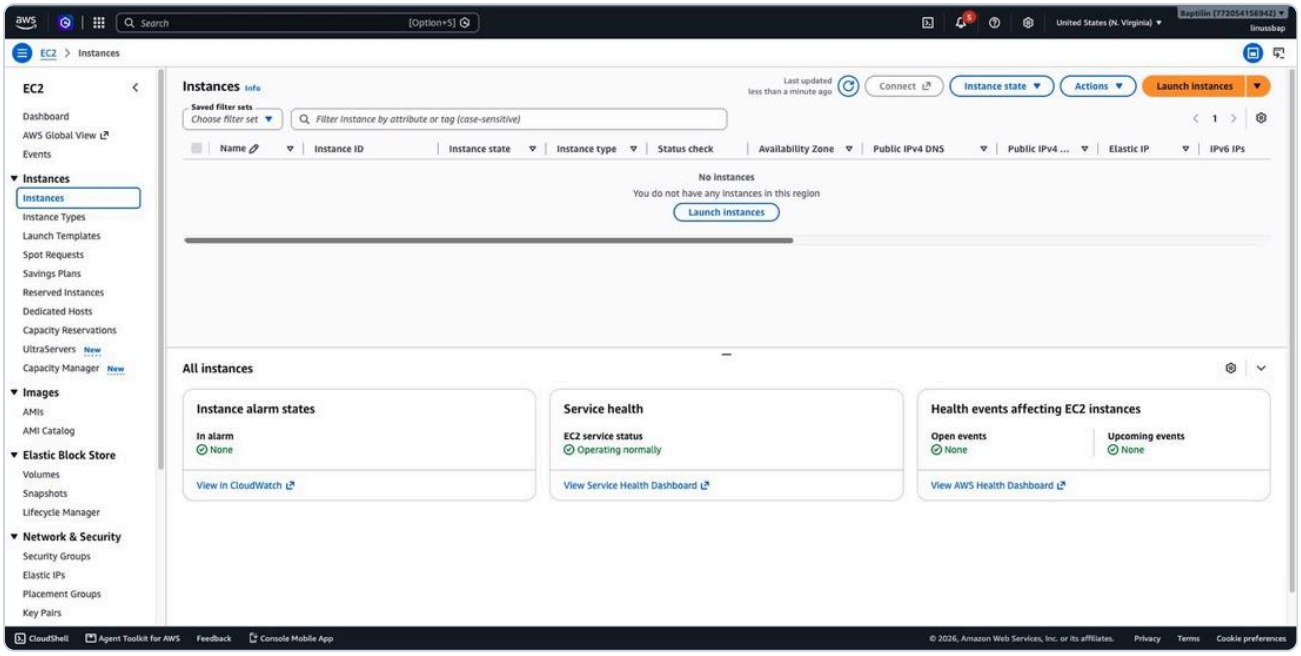


Figure 7.23 27 Oct 2026 · 03:05:37

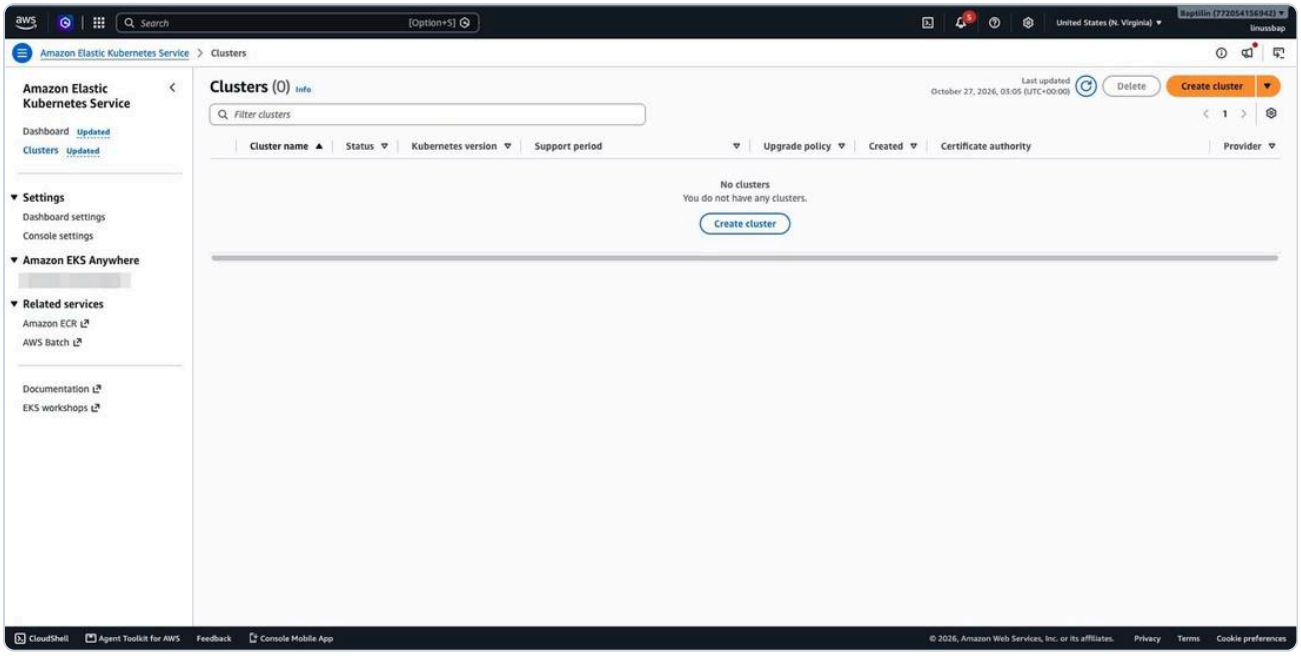


Figure 7.24 27 Oct 2026 · 03:05:56

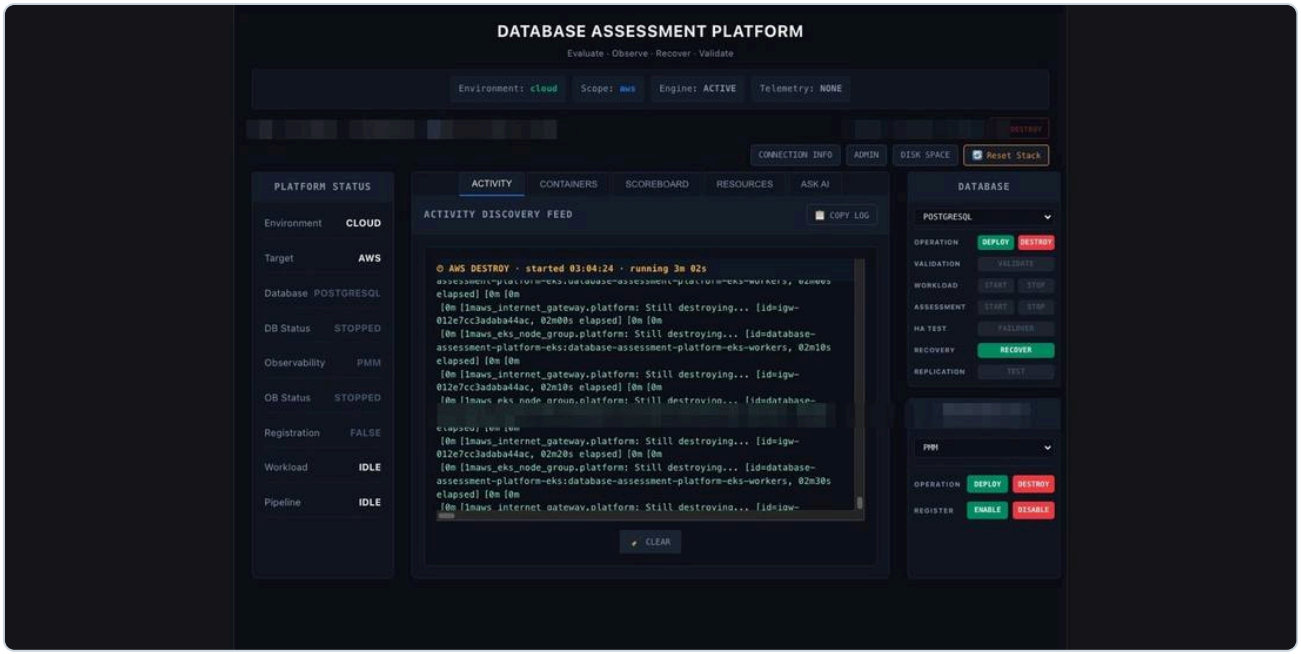


Figure 7.25 27 Oct 2026 · 03:07:27

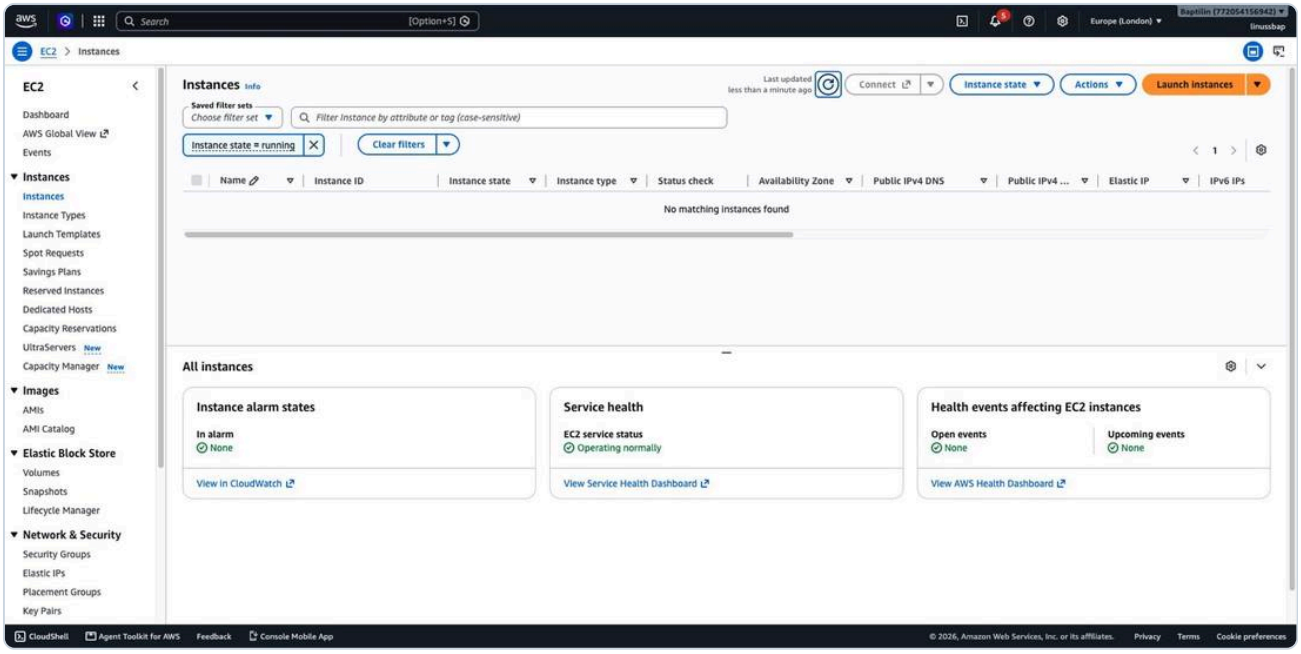


Figure 7.26 27 Oct 2026 · 03:14:32

7.5 More screenshots

Failover

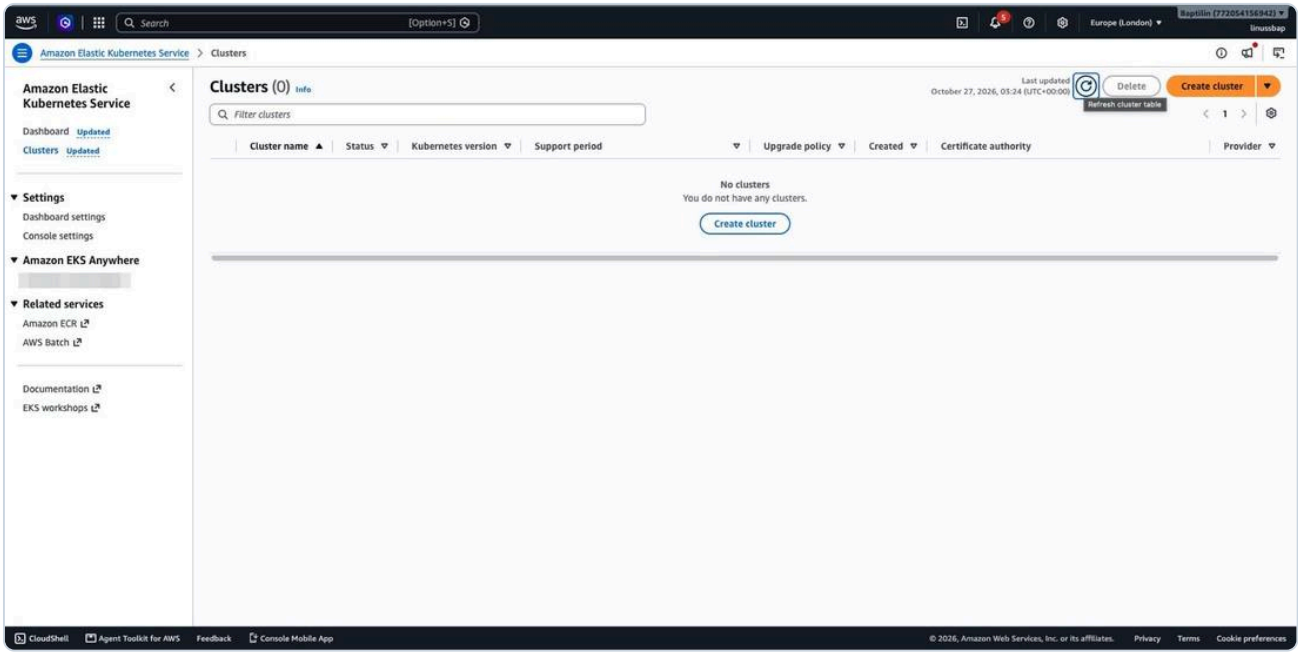


Figure 7.27 27 Oct 2026 · 03:24:40

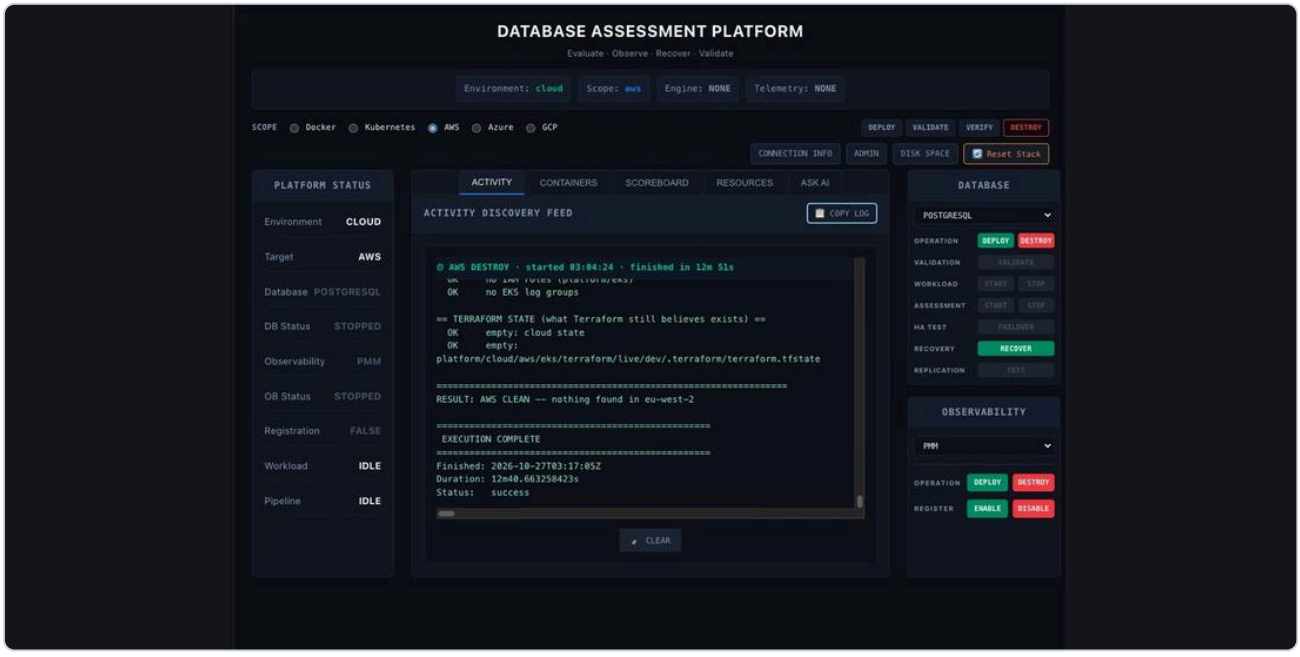


Figure 7.28 27 Oct 2026 · 03:24:56

7.6 Check AWS Status

Information

RunShore checks the cloud sign-in, account and region before building anything.

7.7 Build the AWS Kubernetes cluster

Deploy database **Completed**

RunShore creates the AWS Kubernetes cluster with Terraform and connects to it.

Started 29 Oct 2026, 02:01 · took 8m 52s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh kubernetes
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ AWS eu-west-2 On-Demand Standard instances: 16 vCPU free (limit 16, in use 0) -- cluster needs 12
- ✔ CPU quota sufficient
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated

7.8 Register PostgreSQL with PMM

Register | PMM | PostgreSQL | Completed

PostgreSQL is connected to PMM, so its metrics and dashboards appear there.

Started 28 Oct 2026, 02:57 · took 28s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/register_cnpg_service.sh
```

- ✔ Credentials loaded
- ✔ pmm-0
- ✔ PostgreSQL successfully registered into PMM

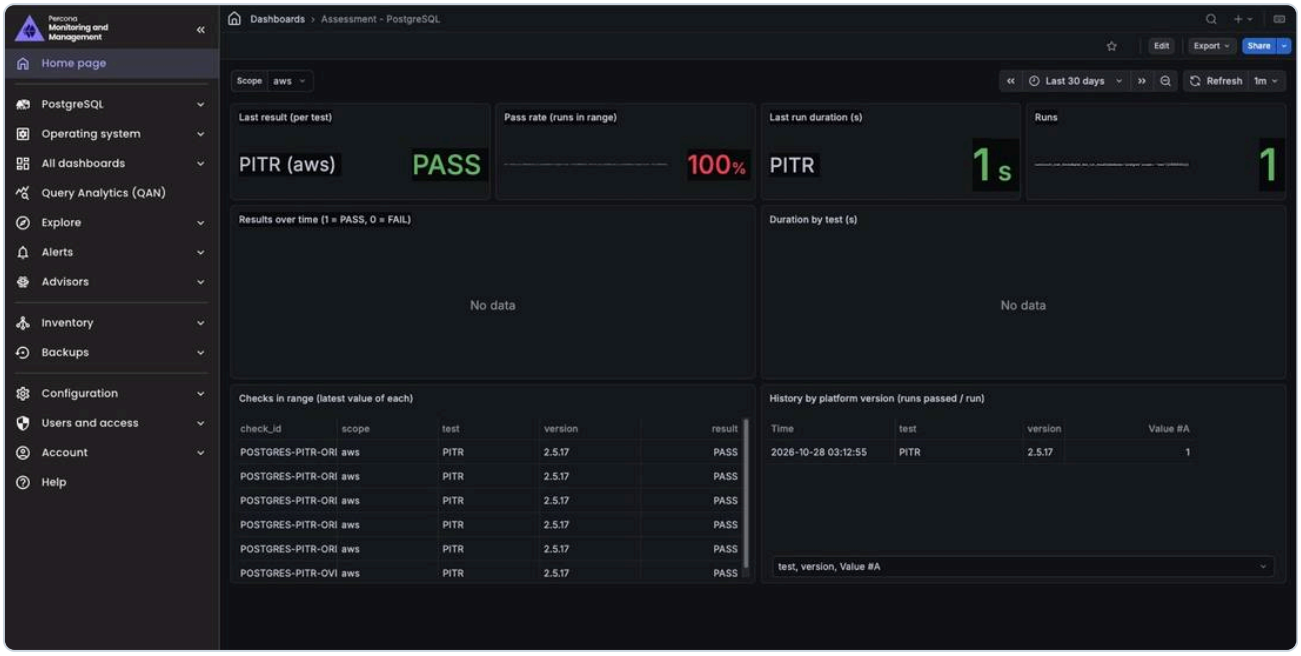


Figure 7.29 28 Oct 2026 · 03:13:31



Figure 7.30 28 Oct 2026 · 03:13:44

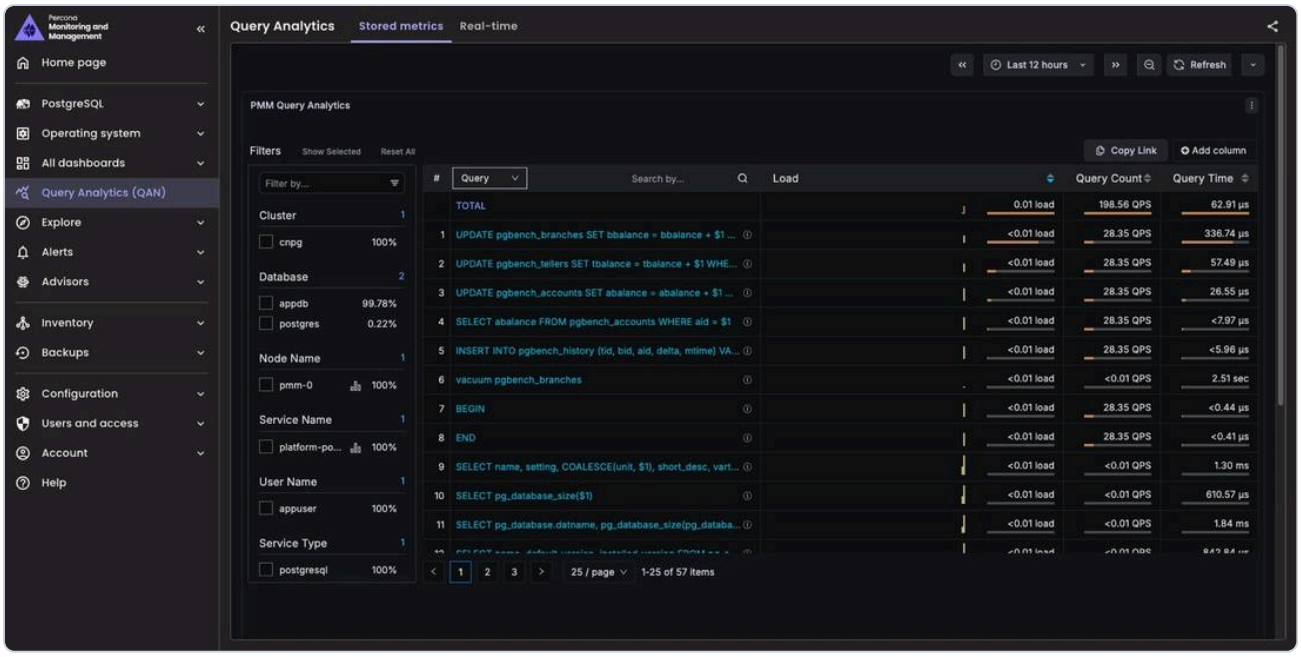


Figure 7.31 28 Oct 2026 · 03:14:11

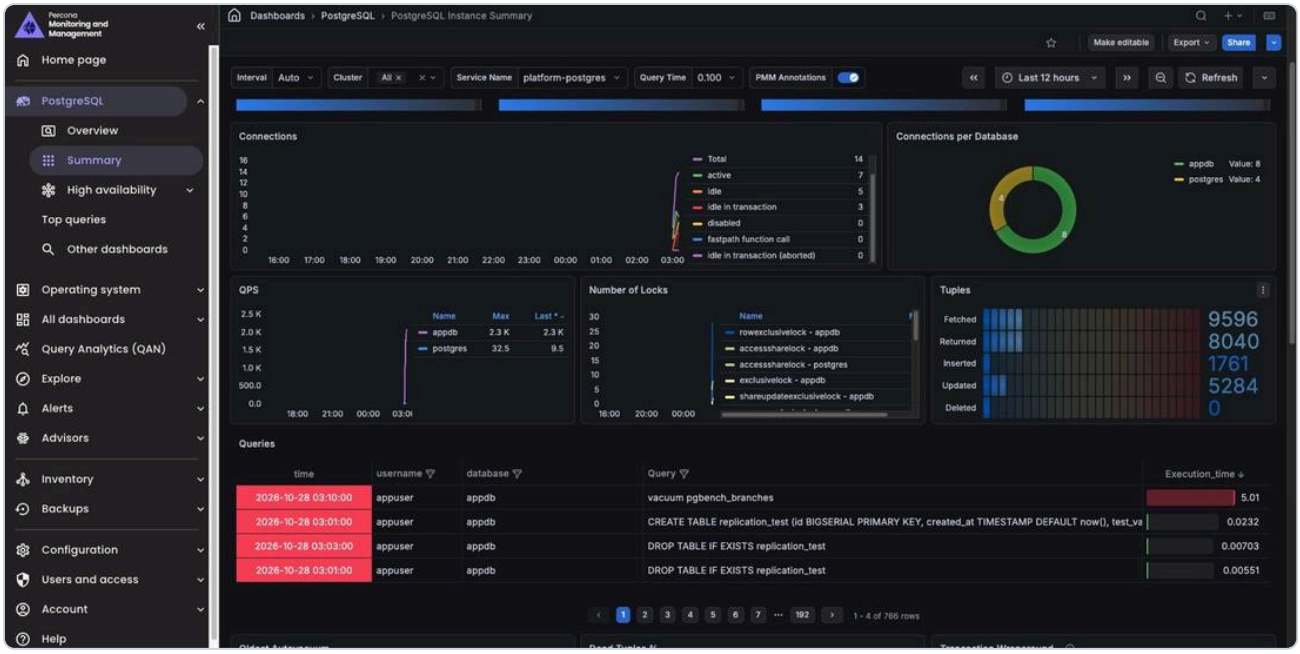


Figure 7.32 28 Oct 2026 · 03:16:07

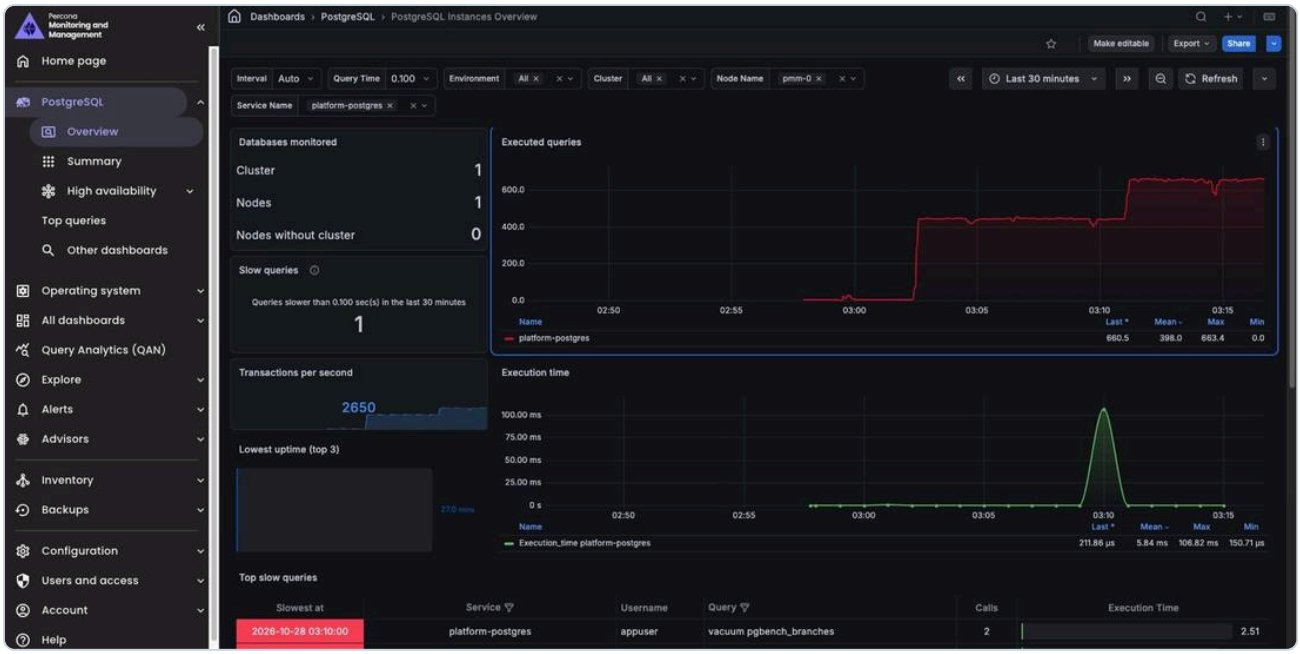


Figure 7.33 28 Oct 2026 · 03:16:51

7.9 Unregister PostgreSQL from PMM

Unregister	PMM	PostgreSQL	Completed

PostgreSQL is removed from PMM; the database itself keeps running.

Started 28 Oct 2026, 03:16 · took 16s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/unregister.sh
```

✔ pmm-0

- ✔ nothing left in PMM inventory -- Registration shows FALSE

7.10 Install PMM

Deploy tool | PMM | Completed

PMM is installed, ready to monitor the databases in this cluster.

Started 28 Oct 2026, 09:47 · took 5m 05s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh pmm
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ PMM Client pod pmm-client-54d5457798-l874q is Ready
- ✔ PMM Client registered and connected to PMM Server

7.11 Register MongoDB with PMM

Register | PMM | MongoDB | Completed

MongoDB is connected to PMM, so its metrics and dashboards appear there.

Started 28 Oct 2026, 10:21 · took 44s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/register_mongodb_replicaset.sh
```

- ✔ MongoDB namespace : mongodb
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ Replica set : rs0
- ✔ Running MongoDB members: 3
- ✔ Cleanup complete
- ✔ MongoDB service present in PMM inventory



Figure 7.34 28 Oct 2026 · 10:42:03

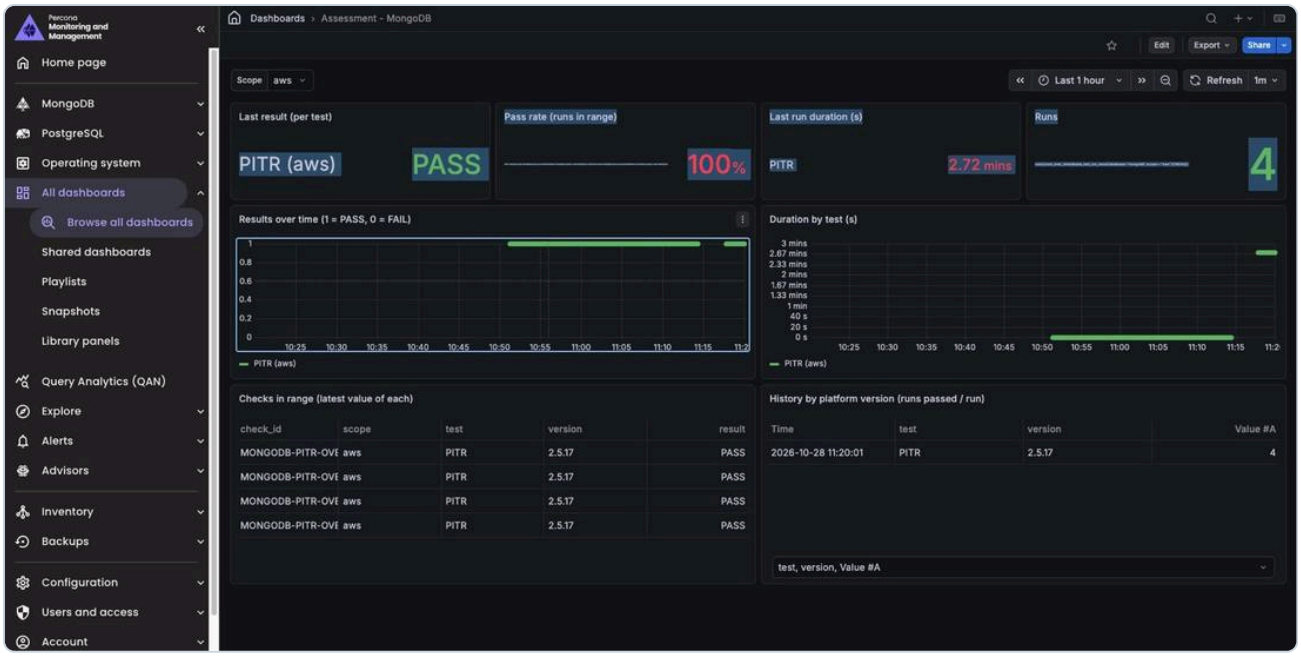


Figure 7.35 28 Oct 2026 · 11:20:07

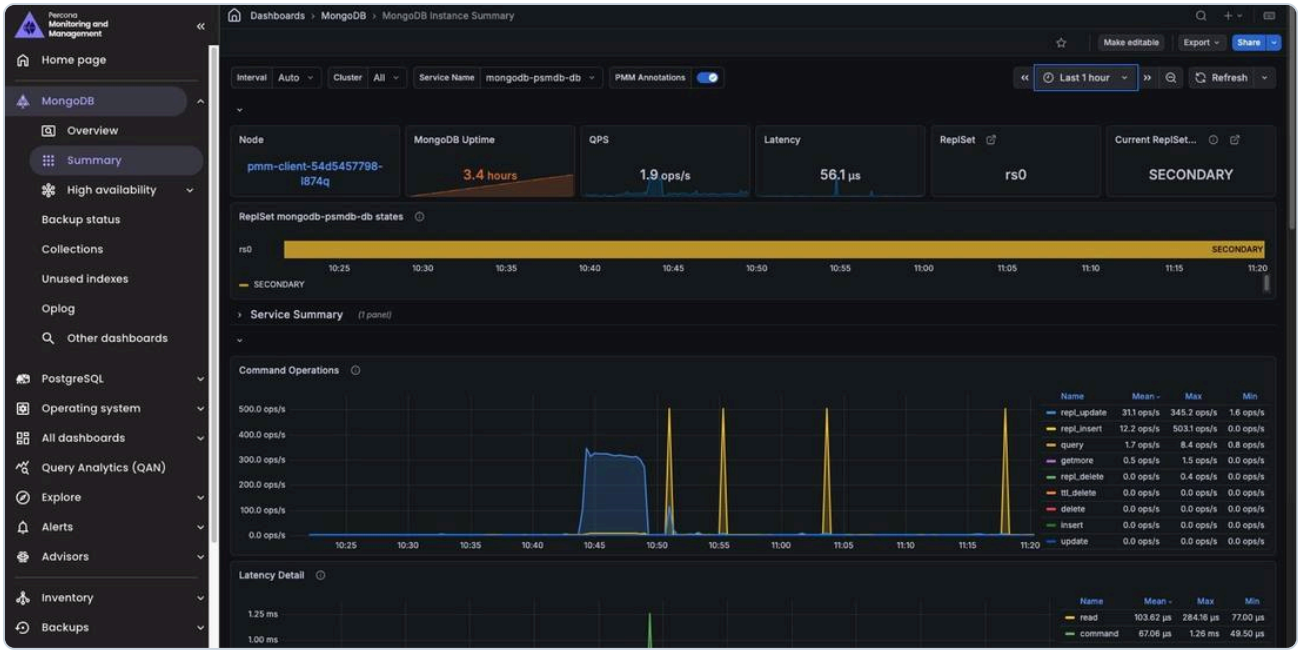


Figure 7.36 28 Oct 2026 · 11:20:29

7.12 Unregister MongoDB from PMM

Unregister	PMM	MongoDB	Completed

MongoDB is removed from PMM; the database itself keeps running.

Started 28 Oct 2026, 11:20 · took 20s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE
- ✔ MongoDB service absent from PMM: mongodb-psmdb-db

7.13 Deploy MySQL

Deploy database	MySQL	Completed

RunShore deploys MySQL on AWS · EKS and waits until every node is ready.

Started 28 Oct 2026, 11:23 · took 4m 52s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh mysql
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks

- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ PXC cluster health converged
- ✔ PXC deployment complete

7.14 Register MySQL with PMM

Register | PMM | MySQL | Completed

MySQL is connected to PMM, so its metrics and dashboards appear there.

Started 28 Oct 2026, 12:24 · took 40s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/register_pxc_service.sh
```

- ✔ pmm-0
- ✔ Credentials loaded
- ✔ PXC successfully registered into PMM

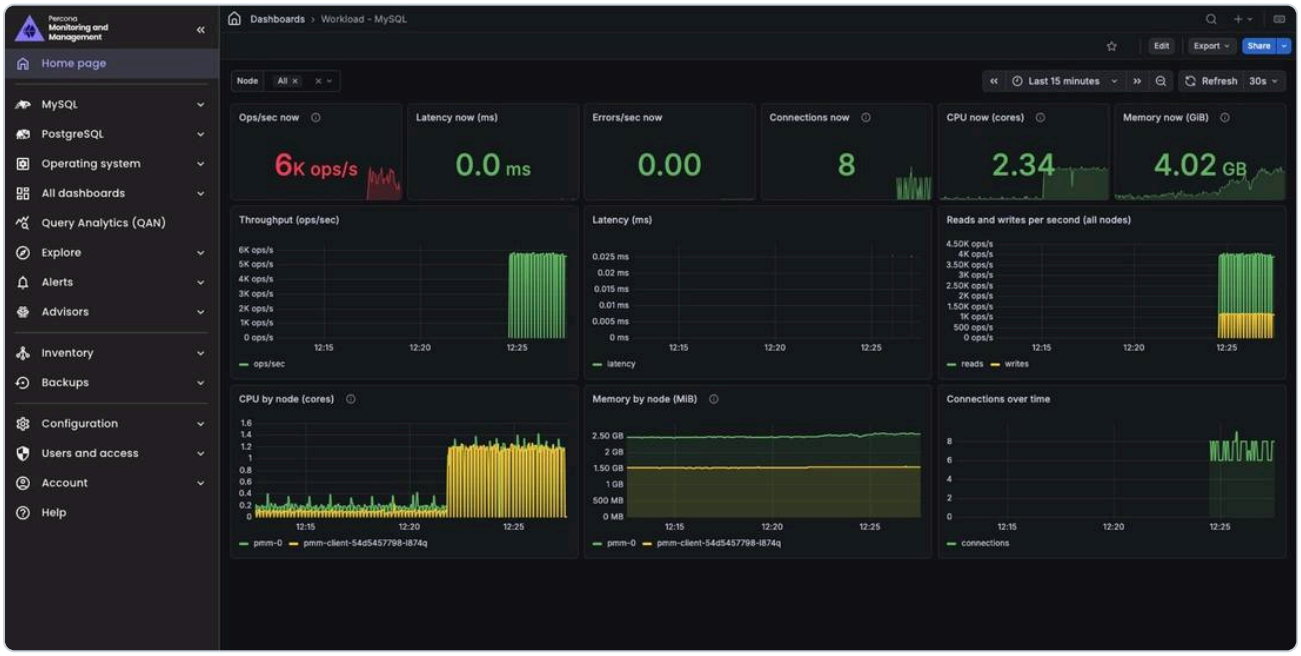


Figure 7.37 28 Oct 2026 · 12:27:59

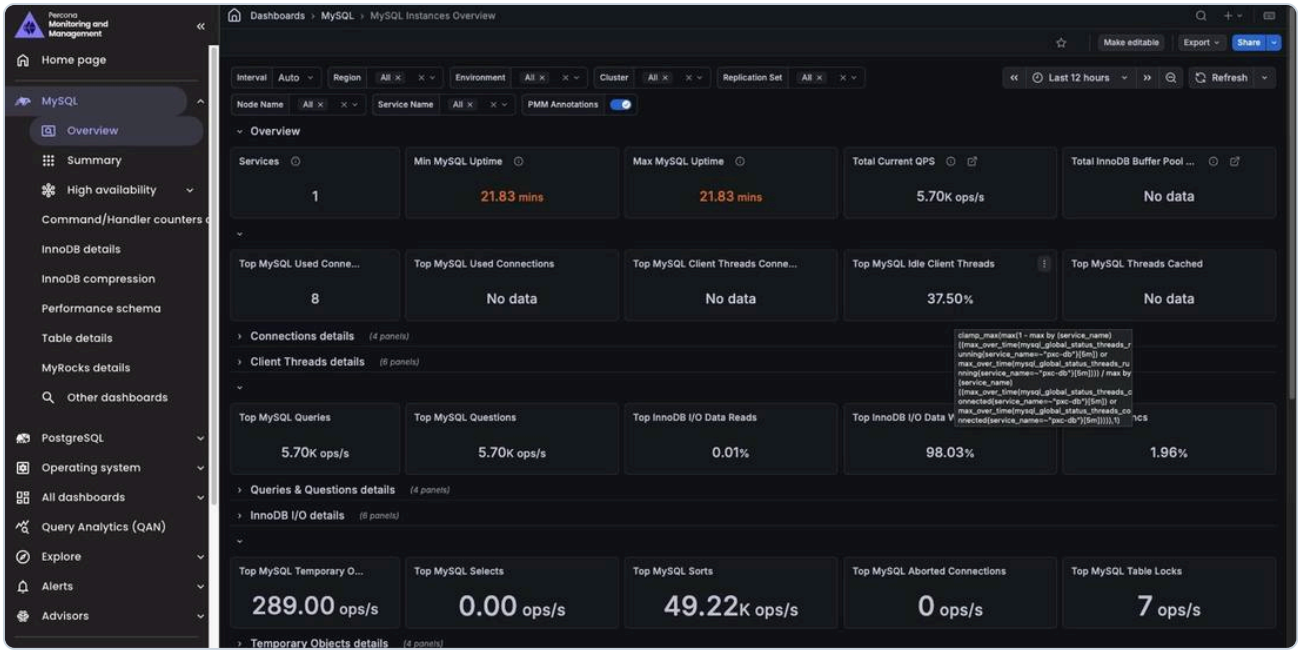


Figure 7.38 28 Oct 2026 · 12:28:46

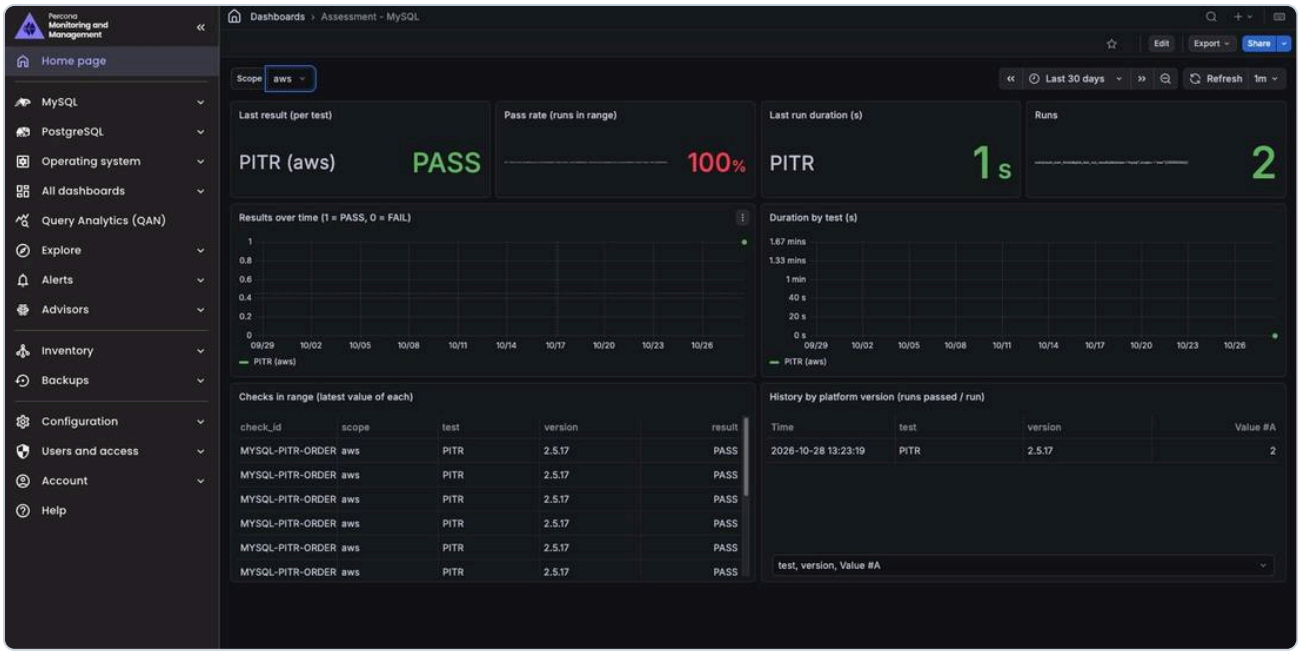


Figure 7.39 28 Oct 2026 · 13:23:33

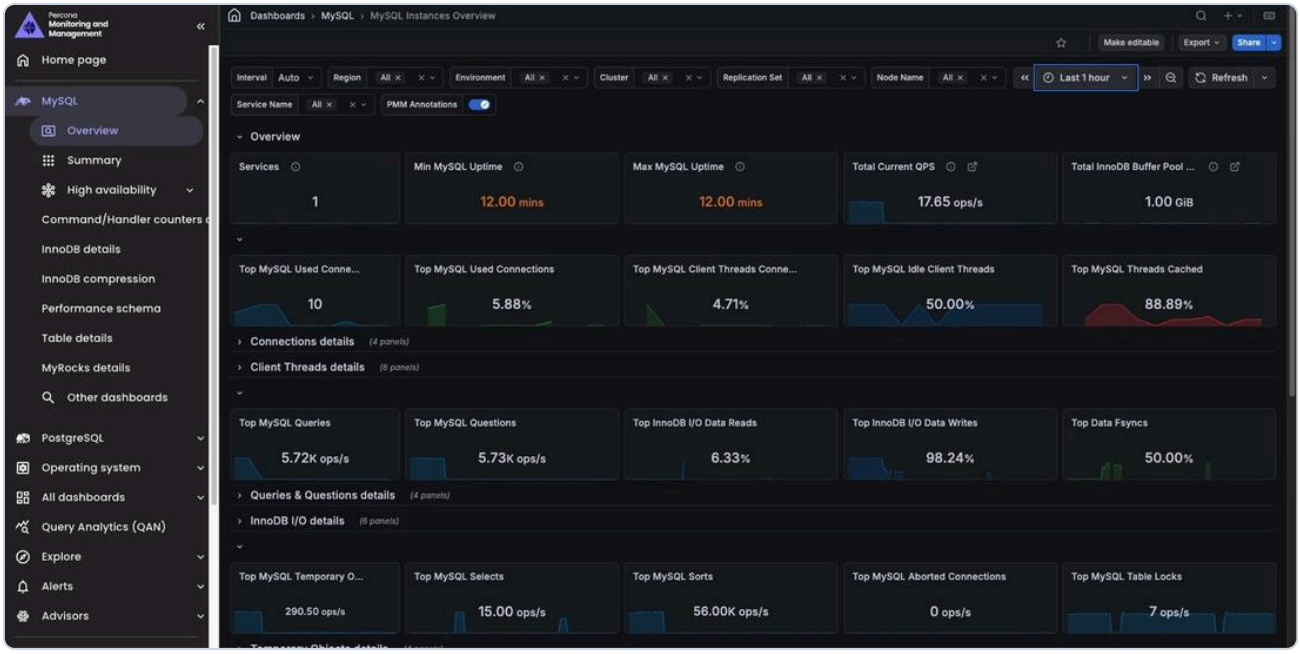


Figure 7.40 28 Oct 2026 · 13:23:56



Figure 7.41 28 Oct 2026 · 13:24:47

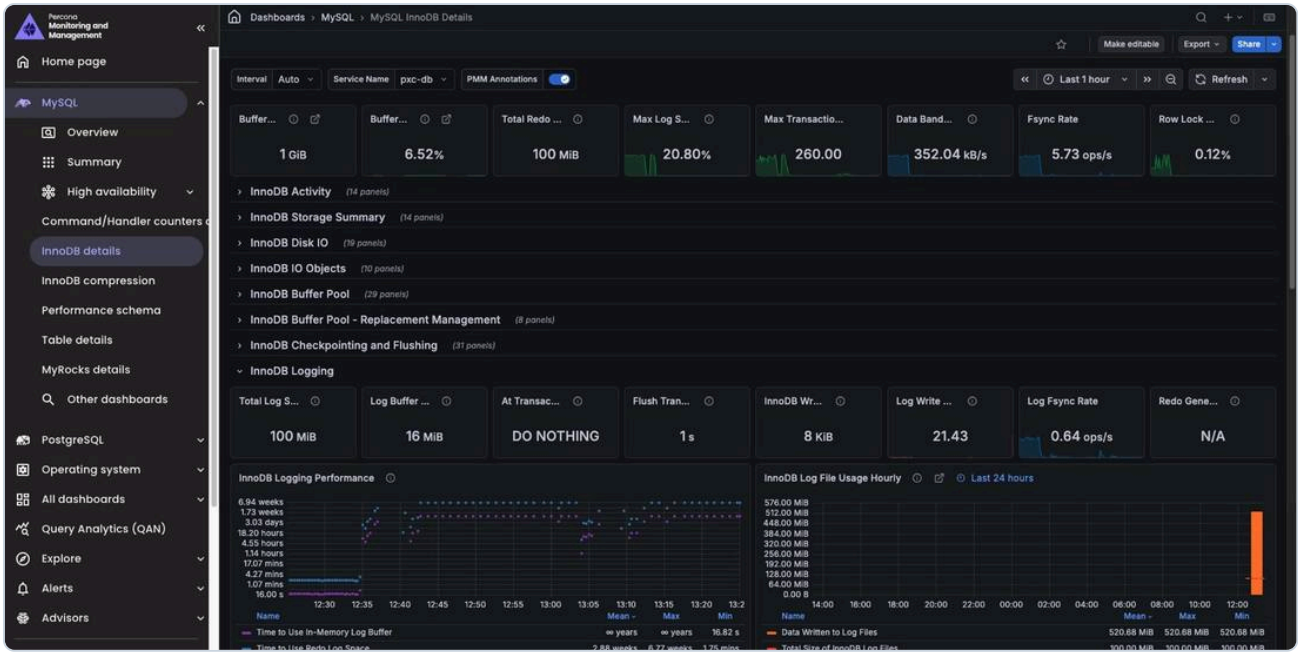


Figure 7.42 28 Oct 2026 · 13:25:14

7.15 Unregister MySQL from PMM

Unregister | PMM | MySQL | Completed

MySQL is removed from PMM; the database itself keeps running.

Started 28 Oct 2026, 13:25 · took 24s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

7.16 Remove PMM

Unregister | PMM | Completed

PMM is removed from the cluster once no database is registered with it.

Started 28 Oct 2026, 13:30 · took 2m 00s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh pmm
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable

- ✔ AWS/EKS context validated
- ✔ PMM Client deployment removed
- ✔ PMM namespace absent

7.17 Deploy PostgreSQL

Deploy database | PostgreSQL | Completed

RunShore deploys PostgreSQL on AWS · EKS and waits until every node is ready.

Started 28 Oct 2026, 13:32 · took 6m 17s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh postgres
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ WAL archive: <backup location>/platform-postgres-20261028133433/ (rendered: runtime/rendered/postgres/cluster.yaml)
- ✔ PostgreSQL platform deployed

7.18 Install New Relic

Deploy tool | New Relic | Completed

New Relic is installed, ready to monitor the databases in this cluster.

Started 28 Oct 2026, 13:39 · took 1m 12s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh newrelic
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ New Relic Helm repository ready
- ✔ New Relic Helm release deployed

7.19 Register PostgreSQL with New Relic

Register | New Relic | PostgreSQL | Completed

PostgreSQL is connected to New Relic, so its metrics and dashboards appear there.

Started 28 Oct 2026, 13:42 · took 40s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/register.sh
```

- ✔ Kubernetes, PostgreSQL and New Relic available
- ✔ Prerequisites available
- ✔ Monitoring secret available: postgres-monitoring
- ✔ Monitoring credentials loaded
- ✔ New Relic agents refreshed
- ✔ nri-postgresql configuration present

7.20 Failover test · PostgreSQL

Failover | PostgreSQL | Completed

The primary PostgreSQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 28 Oct 2026, 13:44 · took 1m 01s

COMMAND

```
SCOPE=aws db-platform/postgres/failover/kubernetes/run_failover.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ Final CNPG phase: Cluster in healthy state
- ✔ Final ready instances: 3

7.21 Workload · PostgreSQL

Workload | PostgreSQL | Completed

A load generator runs against PostgreSQL, so the monitoring shows real traffic. It stops by itself after

a set time.

Started 28 Oct 2026, 13:50 · took 20s

COMMAND

```
bash db-platform/postgres/workload/kubernetes/start.sh
```

- ✔ pgbench schema initialized.
- ✔ pgbench runtime permissions ensured for appuser.
- ✔ pgbench schema validated.

7.22 Point-in-time recovery test · PostgreSQL

Assessment | PostgreSQL | Completed

RunShore restores PostgreSQL to a chosen moment from its backups and checks the restored data is correct.

Started 28 Oct 2026, 13:52 · took 7m 33s

COMMAND

```
SCOPE=aws db-platform/postgres/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ PITR clone submitted
- ✔ PITR clone ready

7.23 Replication test · PostgreSQL

Replication | PostgreSQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 28 Oct 2026, 14:00 · took 1m 12s

COMMAND

```
SCOPE=aws db-platform/postgres/replication/kubernetes/run_replication.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.

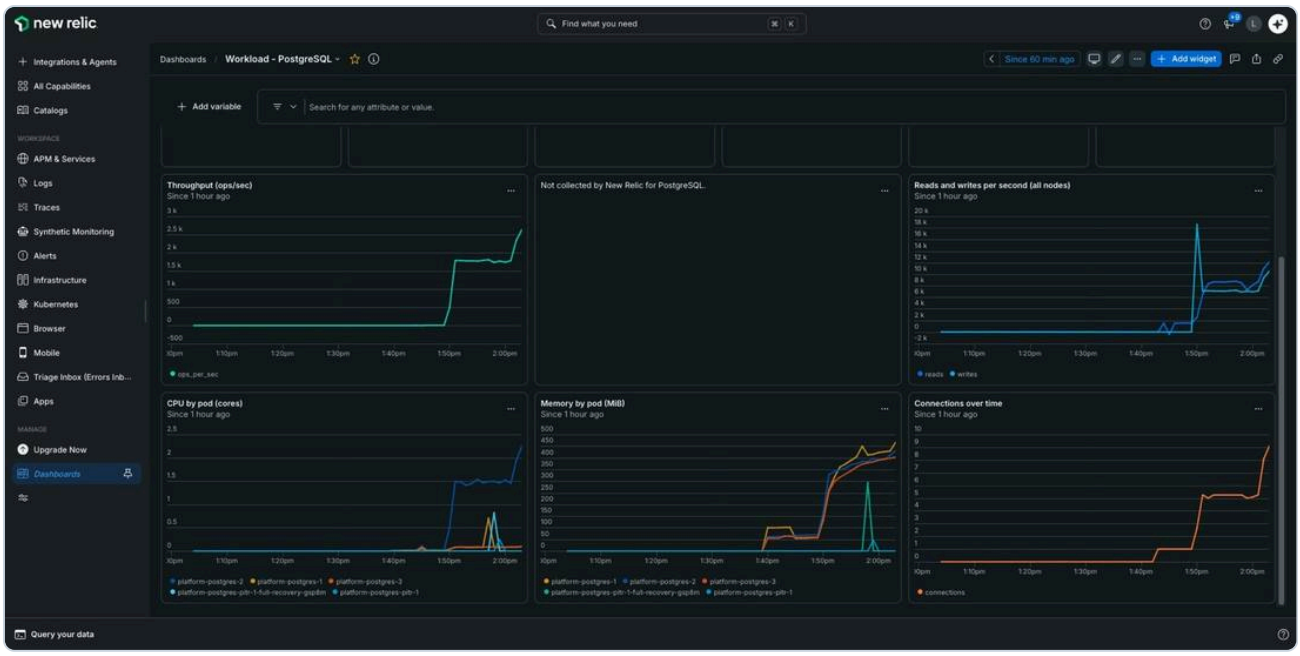


Figure 7.43 28 Oct 2026 · 14:04:08

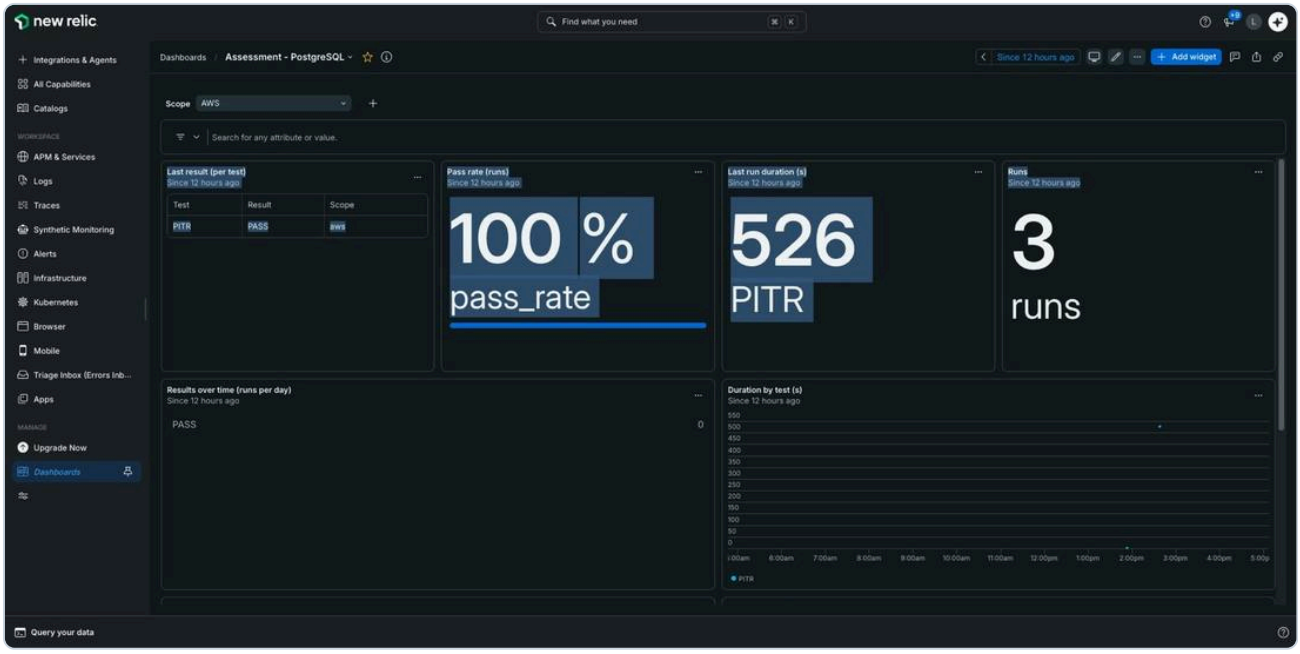


Figure 7.44 28 Oct 2026 · 17:23:51

7.24 Unregister PostgreSQL from New Relic

Unregister | New Relic | PostgreSQL | Completed

PostgreSQL is removed from New Relic; the database itself keeps running.

Started 28 Oct 2026, 17:25 · took 16s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/unregister.sh
```

✔ Kubernetes API reachable

✔ PostgreSQL integration configuration absent

- ✔ New Relic PostgreSQL telemetry secret absent
- ✔ newrelic-postgres-secret absent
- ✔ New Relic agents refreshed
- ✔ postgres-config.yml absent

7.25 Destroy PostgreSQL

Destroy | PostgreSQL | Completed

PostgreSQL is removed and its storage released.

Started 28 Oct 2026, 17:26 · took 1m 08s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh postgres
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ PostgreSQL pods stopped (20s)
- ✔ CloudNativePG operator removed

7.26 Deploy MongoDB

Deploy database | MongoDB | Completed

RunShore deploys MongoDB on AWS · EKS and waits until every node is ready.

Started 28 Oct 2026, 17:27 · took 2m 37s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh mongodb
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ Bound PVCs: 3
- ✔ MongoDB platform validation complete

7.27 Register MongoDB with New Relic

Register | New Relic | MongoDB | Completed

MongoDB is connected to New Relic, so its metrics and dashboards appear there.

Started 28 Oct 2026, 17:41 · took 1m 53s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/register.sh
```

- ✔ Kubernetes API reachable
- ✔ New Relic deployment available
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ MongoDB namespace: mongodb
- ✔ Exporter reachable inside the cluster: http://mongodb-newrelic-exporter.mongodb.svc.cluster.local:9216/metri (mongodb_up 1)
- ✔ New Relic scraped MongoDB with no errors since 2026-10-28T17:42:47Z

7.28 Failover test · MongoDB

Failover | MongoDB | Completed

The primary MongoDB node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 28 Oct 2026, 17:43 · took 1m 16s

COMMAND

```
SCOPE=aws db-platform/mongodb/failover/kubernetes/run_failover.sh
```

- ✔ MONGODB REPLICATION TEST STOPPED
- ✔ Native PSMDB replica-set replication was not modified.
- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ PSMDB cluster is ready.
- ✔ HA validation marker survived failover.
- ✔ MongoDB replica set recovered to 3/3 ready members.

7.29 Replication test · MongoDB

Replication | MongoDB | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 28 Oct 2026, 17:45 · took 1m 25s

COMMAND

```
SCOPE=aws db-platform/mongodb/replication/kubernetes/run_replication.sh
```

- ✔ MONGODB REPLICATION TEST STOPPED
- ✔ Native PSMDB replica-set replication was not modified.
- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Three-member replica-set topology verified.
- ✔ Replicated data matches.
- ✔ MONGODB REPLICATION: PASS

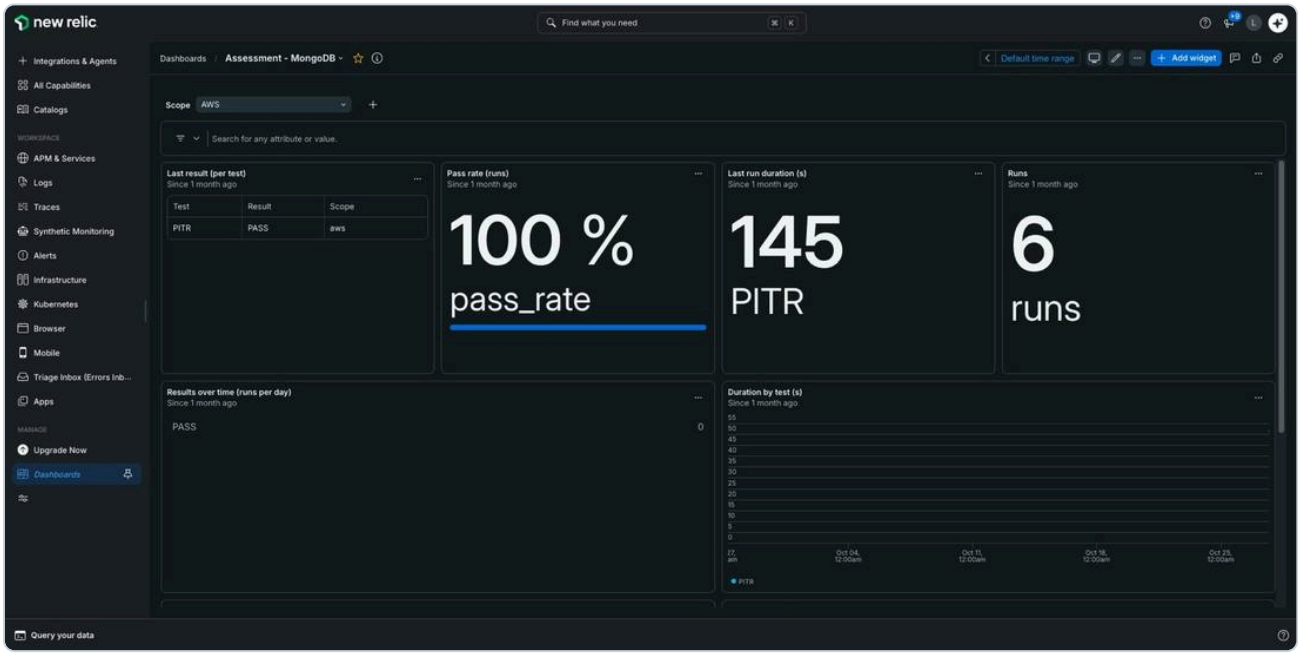


Figure 7.45 28 Oct 2026 · 17:52:20

7.30 Point-in-time recovery test · MongoDB

Assessment | MongoDB | Completed

RunShore restores MongoDB to a chosen moment from its backups and checks the restored data is

correct.

Started 28 Oct 2026, 17:47 · took 2m 27s

COMMAND

```
SCOPE=aws db-platform/mongodb/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ MONGODB REPLICATION TEST STOPPED
- ✔ Native PSMDB replica-set replication was not modified.
- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Cluster: mongodb-psmdb-db
- ✔ Post-marker orders inserted
- ✔ Archived oplog covers the target (2026-10-28T17:48:45 - 2026-10-28T17:49:10) after 10s

7.31 Workload · MongoDB

Workload | MongoDB | Completed

A load generator runs against MongoDB, so the monitoring shows real traffic. It stops by itself after a set time.

Started 28 Oct 2026, 17:52 · took 8s

COMMAND

```
bash db-platform/mongodb/workload/kubernetes/start.sh
```

- ✔ MongoDB YCSB workload Pod is Running.

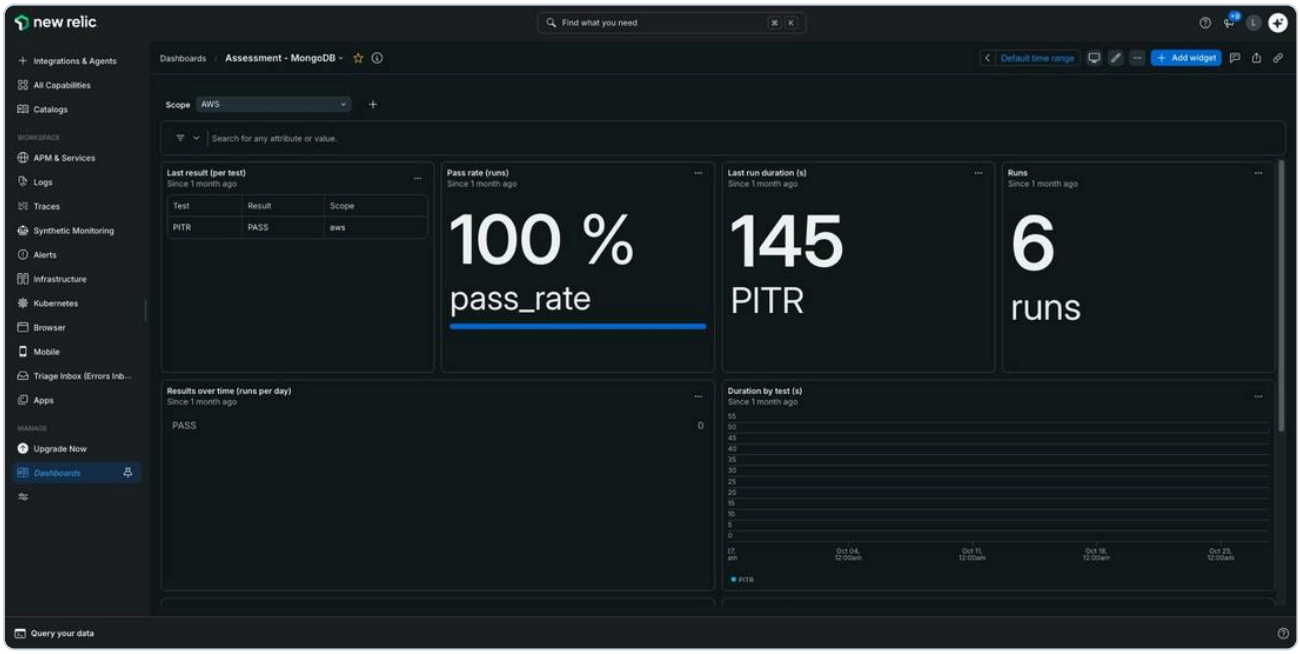


Figure 7.46 28 Oct 2026 · 17:54:34

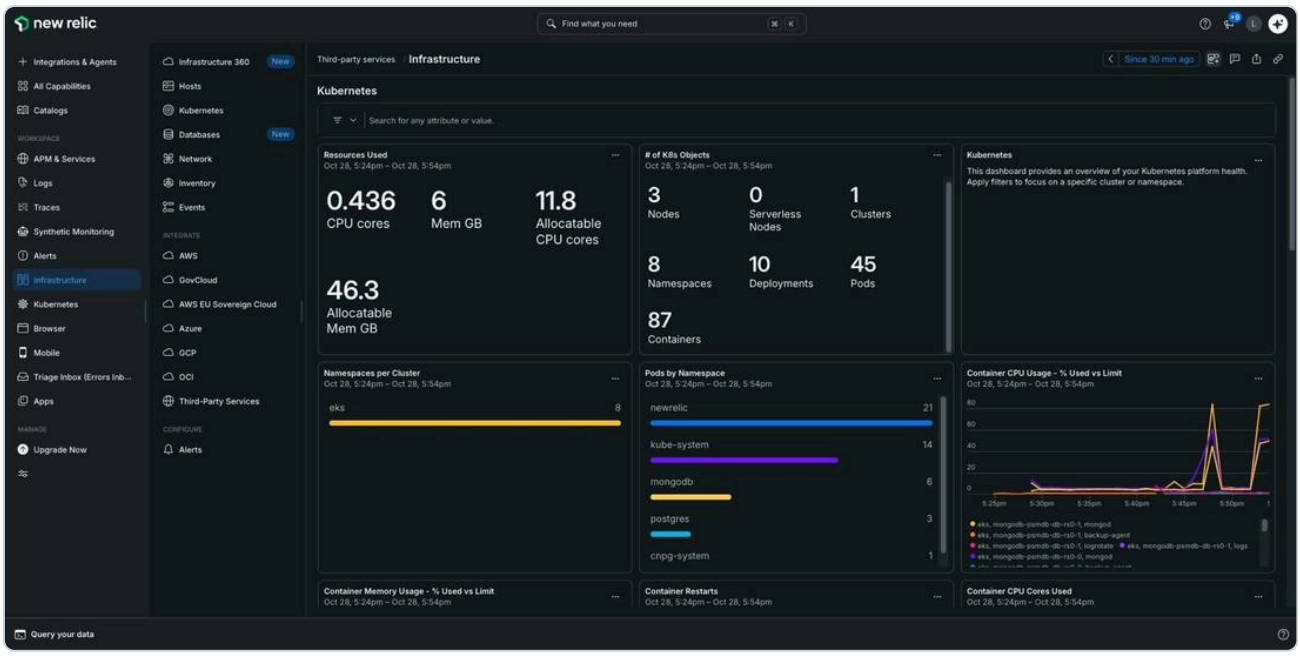


Figure 7.47 28 Oct 2026 · 17:55:07

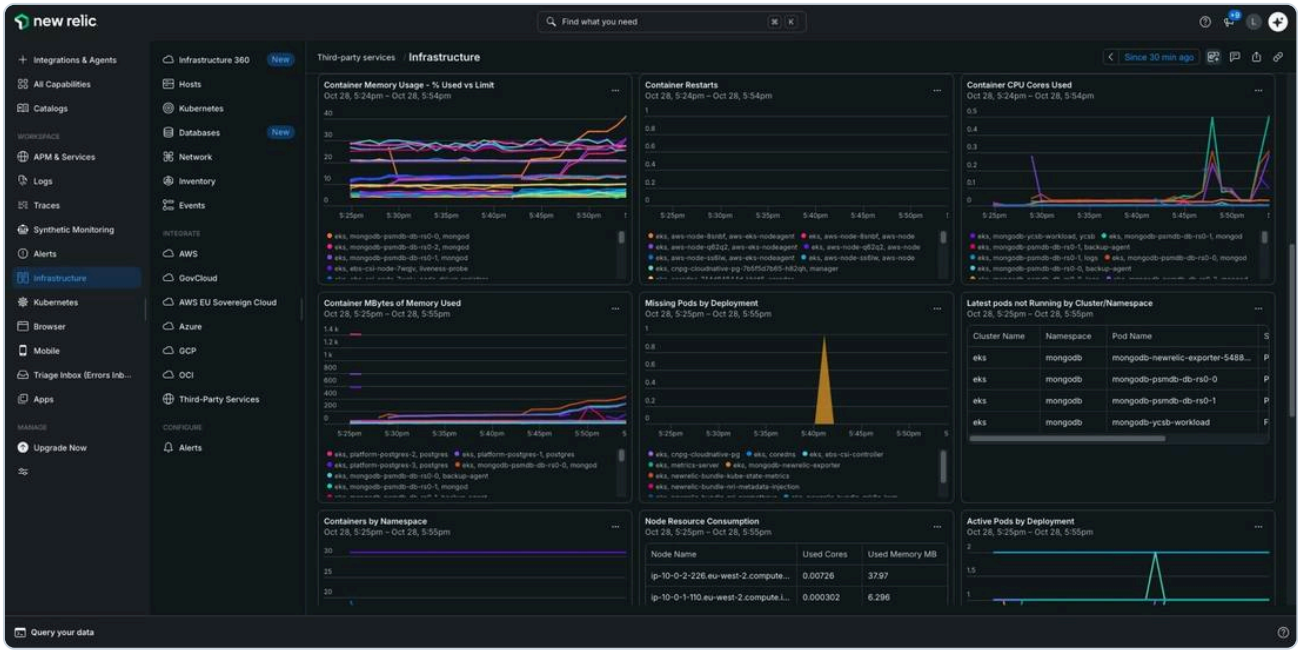


Figure 7.48 28 Oct 2026 · 17:55:53

7.32 Unregister MongoDB from New Relic

Unregister	New Relic	MongoDB	Completed

MongoDB is removed from New Relic; the database itself keeps running.

Started 28 Oct 2026, 18:11 · took 24s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/unregister.sh
```

✔ Kubernetes API reachable

- ✔ MongoDB New Relic registration state removed
- ✔ Removed Deployment/mongodb-newrelic-exporter (namespace mongodb)
- ✔ Removed Service/mongodb-newrelic-exporter (namespace mongodb)
- ✔ New Relic kubelet agents refreshed
- ✔ mongodb-config.yml absent

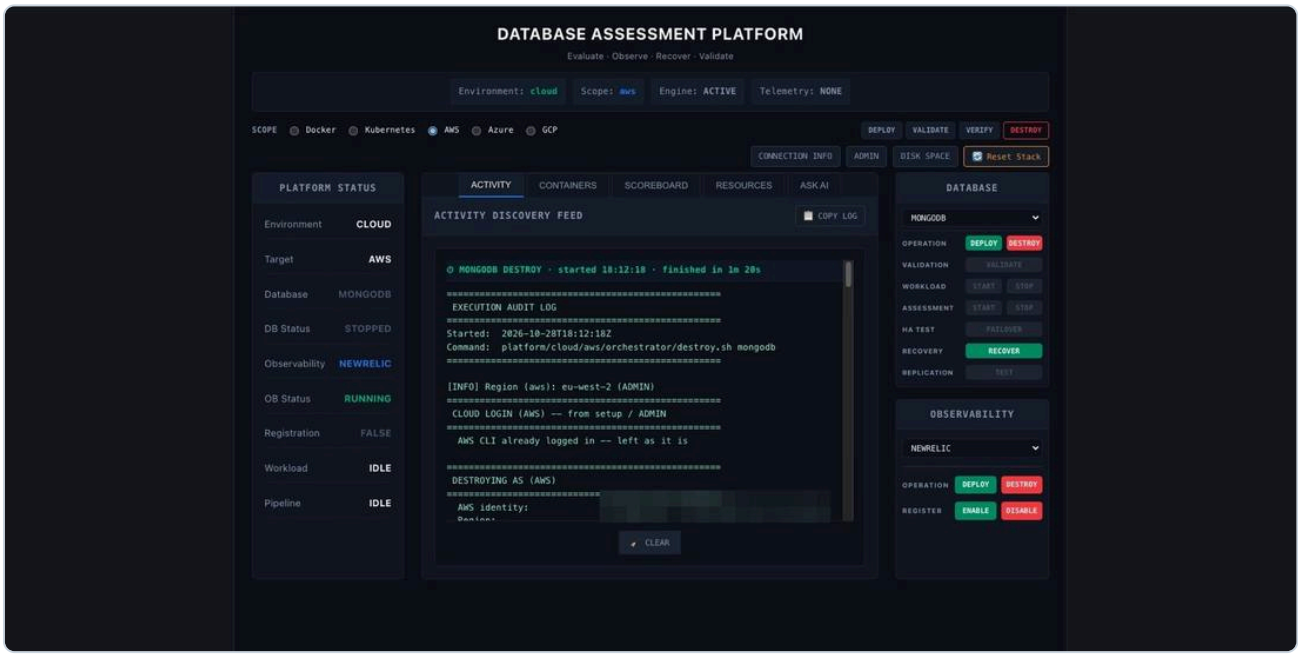


Figure 7.49 28 Oct 2026 · 18:13:53

7.33 Destroy MongoDB

Destroy | MongoDB | Completed

MongoDB is removed and its storage released.

Started 28 Oct 2026, 18:12 · took 1m 20s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh mongodb
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ MongoDB namespace absent
- ✔ Bucket deleted: mongodb-db-backups-<account-id>-1785249625

7.34 Deploy MySQL

Deploy database | MySQL | Completed

RunShore deploys MySQL on AWS · EKS and waits until every node is ready.

Started 28 Oct 2026, 18:17 · took 4m 25s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh mysql
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ PXC cluster health converged
- ✔ PXC deployment complete

7.35 Register MySQL with New Relic

Register | New Relic | MySQL | Completed

MySQL is connected to New Relic, so its metrics and dashboards appear there.

Started 28 Oct 2026, 18:24 · took 32s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/register.sh
```

- ✔ Kubernetes, PXC and New Relic available
- ✔ PXC cluster ready
- ✔ MySQL endpoint discovered
- ✔ PXC administrative credential available
- ✔ PXC HAProxy endpoint present
- ✔ MySQL telemetry secret present

7.36 Failover test · MySQL

Failover | MySQL | Completed

The primary MySQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 28 Oct 2026, 18:25 · took 3m 05s

COMMAND

```
SCOPE=aws db-platform/mysql/failover/kubernetes/run_failover.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.

- ✔ PXC administrative credential available
- ✔ All three PXC members are Kubernetes-Ready and Primary/Synced/Ready
- ✔ Assessment data verified on pxc-db-pxc-2
- ✔ Final three-member cluster is Kubernetes-Ready and Primary/Synced/Ready

7.37 Replication test · MySQL

Replication | MySQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 28 Oct 2026, 18:32 · took 2m 01s

COMMAND

```
SCOPE=aws db-platform/mysql/replication/kubernetes/run_replication.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.

7.38 Unregister MySQL from New Relic

Unregister | New Relic | MySQL | Completed

MySQL is removed from New Relic; the database itself keeps running.

Started 28 Oct 2026, 20:23 · took 20s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/unregister.sh
```

- ✔ Kubernetes and New Relic available
- ✔ MySQL registration state cleared
- ✔ Removed mysql-config.yml
- ✔ Removed newrelic-mysql-secret
- ✔ mysql-config.yml absent
- ✔ newrelic-mysql-secret absent

7.39 Remove New Relic

Unregister | New Relic | Completed

New Relic is removed from the cluster once no database is registered with it.

Started 28 Oct 2026, 20:42 · took 36s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh newrelic
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ Helm release removed
- ✔ Namespace removed

7.40 Install Splunk

Deploy tool | Splunk | Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 28 Oct 2026, 20:44 · took 5m 37s

COMMAND

```
platform/cloud/aws/orchestrator/deploy.sh splunk
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

7.41 Register MySQL with Splunk

Register | Splunk | MySQL | Completed

MySQL is connected to Splunk, so its metrics and dashboards appear there.

Started 28 Oct 2026, 20:53 · took 3m 33s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/register.sh
```

- ✔ PXC: pxc-db
- ✔ Splunk HEC service: splunk-s1-standalone-service

- ✔ Splunk runtime Secret: splunk-splunk-secret
- ✔ mysql/mysql-splunk-hec converged
- ✔ Percona logrotate container present
- ✔ MySQL → Splunk OTel registration complete

7.42 Workload · MySQL

Workload | MySQL | Completed

A load generator runs against MySQL, so the monitoring shows real traffic. It stops by itself after a set time.

Started 28 Oct 2026, 21:07 · took 37s

COMMAND

```
bash db-platform/mysql/workload/start.sh
```

- ✔ MySQL assessment database available: assessmentdb

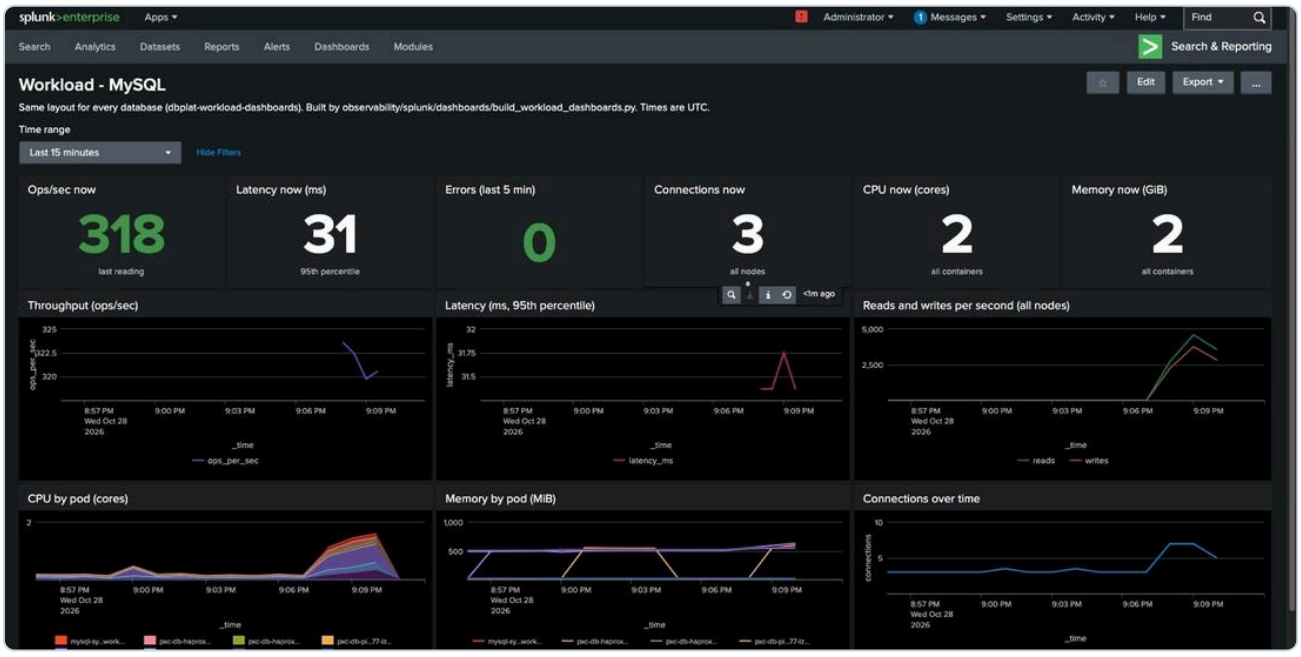


Figure 7.50 28 Oct 2026 · 21:11:43

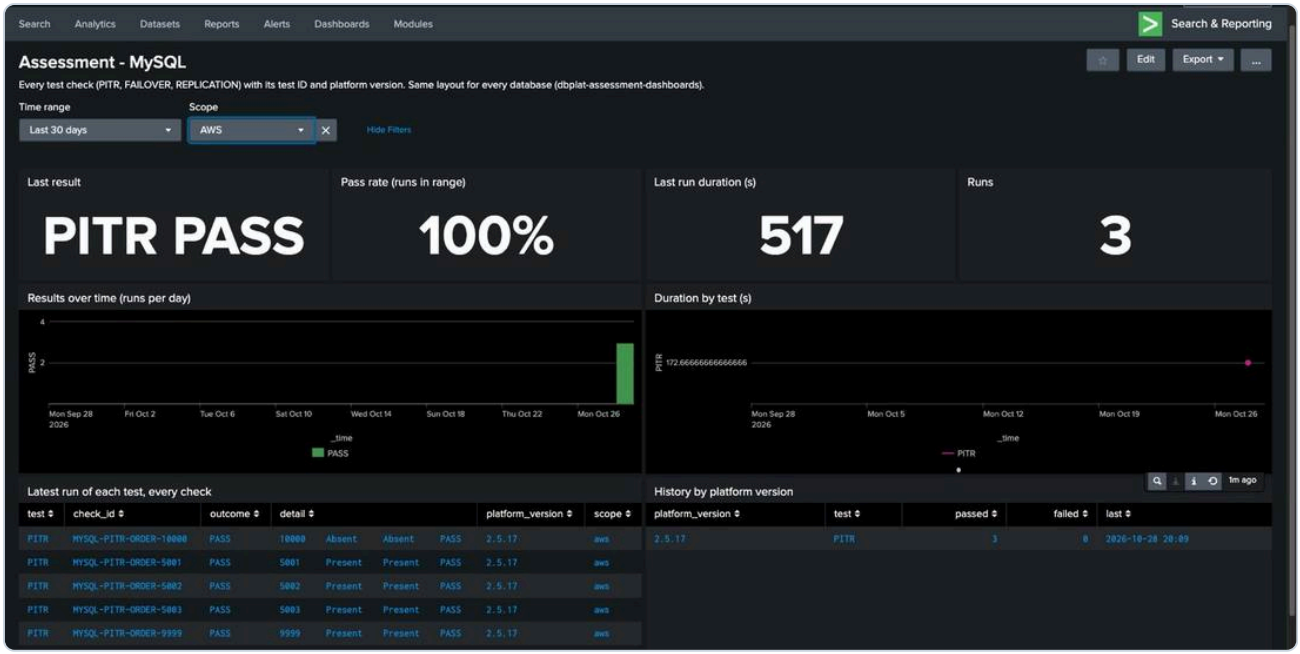


Figure 7.51 28 Oct 2026 · 21:12:22



Figure 7.52 28 Oct 2026 · 21:15:26

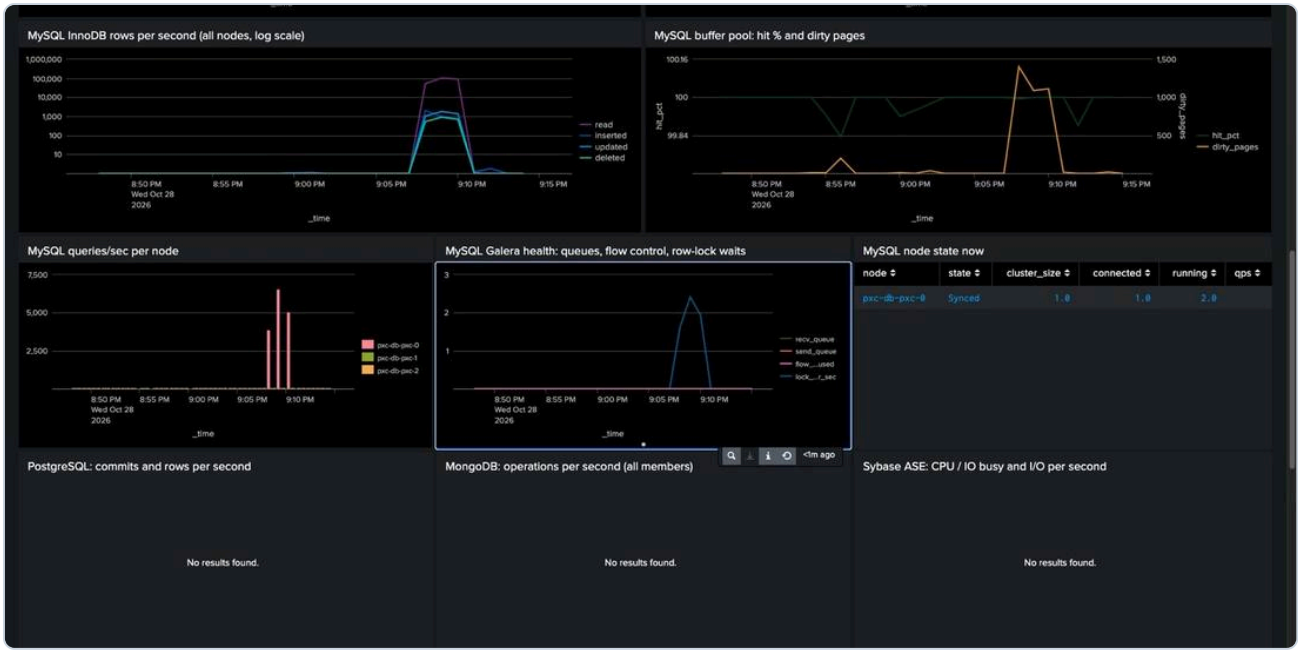


Figure 7.53 28 Oct 2026 · 21:16:41

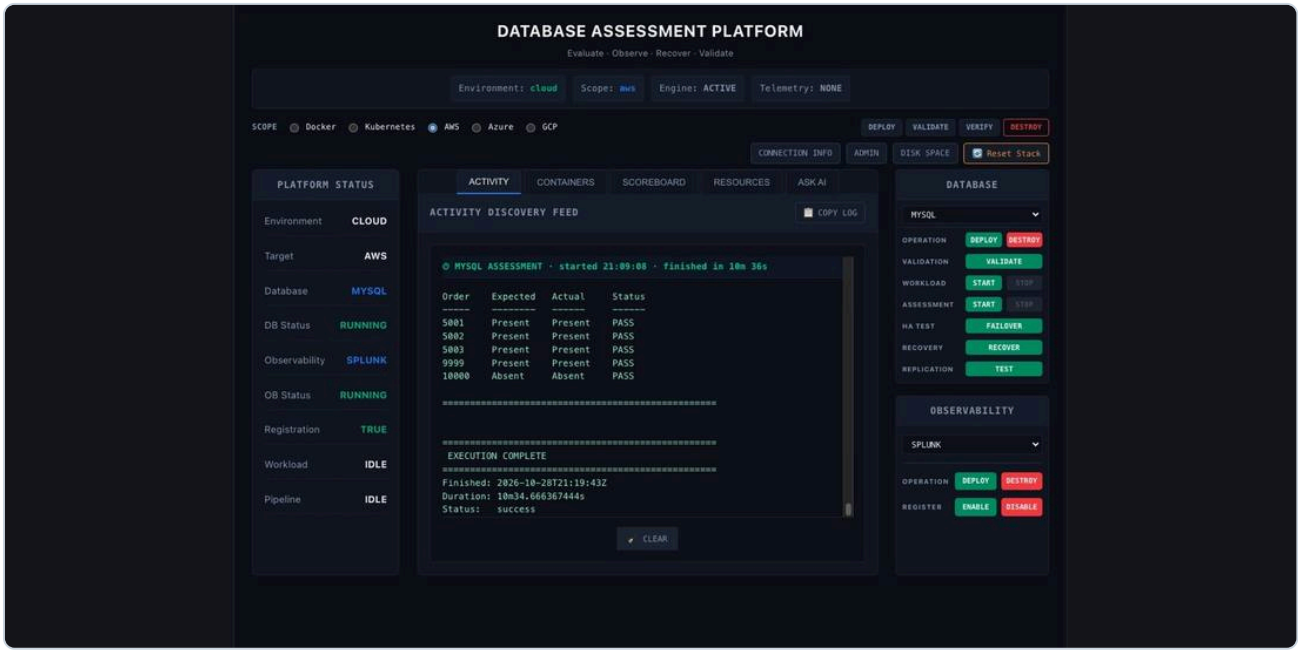


Figure 7.54 28 Oct 2026 · 21:20:59

7.43 Point-in-time recovery test · MySQL

Assessment	MySQL	Completed with warnings

RunShore restores MySQL to a chosen moment from its backups and checks the restored data is

correct.

Started 28 Oct 2026, 21:09 · took 10m 36s

COMMAND

```
SCOPE=aws db-platform/mysql/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ Live root credential resolved from Kubernetes Secret pxc-db-secrets
- ✔ Prerequisites available
- ✔ Restore completed: pitr-demo-restore-1793222028
- ✔ PXC cluster ready

7.44 Destroy MySQL

Destroy | MySQL | Completed

MySQL is removed and its storage released.

Started 28 Oct 2026, 21:22 · took 3m 08s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh mysql
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable
- ✔ AWS/EKS context validated
- ✔ Namespace database fully removed
- ✔ Bucket deleted: mysql-db-backups-<account-id>-1785238733

7.45 Remove Splunk

Unregister | Splunk | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 28 Oct 2026, 21:47 · took 2m 18s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh splunk
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ Kubernetes API reachable

- ✔ AWS/EKS context validated
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

7.46 Destroy the AWS environment

Destroy | **Completed**

Everything RunShore created on AWS is removed, then checked: nothing that can still be billed is left behind.

Started 28 Oct 2026, 21:50 · took 11m 51s

COMMAND

```
platform/cloud/aws/orchestrator/destroy.sh kubernetes
```

- ✔ Terraform state: S3 bucket dbplat-tfstate-<account-id>-eu-west-2
- ✔ kubectl connected to database-assessment-platform-eks
- ✔ AWS/EKS context validated
- ✔ database-assessment-platform-eks-cluster-role: already gone
- ✔ Terraform holds no VPC that still exists
- ✔ delete S3 bucket postgres-db-backups-<account-id>-1785225079

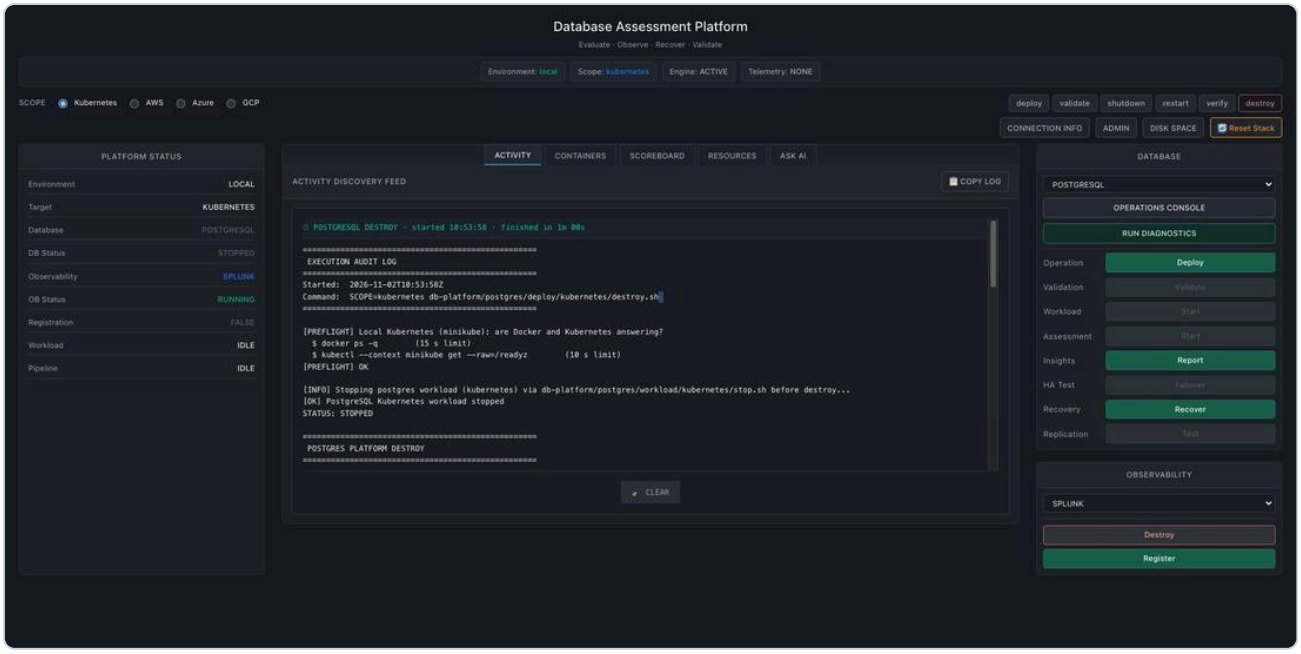


Figure 7.55 2 Nov 2026 · 10:59:56

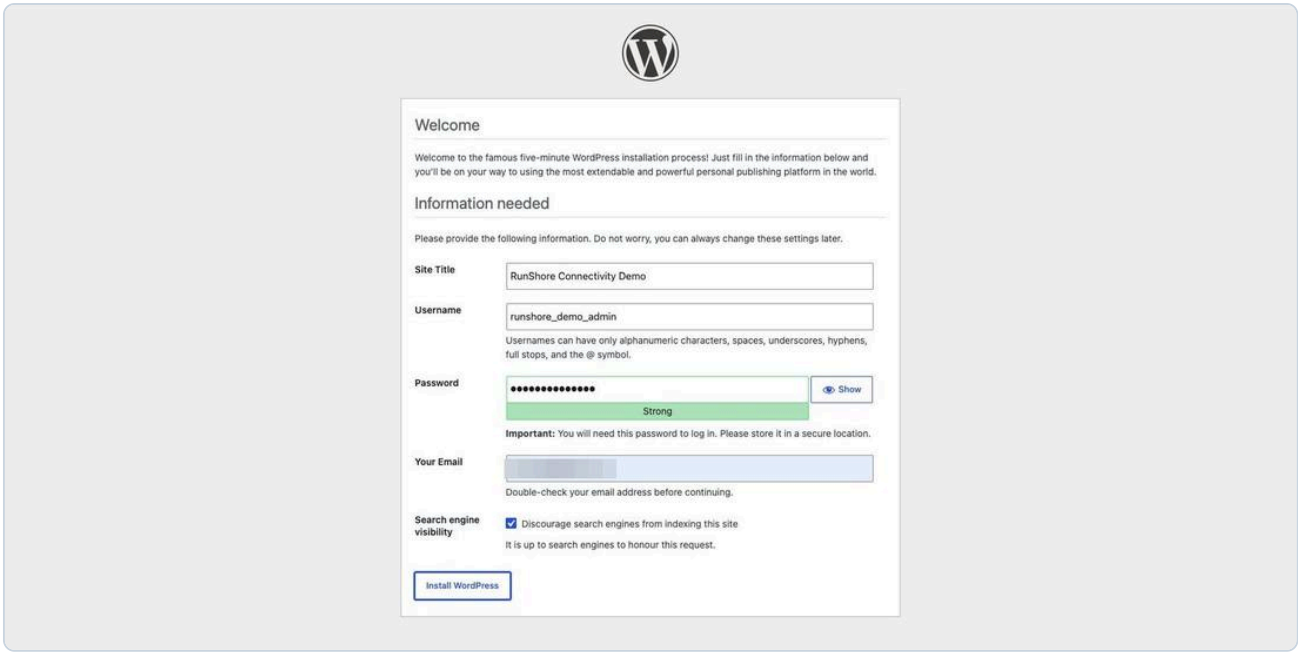


Figure 7.56 2 Nov 2026 · 17:02:19

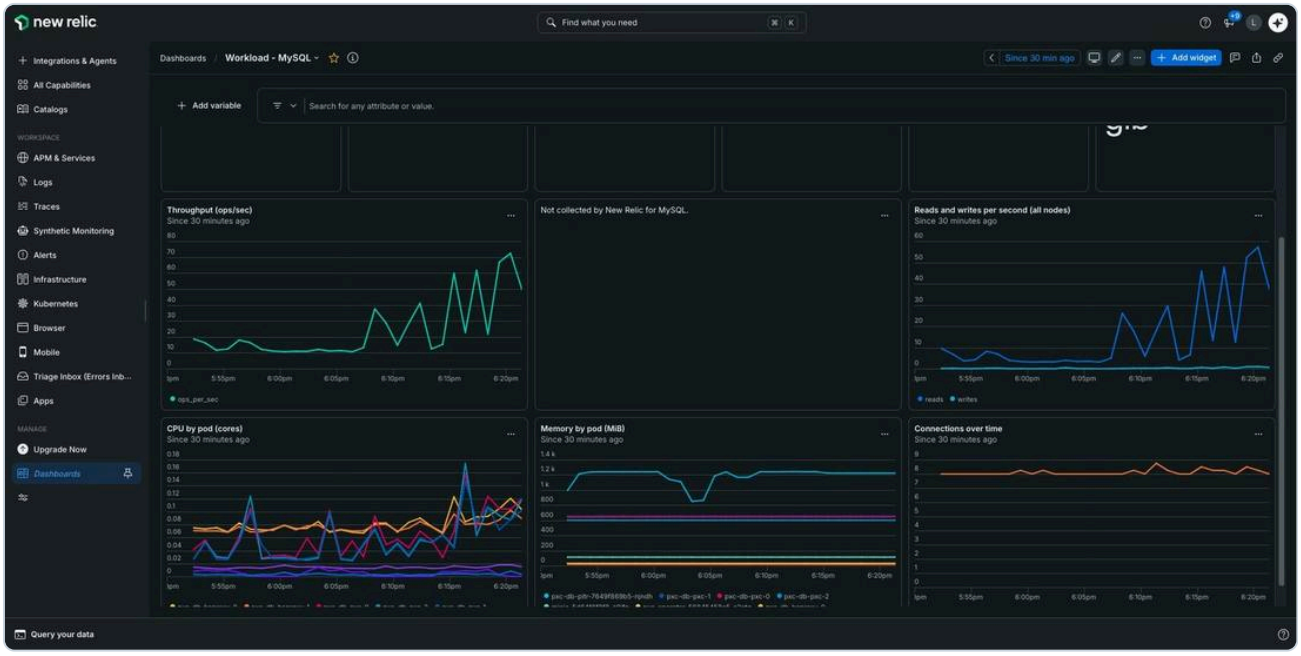


Figure 7.57 2 Nov 2026 · 18:22:30

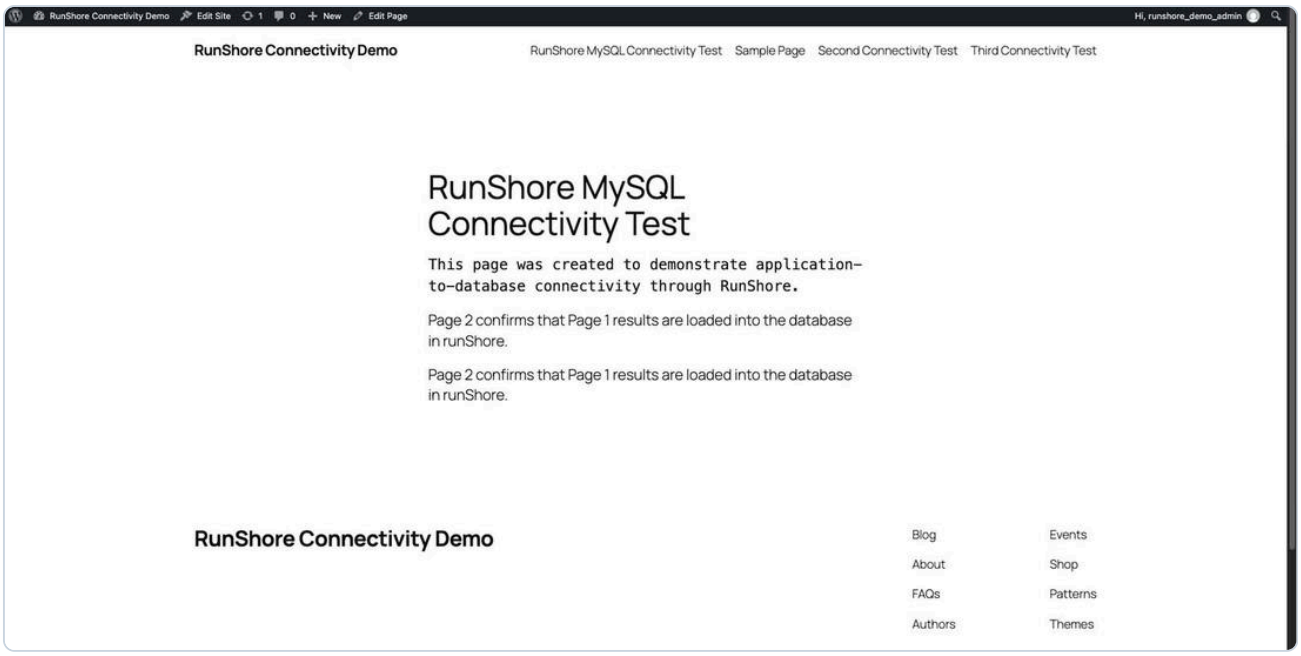


Figure 7.58 3 Nov 2026 · 06:56:00

08 Azure AKS

Microsoft Azure · AKS

The same workflow on Microsoft Azure, using an AKS cluster built with Terraform, ending with a verified destroy.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Microsoft Azure · AKS	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	34 screens · 43 logs	27 Oct 2026

8.1 Screenshots

What the console and dashboards showed at this point.

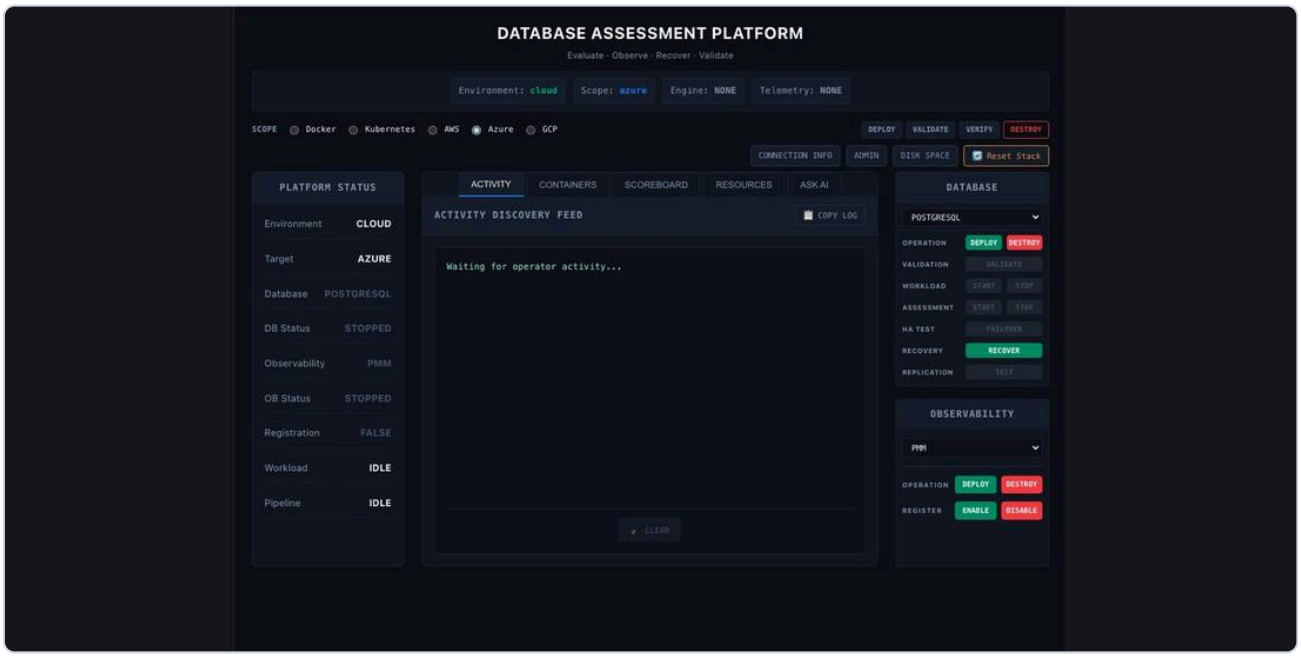


Figure 8.1 27 Oct 2026 · 03:26:58

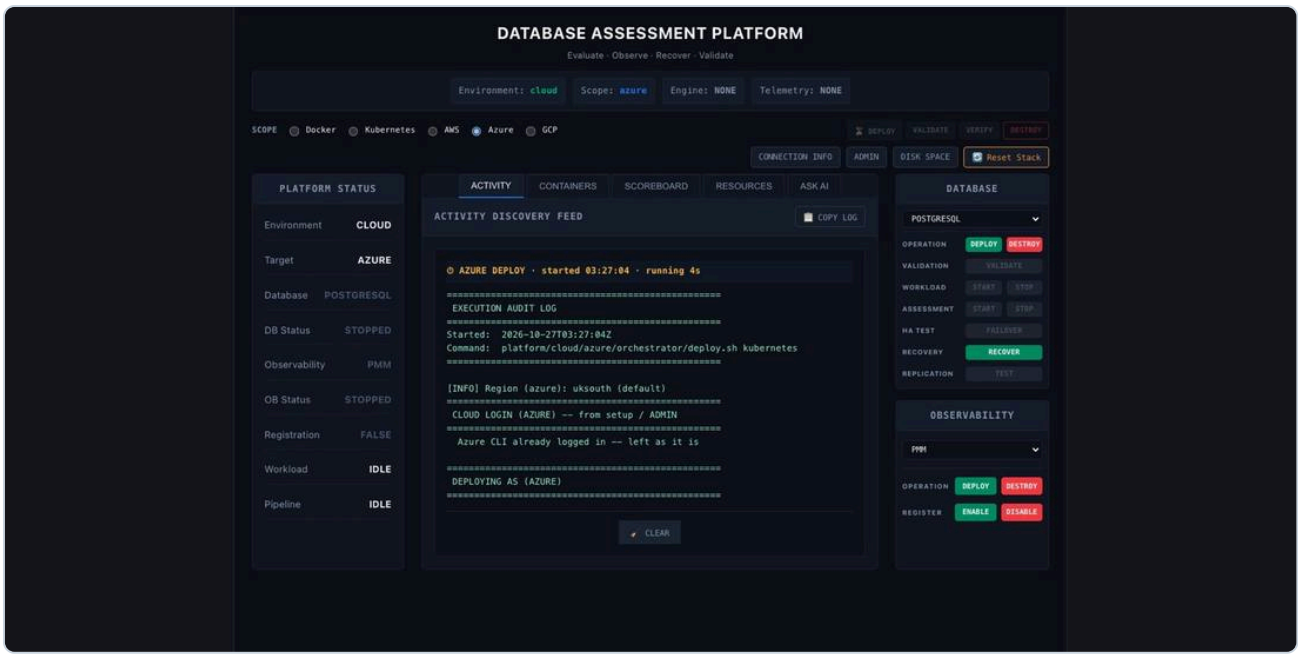


Figure 8.2 27 Oct 2026 · 03:27:09

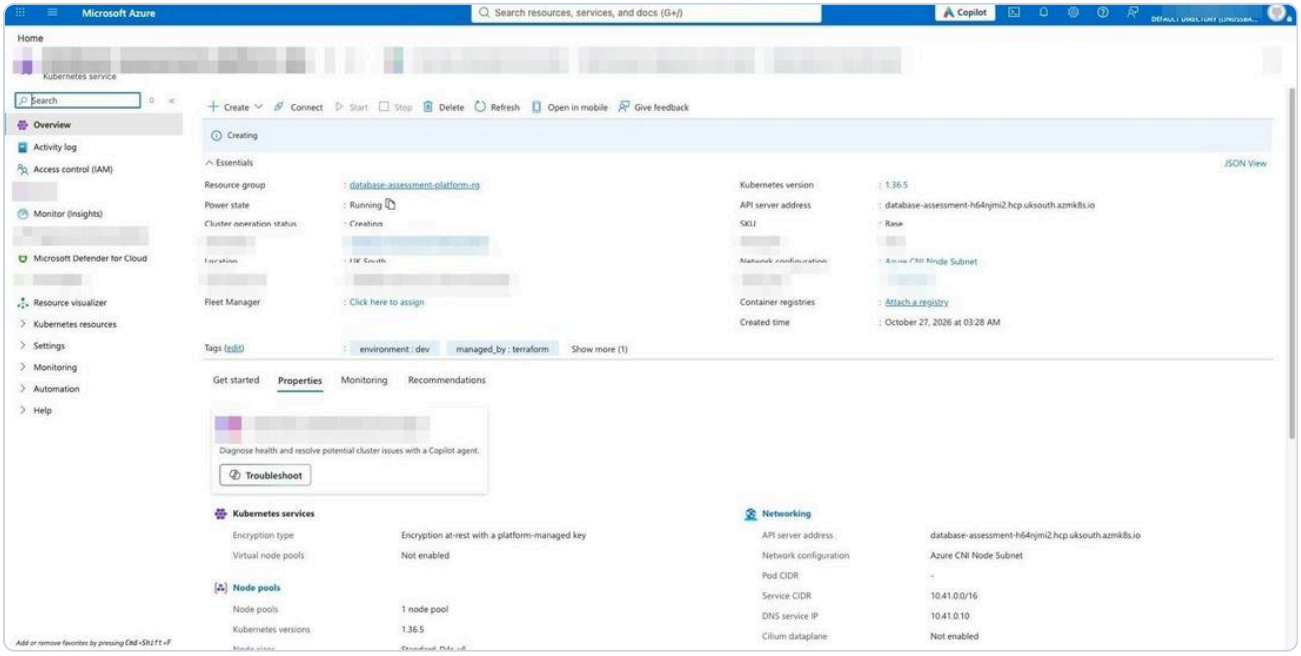


Figure 8.3 27 Oct 2026 · 03:29:36

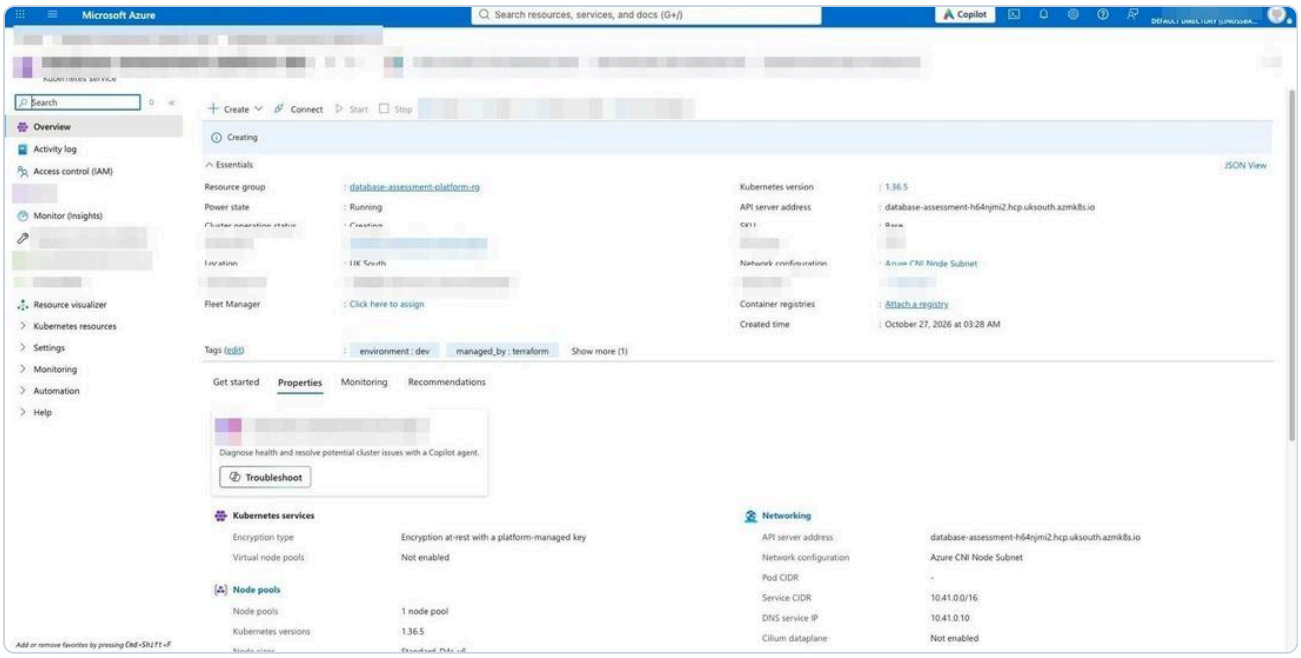


Figure 8.4 27 Oct 2026 · 03:30:12

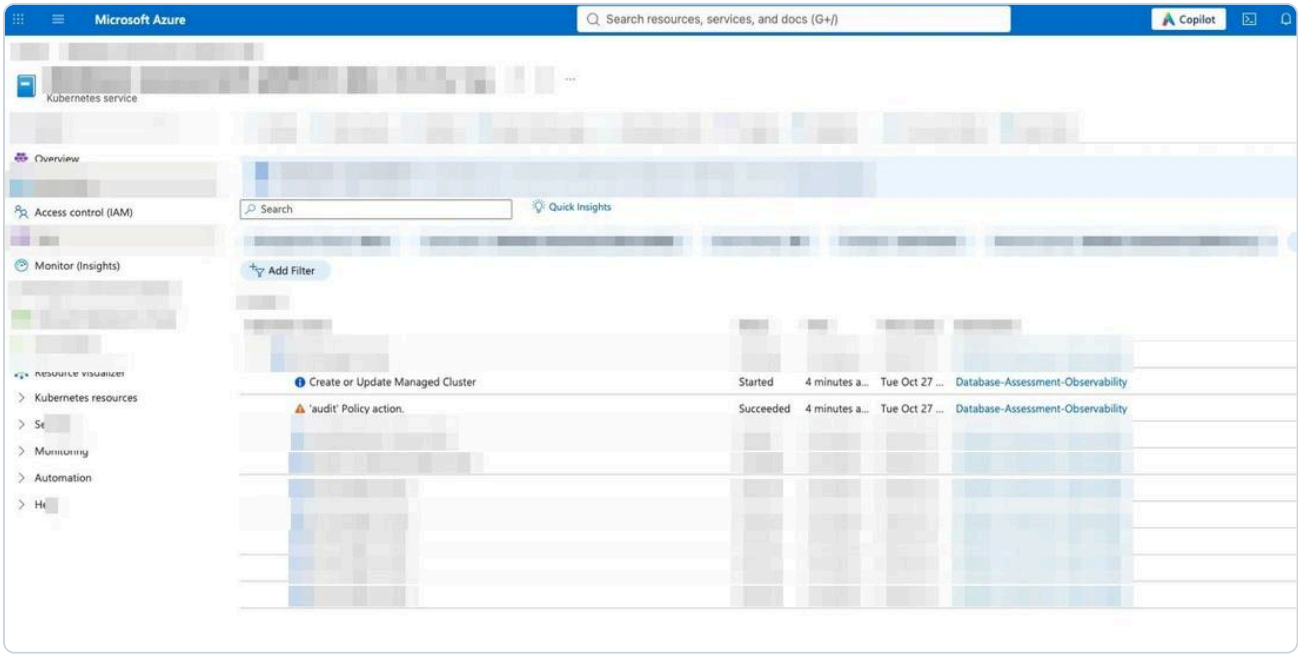


Figure 8.5 27 Oct 2026 · 03:34:12

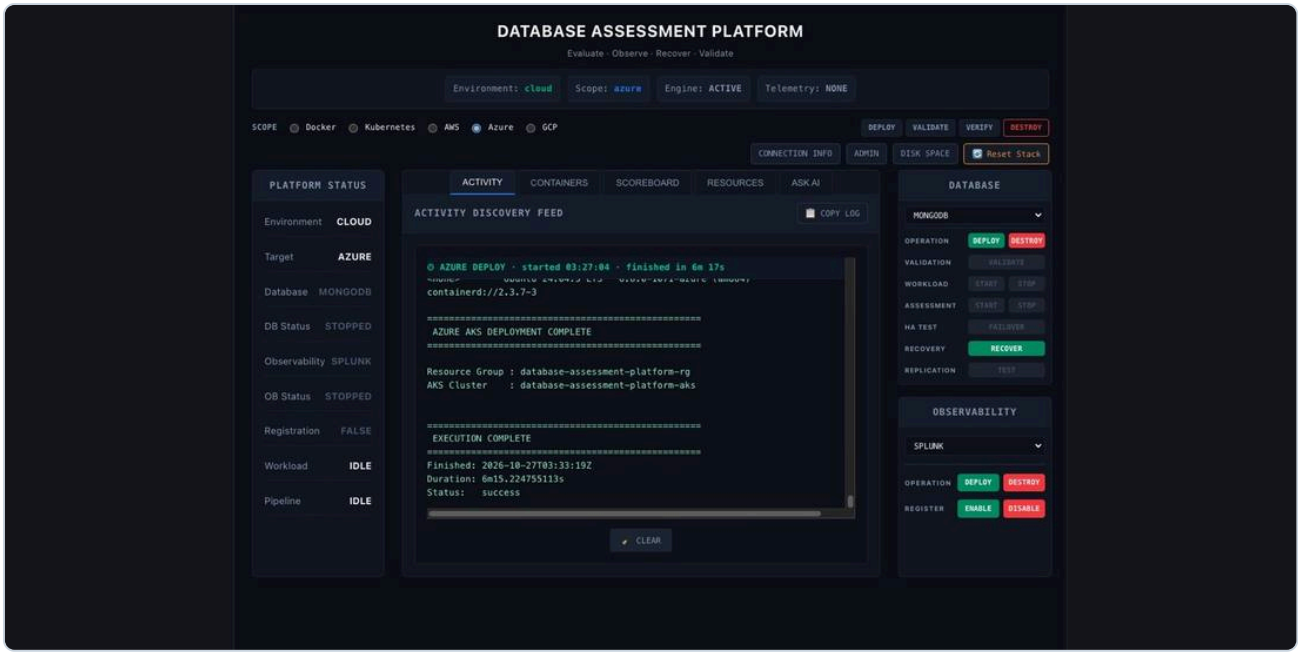


Figure 8.6 27 Oct 2026 · 03:37:53

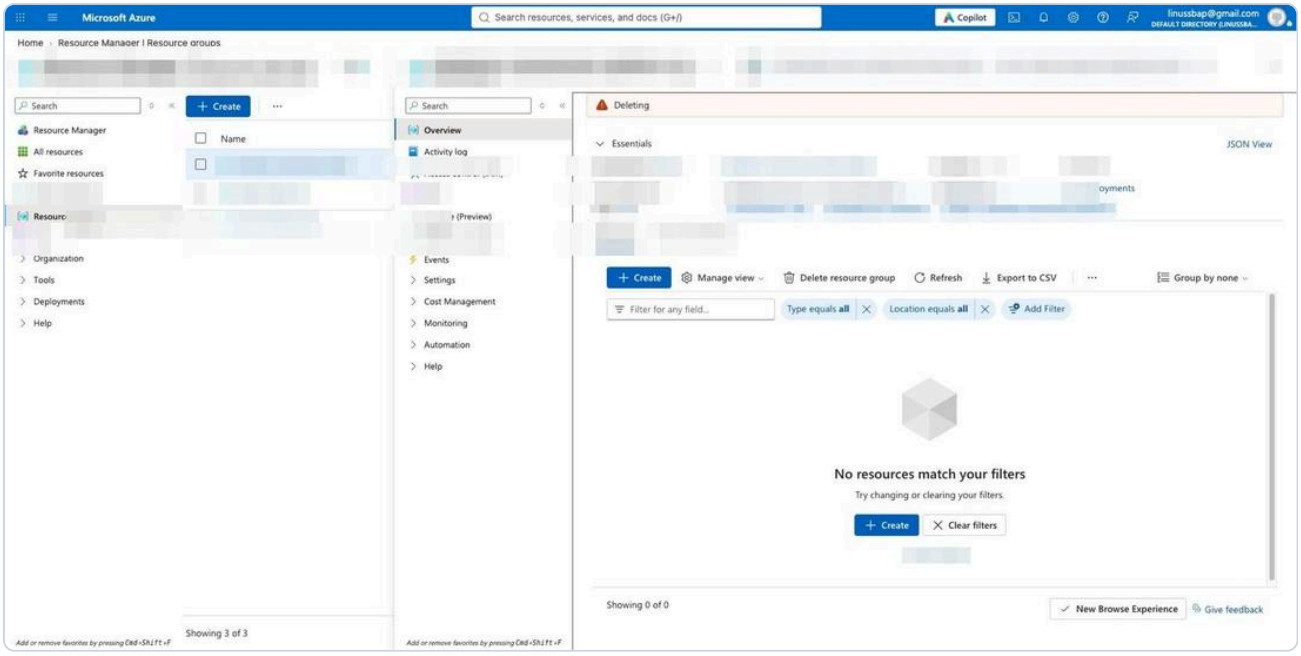


Figure 8.7 27 Oct 2026 · 03:46:22

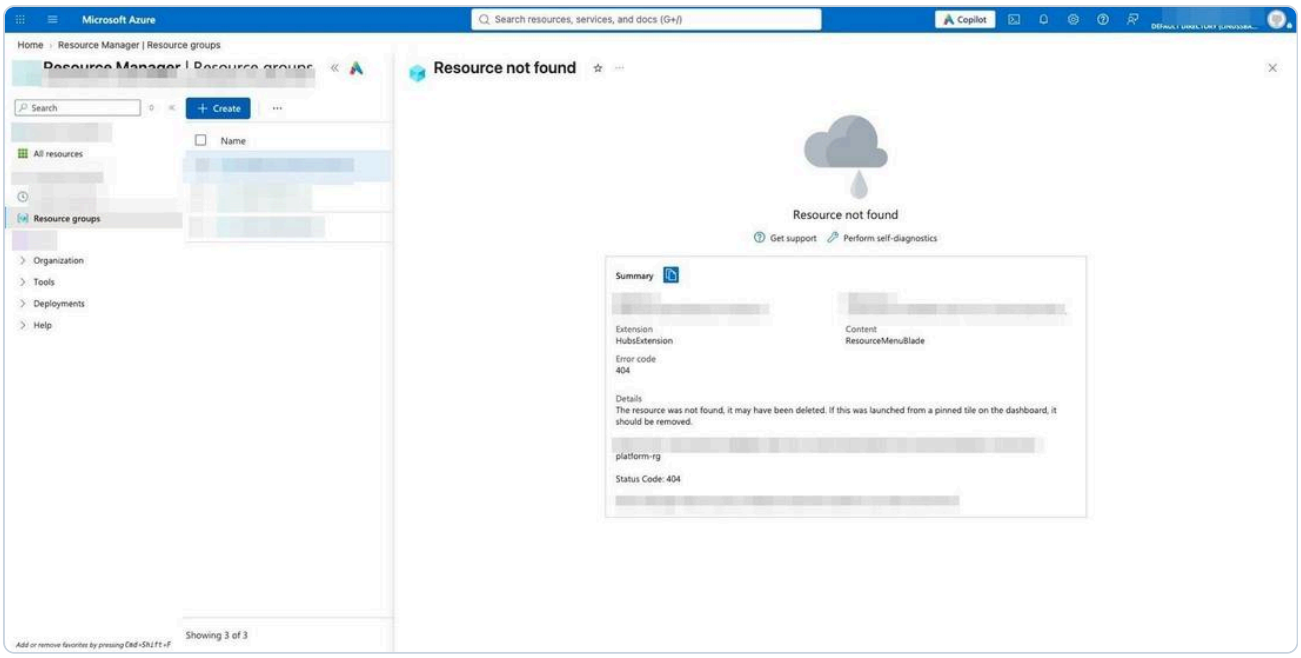


Figure 8.8 27 Oct 2026 · 03:48:43

8.2 Build the Azure Kubernetes cluster

Deploy database Completed

RunShore creates the Azure Kubernetes cluster with Terraform and connects to it.

Started 27 Oct 2026, 08:26 · took 5m 20s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh kubernetes
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ Azure uksouth total regional vCPUs: 32 vCPU free (limit 32, in use 0) -- cluster needs 12
- ✔ Azure uksouth DSv5 family vCPUs (Standard_D4s_v5): 16 vCPU free (limit 16, in use 0) -- cluster needs 12
- ✔ CPU quota sufficient
- ✔ AKS infrastructure converged
- ✔ kubectl context configured

8.3 Install Splunk

Deploy tool	Splunk	Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 27 Oct 2026, 09:26 · took 3m 15s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh splunk
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

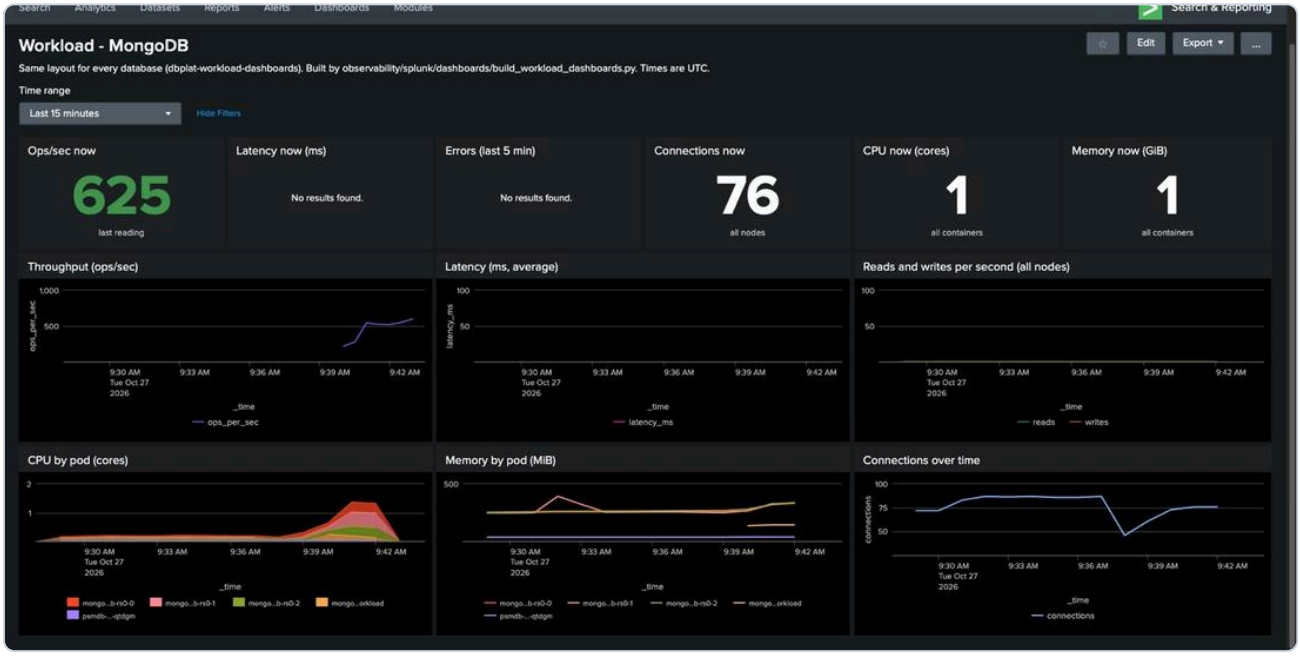


Figure 8.9 27 Oct 2026 · 09:43:45

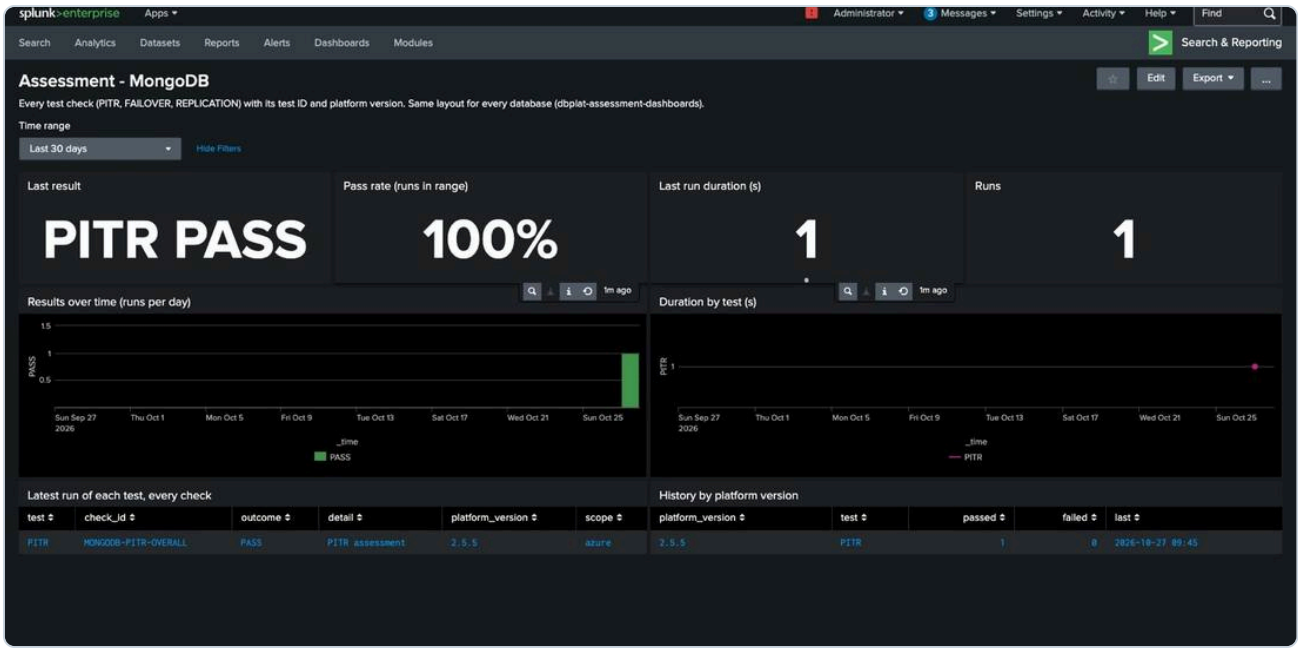


Figure 8.10 27 Oct 2026 · 09:47:41

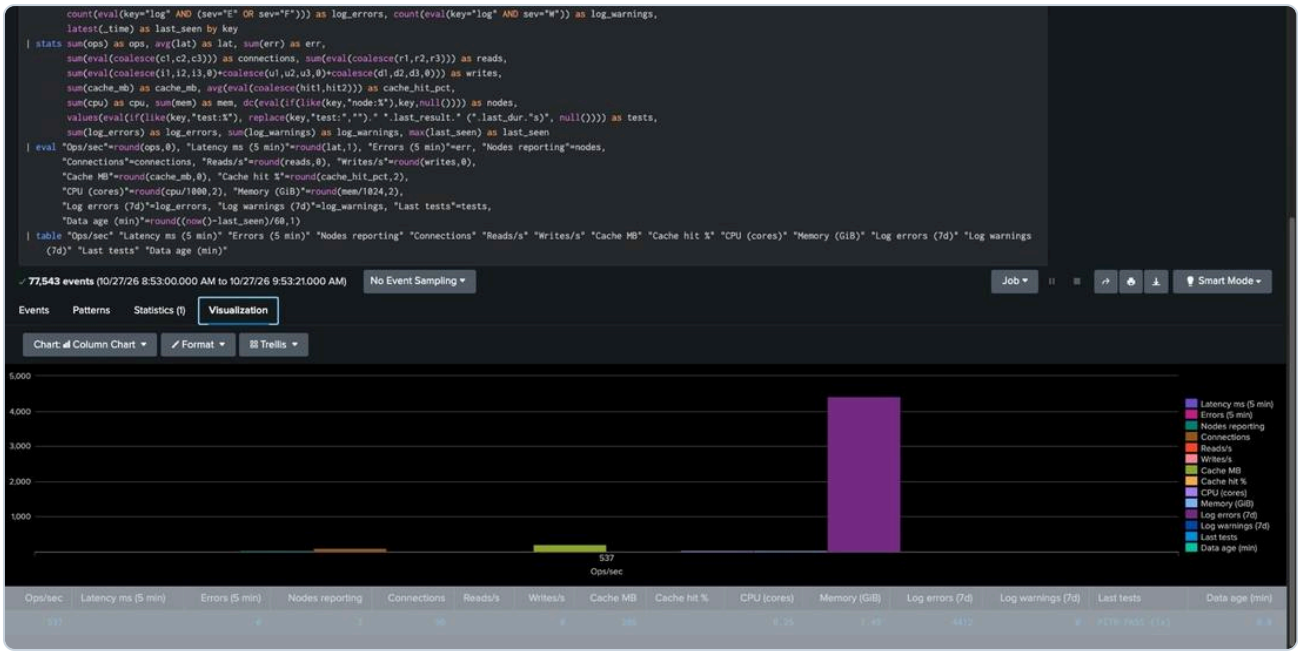


Figure 8.11 27 Oct 2026 · 09:53:49

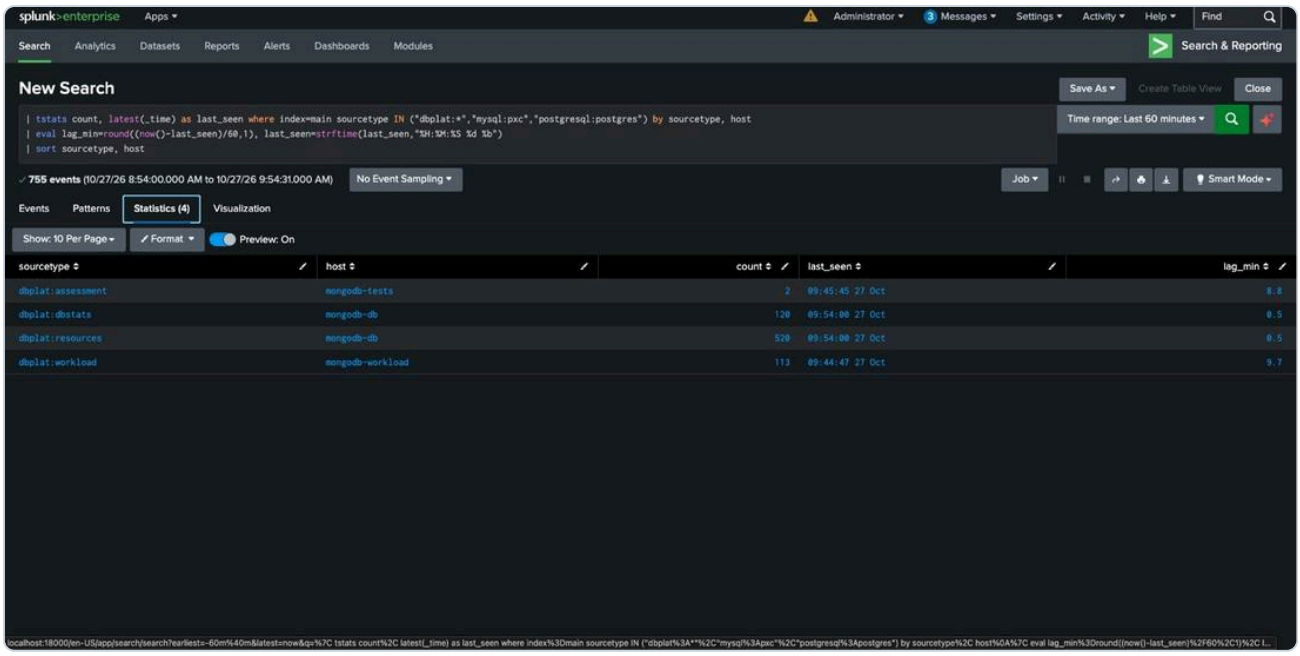


Figure 8.12 27 Oct 2026 · 09:55:07

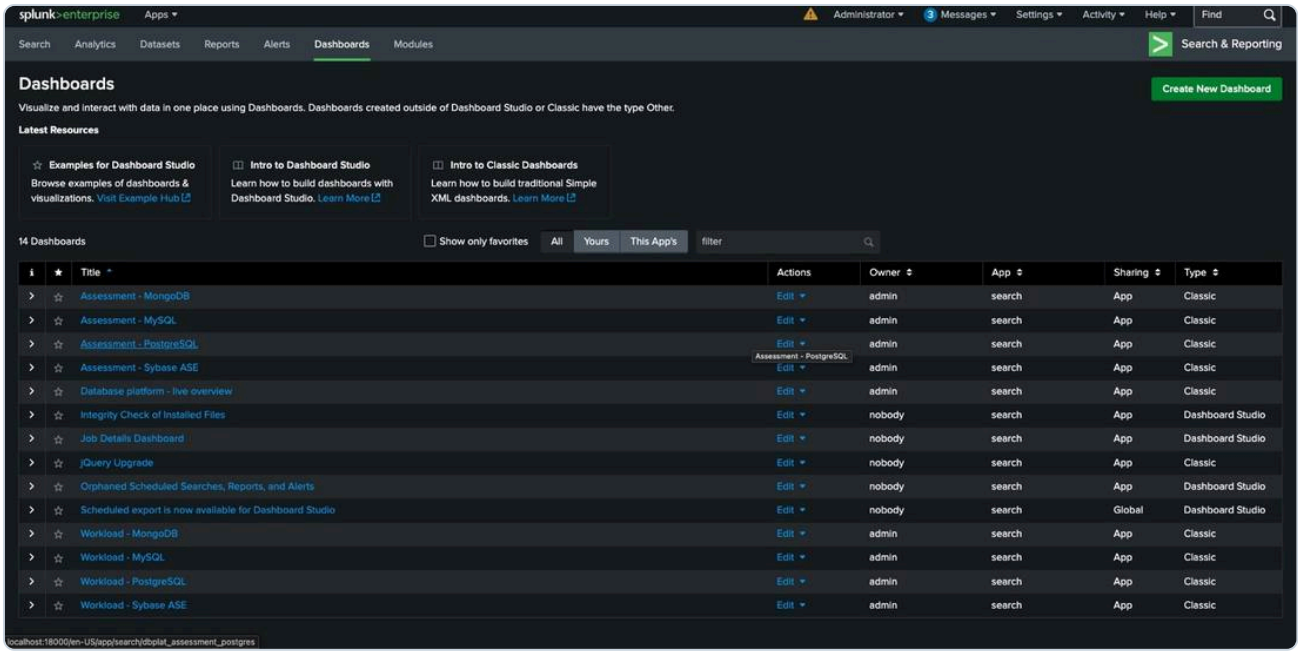


Figure 8.13 27 Oct 2026 · 09:56:24

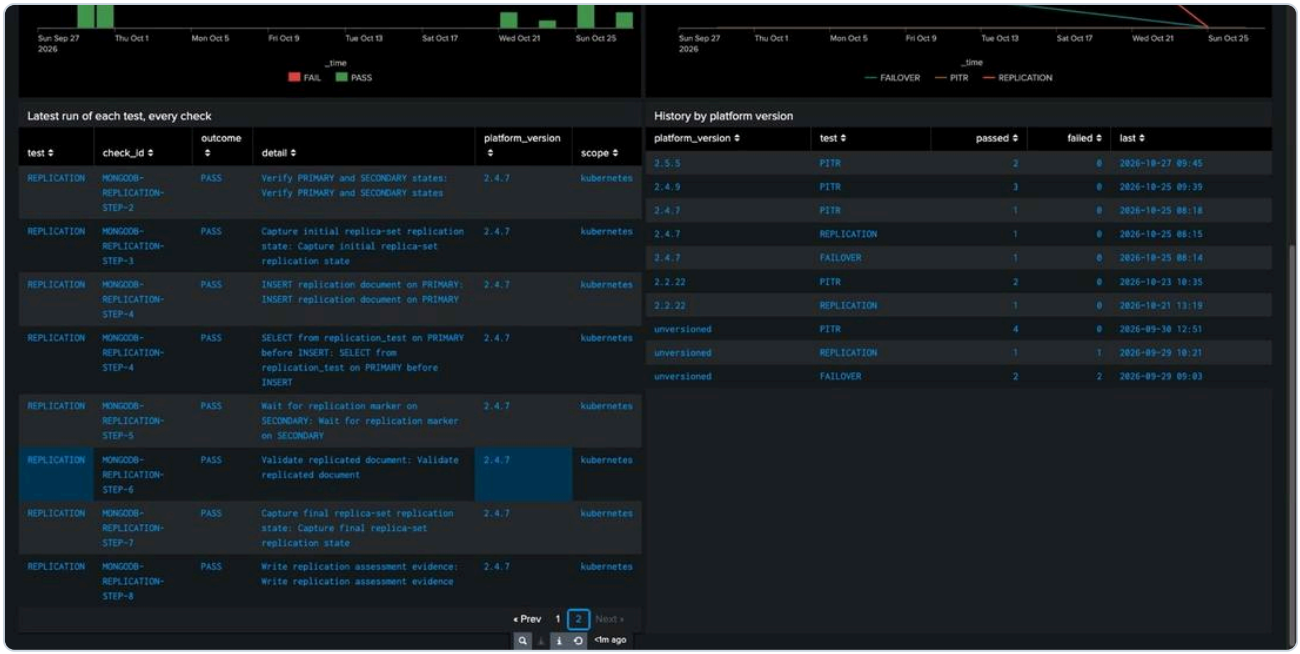


Figure 8.14 27 Oct 2026 · 09:57:16

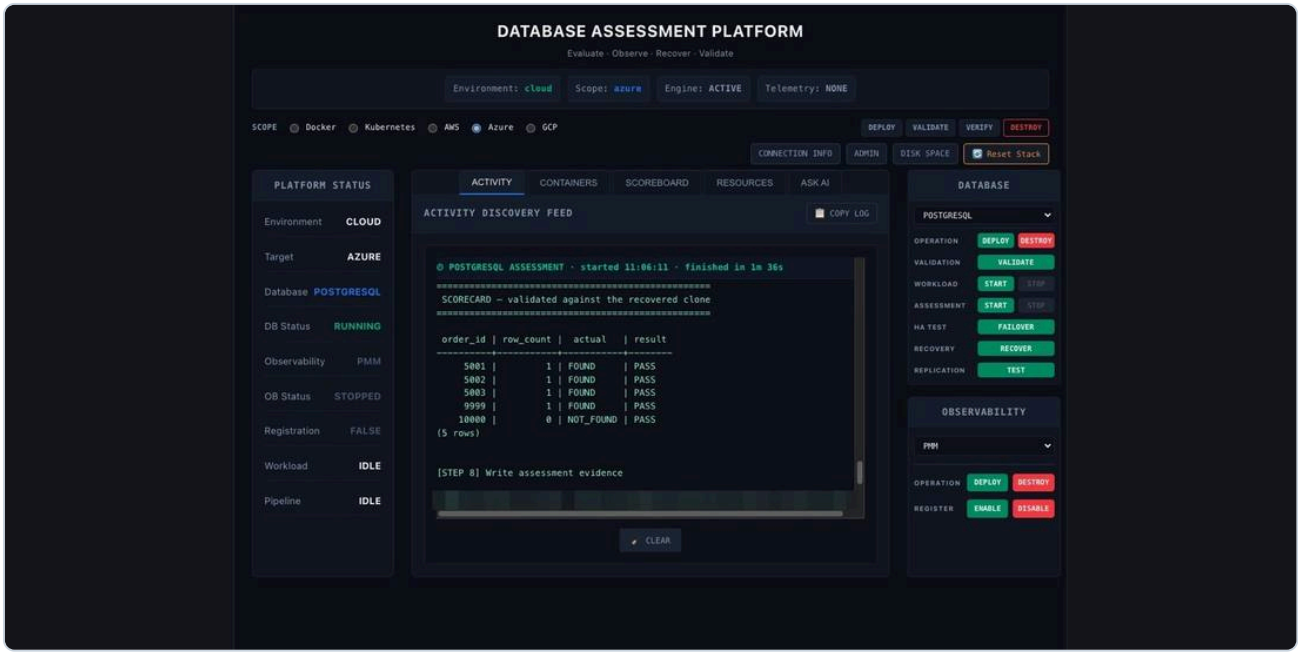


Figure 8.15 27 Oct 2026 · 11:08:30

8.4 Register PostgreSQL with Splunk

Register | Splunk | PostgreSQL | Completed

PostgreSQL is connected to Splunk, so its metrics and dashboards appear there.

Started 27 Oct 2026, 11:10 · took 16s

COMMAND

```
bash observability/splunk/registration/kubernetes/postgres/register.sh
```

- ✔ Resolved Splunk admin credentials from secret: splunk-s1-standalone-secret-v1
- ✔ Created HEC token 'postgres_hec_token'
- ✔ HEC token resolved
- ✔ No known pipeline errors detected
- ✔ Registration event confirmed in Splunk (index=main sourcetype="postgresql:postgres")
- ✔ Postgres logs successfully registered into Splunk

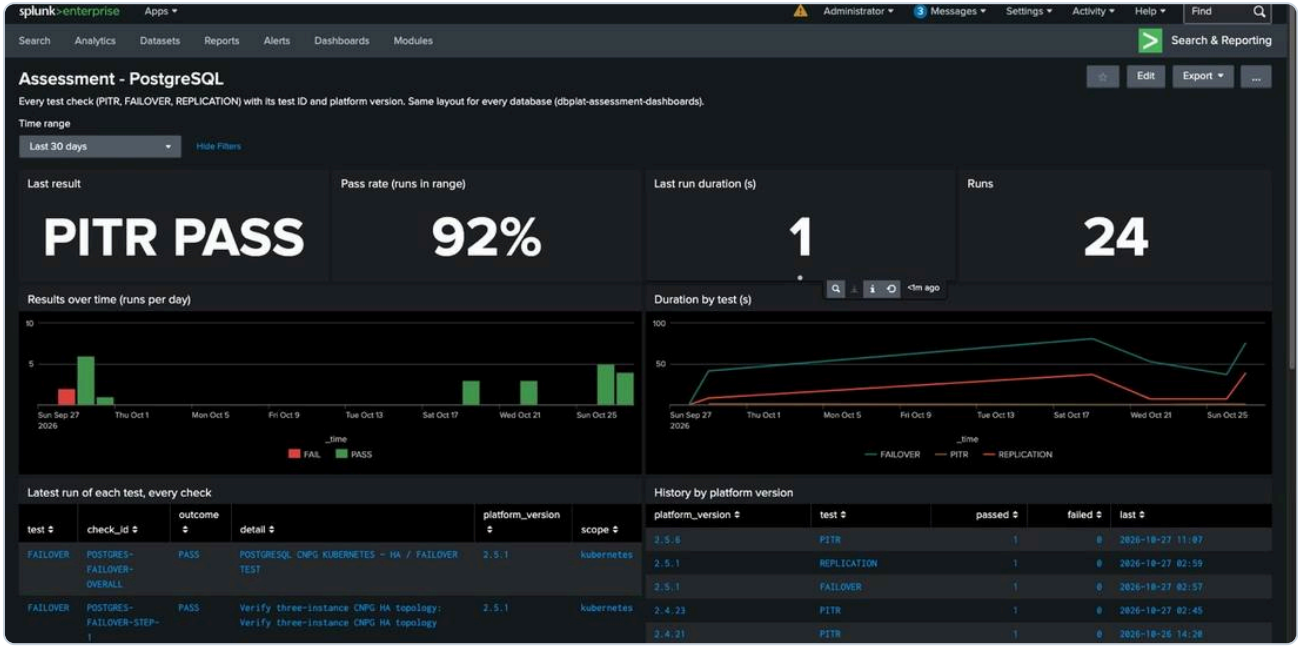


Figure 8.16 27 Oct 2026 · 11:12:16

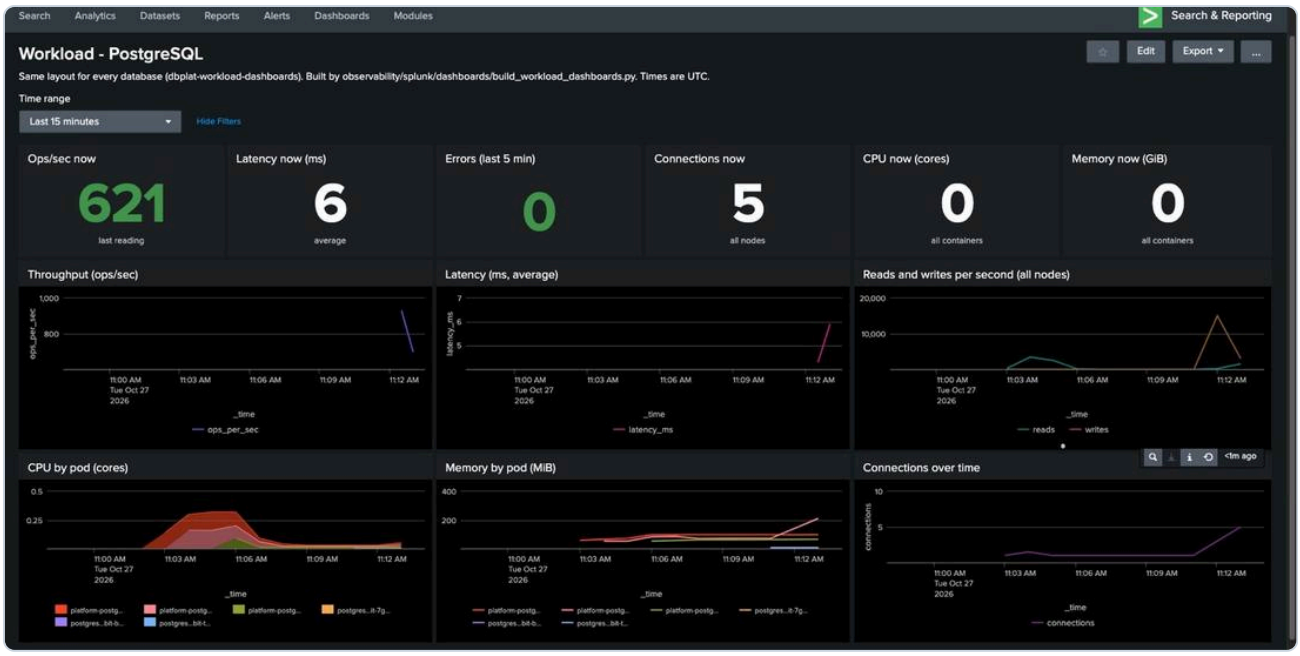


Figure 8.17 27 Oct 2026 · 11:13:31

8.5 Unregister PostgreSQL from Splunk

Unregister	Splunk	PostgreSQL	Completed

PostgreSQL is removed from Splunk; the database itself keeps running.

Started 27 Oct 2026, 11:13 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/postgres/unregister.sh
```

- ✔ Registration state cleared
- ✔ Postgres Splunk log forwarding removed

8.6 Register MySQL with Splunk

Register	Splunk	MySQL	Completed

MySQL is connected to Splunk, so its metrics and dashboards appear there.

Started 27 Oct 2026, 11:23 · took 3m 52s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/register.sh
```

- ✔ PXC: pxc-db
- ✔ Splunk HEC service: splunk-s1-standalone-service
- ✔ Splunk runtime Secret: splunk-splunk-secret
- ✔ mysql/mysql-splunk-hec converged
- ✔ Percona logrotate container present
- ✔ MySQL → Splunk OTel registration complete

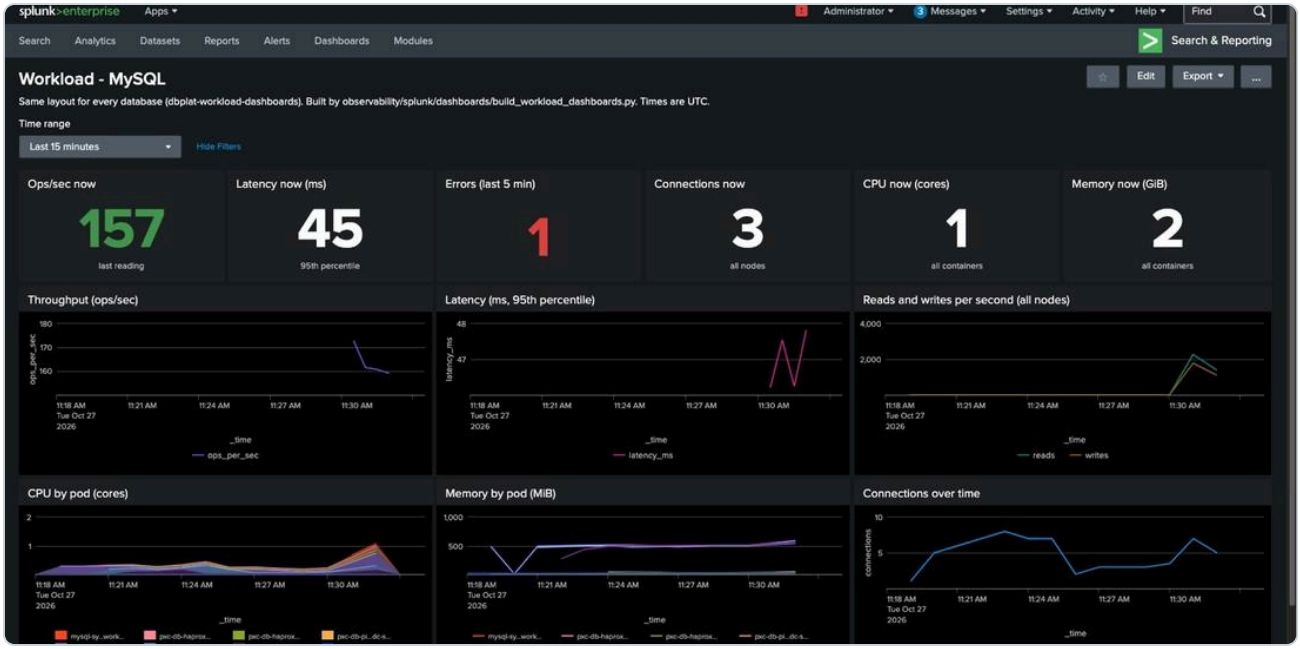


Figure 8.18 27 Oct 2026 · 11:33:16

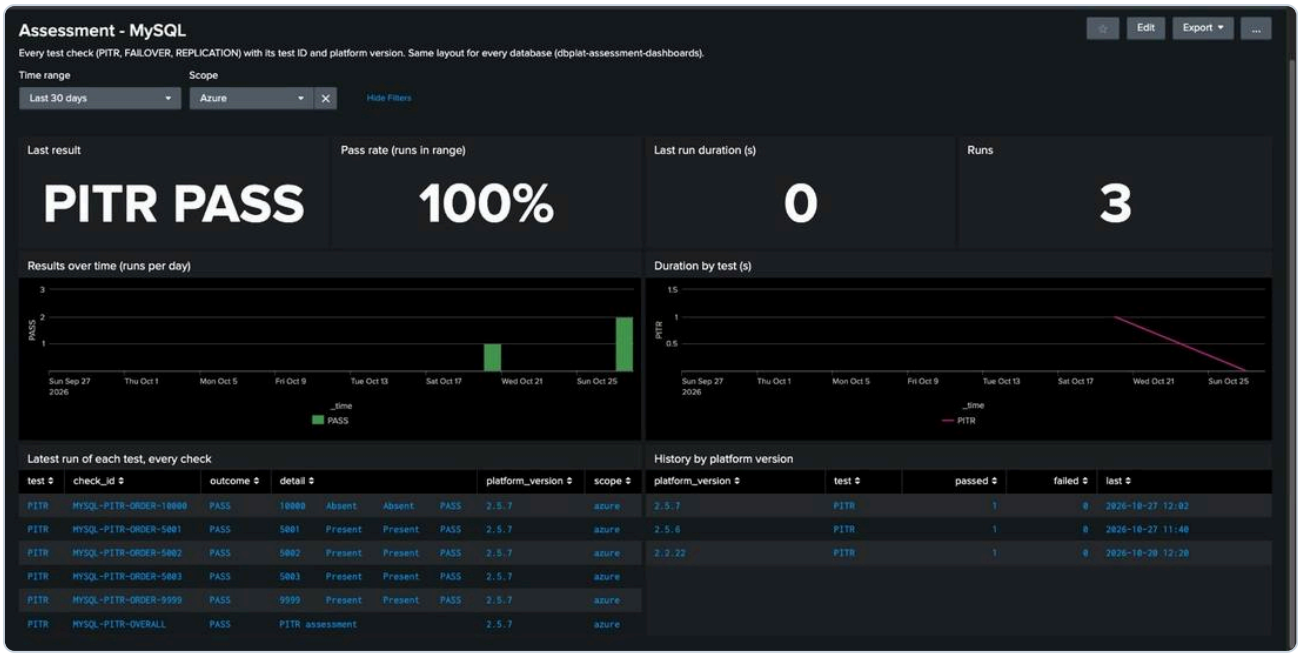


Figure 8.19 27 Oct 2026 · 12:06:35

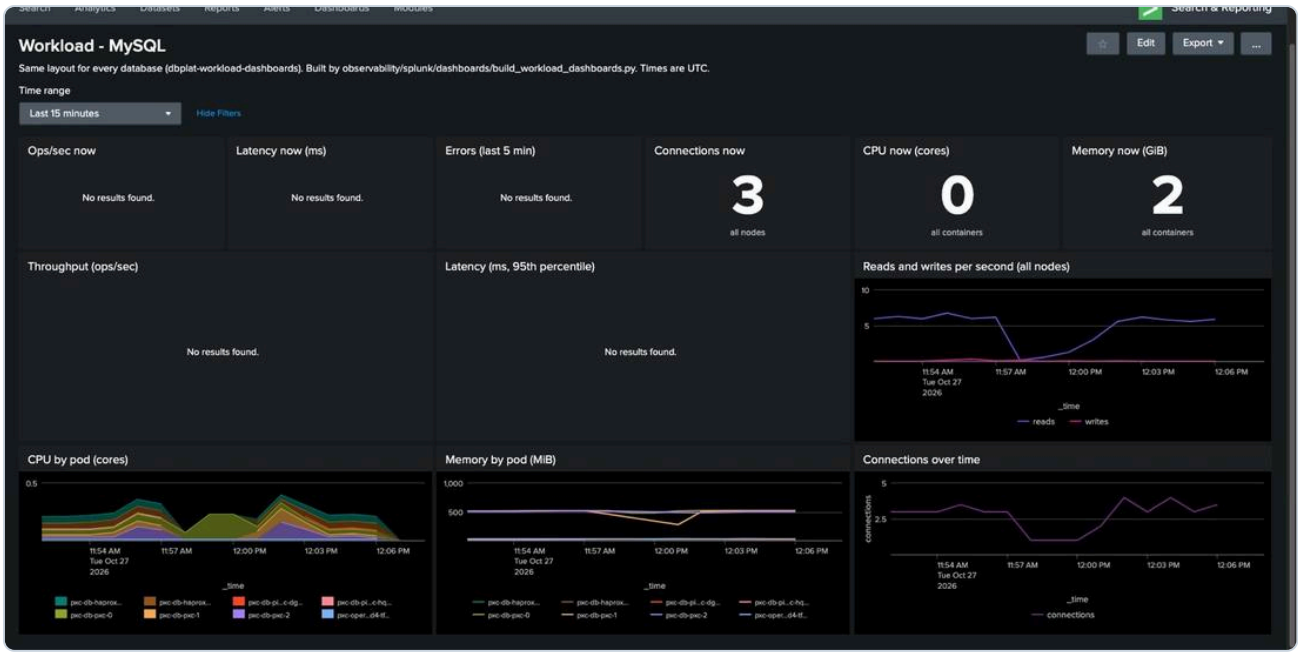


Figure 8.20 27 Oct 2026 · 12:07:05

8.7 Unregister MySQL from Splunk

Unregister	Splunk	MySQL	Completed

MySQL is removed from Splunk; the database itself keeps running.

Started 27 Oct 2026, 12:07 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/unregister.sh
```

- ✔ Percona LogCollector configuration disabled
- ✔ MySQL-local Splunk HEC Secret removed
- ✔ MySQL → Splunk OTel registration removed
- ✔ Registration state cleared

8.8 Register MongoDB with Splunk

Register	Splunk	MongoDB	Completed

MongoDB is connected to Splunk, so its metrics and dashboards appear there.

Started 27 Oct 2026, 12:15 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/register.sh
```

- ✔ PSMDB: mongodb-psmdb-db
- ✔ Splunk HEC service: splunk-s1-standalone-service
- ✔ Backup written:
- ✔ MongoDB Splunk HEC secret converged
- ✔ MongoDB Splunk HEC secret validated

- ✔ MongoDB logs successfully registered into Splunk

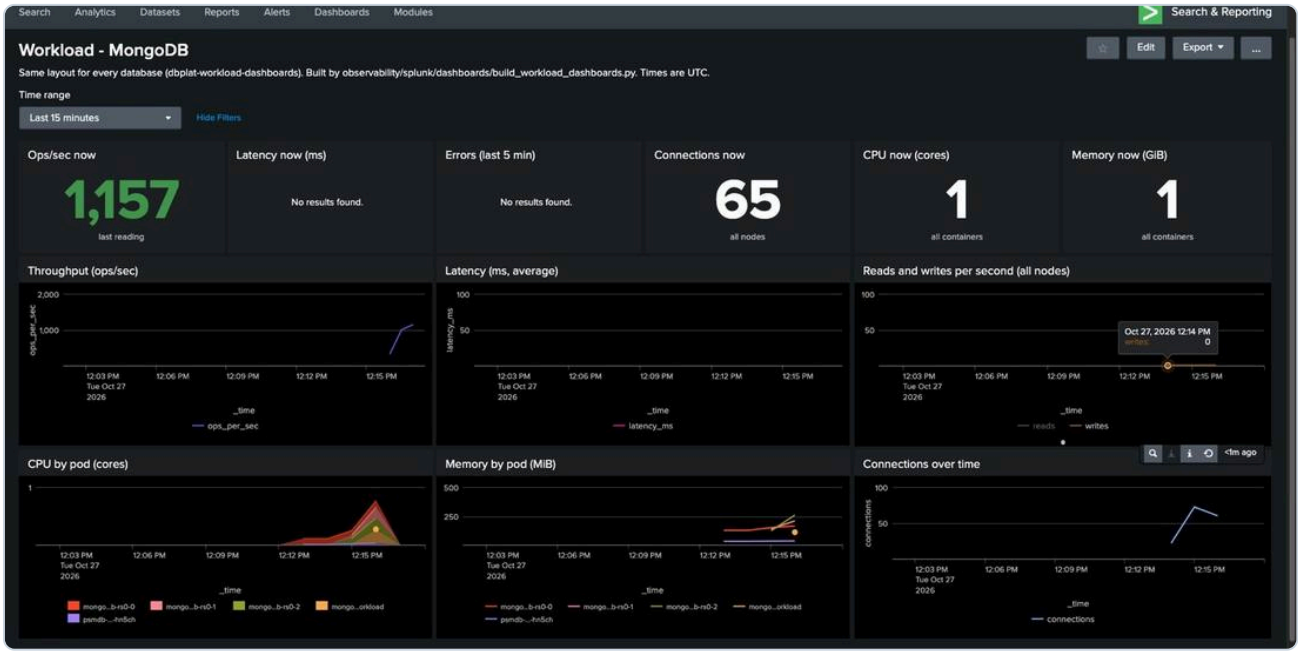


Figure 8.21 27 Oct 2026 · 12:17:32

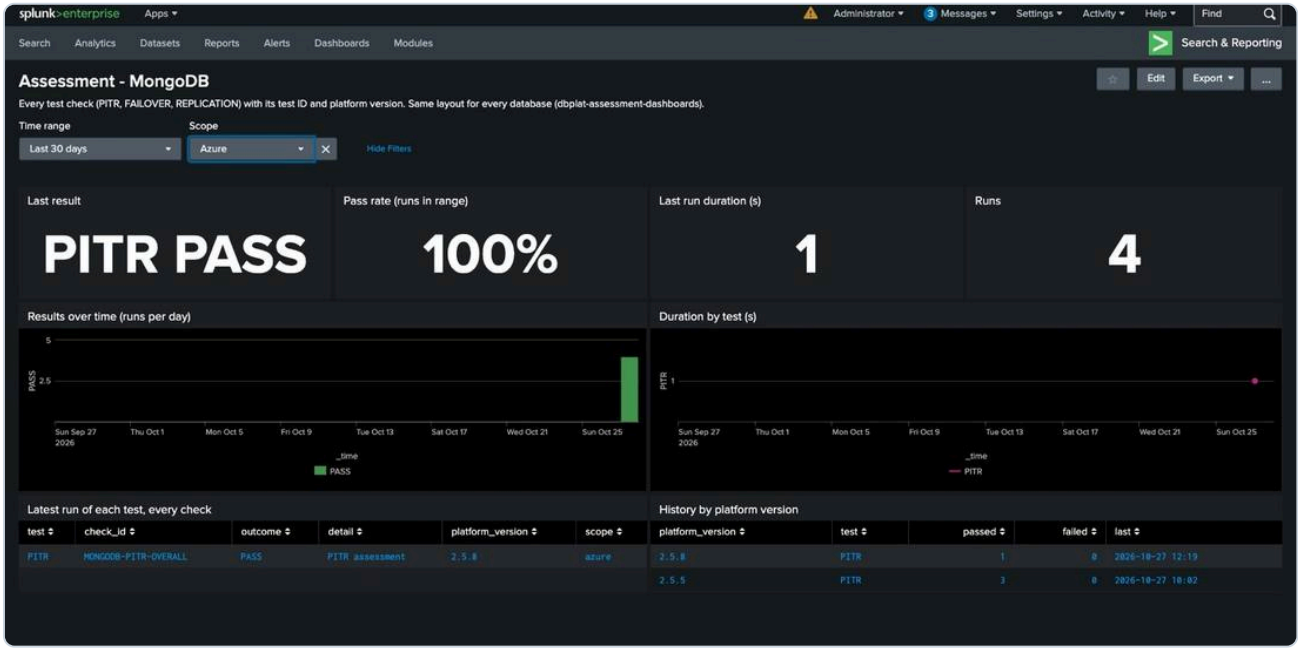


Figure 8.22 27 Oct 2026 · 12:20:53

8.9 Unregister MongoDB from Splunk

Unregister | Splunk | MongoDB | Completed

MongoDB is removed from Splunk; the database itself keeps running.

Started 27 Oct 2026, 12:23 · took 1m 53s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/unregister.sh
```

- ✔ Splunk logcollector configuration cleared
- ✔ Confirmed: no Splunk configuration or envFrom remains on the CR
- ✔ All pods have rolled — Splunk forwarding should now be fully stopped
- ✔ MongoDB Splunk log forwarding removed
- ✔ Registration state cleared

8.10 Remove Splunk

Unregister | Splunk | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 27 Oct 2026, 12:26 · took 1m 32s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh splunk
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

8.11 Install PMM

Deploy tool	PMM	Completed with warnings

PMM is installed, ready to monitor the databases in this cluster.

Started 27 Oct 2026, 12:28 · took 4m 20s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh pmm
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ PMM Client pod pmm-client-6755cf5b44-zrxsn is Ready
- ✔ PMM Client registered and connected to PMM Server

8.12 Register MongoDB with PMM

Register	PMM	MongoDB	Completed

MongoDB is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 12:54 · took 20s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/register_mongodb_replicaset.sh
```

- ✔ MongoDB namespace : mongodb
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ Replica set : rs0
- ✔ Running MongoDB members: 3
- ✔ Cleanup complete
- ✔ MongoDB service present in PMM inventory



Figure 8.23 27 Oct 2026 · 13:04:14

8.13 Replication test · MongoDB

Replication | MongoDB | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 27 Oct 2026, 13:08

COMMAND

```
SCOPE=azure db-platform/mongodb/replication/kubernetes/run_replication.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Three-member replica-set topology verified.
- ✔ PRIMARY is writable.
- ✔ SECONDARY is in secondary state.
- ✔ Replicated data matches.
- ✔ MONGODB REPLICATION: PASS

8.14 Unregister MongoDB from PMM

Unregister | PMM | MongoDB | Completed

MongoDB is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 13:11

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/unregister.sh
```

- ✔ pmm-0

- ✔ nothing left in PMM inventory -- Registration shows FALSE
- ✔ MongoDB service absent from PMM: mongodb-psmdb-db

8.15 Deploy MySQL

Deploy database | MySQL | Completed

RunShore deploys MySQL on Microsoft Azure · AKS and waits until every node is ready.

Started 27 Oct 2026, 13:15 · took 4m 48s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh mysql
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ PXC cluster health converged
- ✔ PXC deployment complete

8.16 Register MySQL with PMM

Register | PMM | MySQL | Completed

MySQL is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 13:21 · took 16s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/register_pxc_service.sh
```

- ✔ pmm-0
- ✔ Credentials loaded
- ✔ PXC successfully registered into PMM

8.17 Workload · MySQL

Workload | MySQL | Completed

A load generator runs against MySQL, so the monitoring shows real traffic. It stops by itself after a

set time.

Started 27 Oct 2026, 13:21 · took 21s

COMMAND

```
bash db-platform/mysql/workload/start.sh
```

- ✔ MySQL assessment database available: assessmentdb

8.18 Point-in-time recovery test · MySQL

Assessment | MySQL | Completed with warnings

RunShore restores MySQL to a chosen moment from its backups and checks the restored data is correct.

Started 27 Oct 2026, 13:22 · took 7m 33s

COMMAND

```
SCOPE=azure db-platform/mysql/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ Live root credential resolved from Kubernetes Secret pxc-db-secrets
- ✔ Prerequisites available
- ✔ Restore completed: pitr-demo-restore-1793107472
- ✔ PXC cluster ready

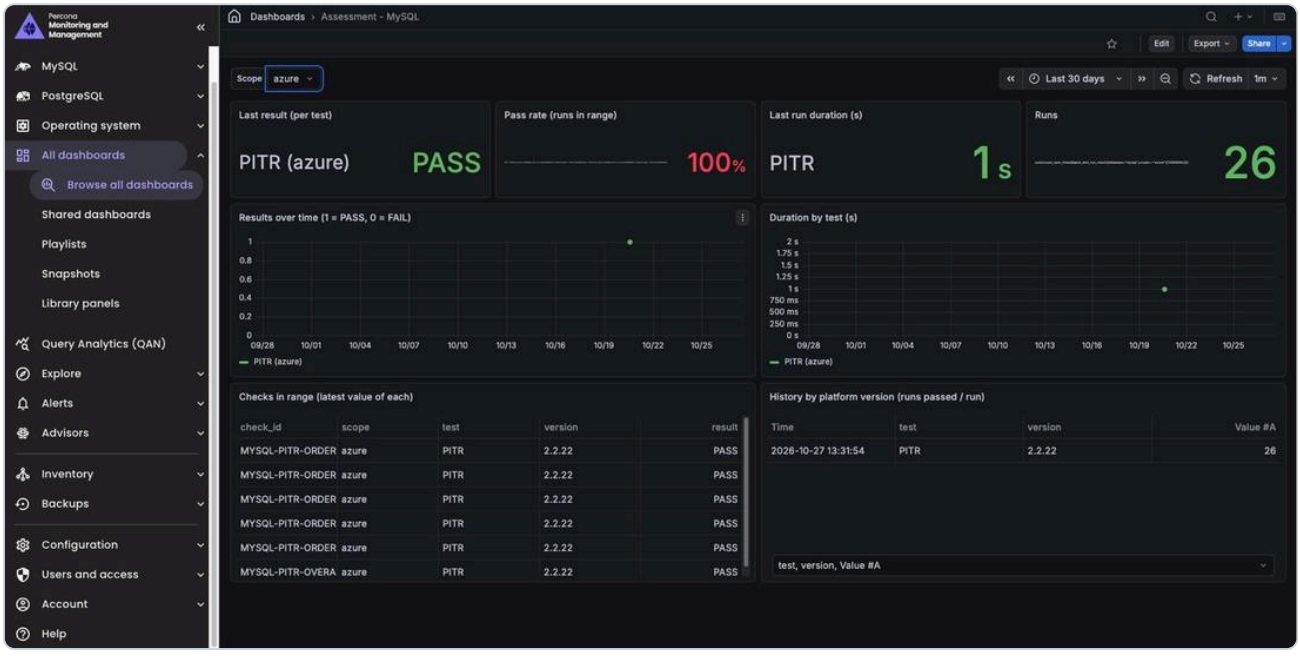


Figure 8.24 27 Oct 2026 · 13:32:07

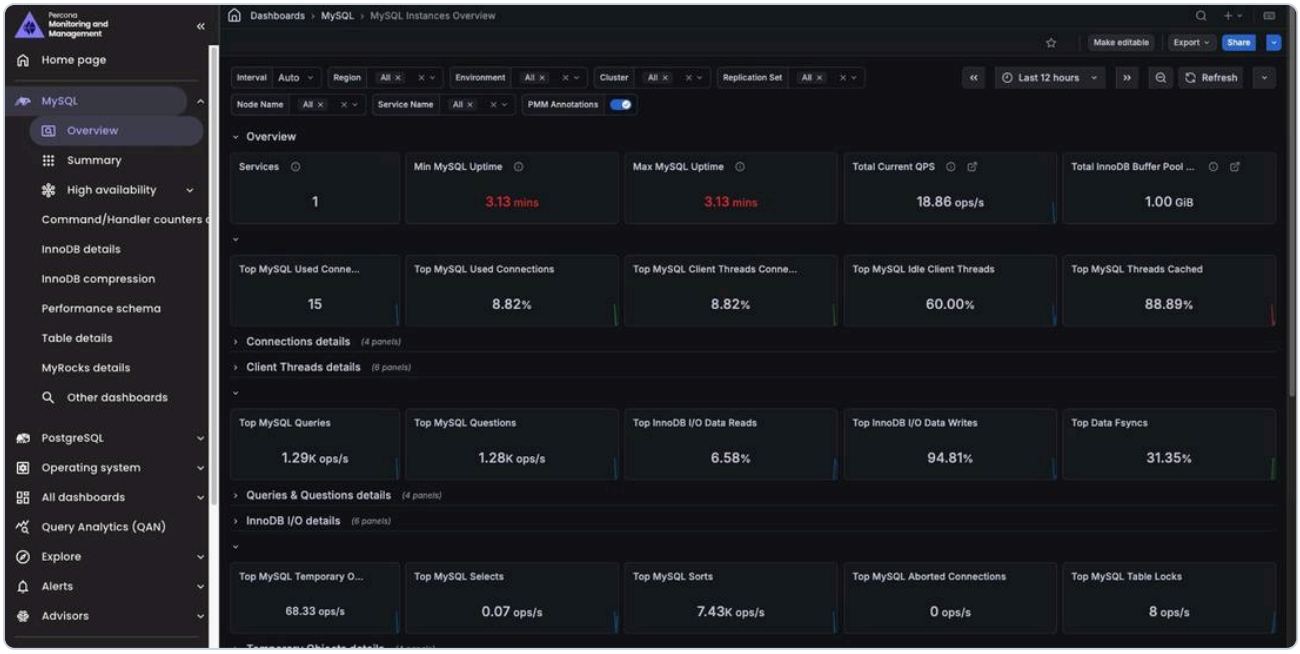


Figure 8.25 27 Oct 2026 · 13:37:26



Figure 8.26 27 Oct 2026 · 13:50:32

8.19 Replication test · MySQL

Replication | MySQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 27 Oct 2026, 13:52 · took 1m 52s

COMMAND

```
SCOPE=azure db-platform/mysql/replication/kubernetes/run_replication.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.

8.20 Unregister MySQL from PMM

Unregister | PMM | MySQL | Completed

MySQL is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 13:54 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

8.21 Destroy the Azure environment

Destroy | Completed

Everything RunShore created on Azure is removed, then checked: nothing that can still be billed is left behind.

Started 27 Oct 2026, 14:49 · took 6m 57s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh kubernetes
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ no LoadBalancer services
- ✔ namespaces deleted
- ✔ Resource group already absent: database-assessment-platform-rg
- ✔ removal requested -- the verify step waits until Azure has finished

8.22 Register PostgreSQL with PMM

Register | PMM | PostgreSQL | Completed

PostgreSQL is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 15:26 · took 24s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/register_cnpg_service.sh
```

- ✔ Credentials loaded
- ✔ pmm-0
- ✔ PostgreSQL successfully registered into PMM

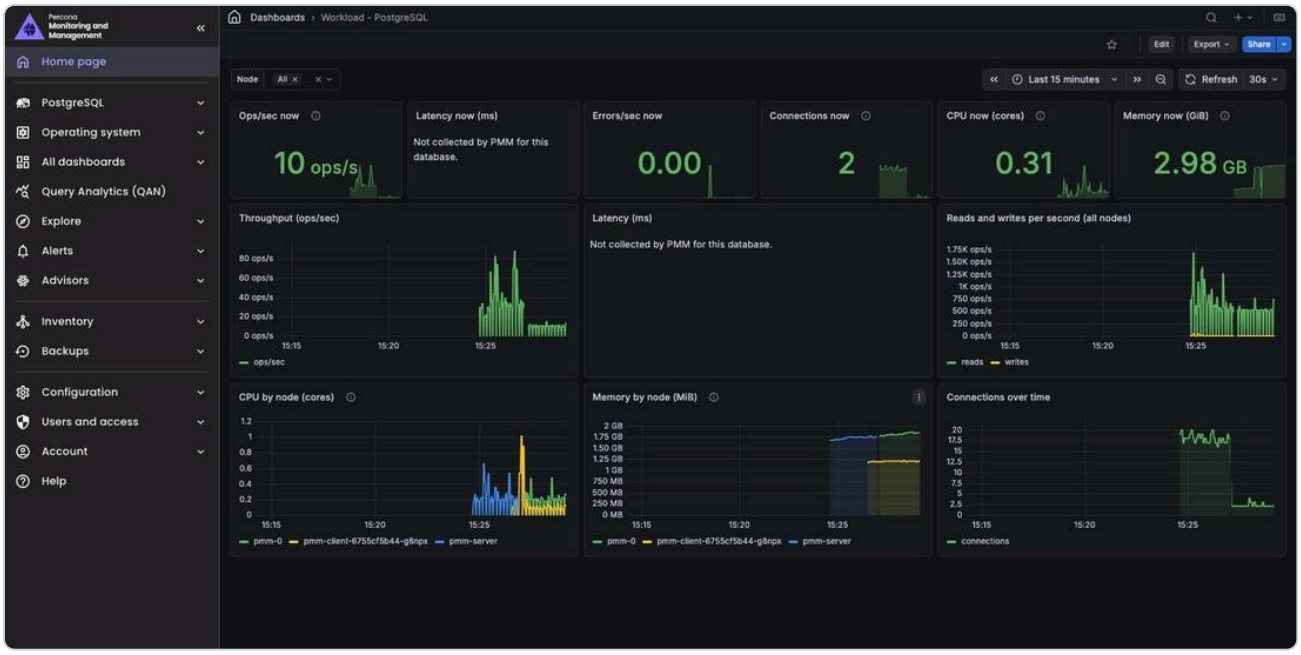


Figure 8.27 27 Oct 2026 · 15:29:25

8.23 Unregister PostgreSQL from PMM

Unregister | PMM | PostgreSQL | Completed

PostgreSQL is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 15:33 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

8.24 Deploy PostgreSQL

Deploy database | PostgreSQL | Completed

RunShore deploys PostgreSQL on Microsoft Azure · AKS and waits until every node is ready.

Started 27 Oct 2026, 15:37 · took 3m 24s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh postgres
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ Manifest rendered with Azure storage account: pgbackup7d96846f32d44ccb
- ✔ PostgreSQL platform deployed

8.25 Remove PMM

Unregister | PMM | Completed

PMM is removed from the cluster once no database is registered with it.

Started 27 Oct 2026, 15:45 · took 44s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh pmm
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ PMM Client deployment removed
- ✔ PMM namespace absent

8.26 Install New Relic

Deploy tool | New Relic | Completed

New Relic is installed, ready to monitor the databases in this cluster.

Started 27 Oct 2026, 15:47 · took 1m 16s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh newrelic
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ New Relic Helm repository ready
- ✔ New Relic Helm release deployed

8.27 Register PostgreSQL with New Relic

Register | New Relic | PostgreSQL | Completed

PostgreSQL is connected to New Relic, so its metrics and dashboards appear there.

Started 27 Oct 2026, 15:49 · took 16s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/register.sh
```

- ✔ Kubernetes, PostgreSQL and New Relic available
- ✔ Prerequisites available
- ✔ Monitoring secret available: postgres-monitoring
- ✔ Monitoring credentials loaded
- ✔ New Relic agents refreshed
- ✔ nri-postgresql configuration present

8.28 Workload · PostgreSQL

Workload | PostgreSQL | Completed

A load generator runs against PostgreSQL, so the monitoring shows real traffic. It stops by itself after

a set time.

Started 27 Oct 2026, 15:50 · took 12s

COMMAND

```
bash db-platform/postgres/workload/kubernetes/start.sh
```

- ✔ pgbench schema initialized.
- ✔ pgbench runtime permissions ensured for appuser.
- ✔ pgbench schema validated.

8.29 Failover test · PostgreSQL

Failover | PostgreSQL | Completed

The primary PostgreSQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 27 Oct 2026, 16:41 · took 1m 40s

COMMAND

```
SCOPE=azure db-platform/postgres/failover/kubernetes/run_failover.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ Final CNPG phase: Cluster in healthy state
- ✔ Final ready instances: 3

8.30 Point-in-time recovery test · PostgreSQL

Assessment | PostgreSQL | Completed

RunShore restores PostgreSQL to a chosen moment from its backups and checks the restored data is correct.

Started 27 Oct 2026, 16:39 · took 1m 24s

COMMAND

```
SCOPE=azure db-platform/postgres/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ PITR clone submitted
- ✔ PITR clone ready

8.31 Replication test · PostgreSQL

Replication | PostgreSQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 27 Oct 2026, 16:45 · took 16s

COMMAND

```
SCOPE=azure db-platform/postgres/replication/kubernetes/run_replication.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.



Figure 8.28 27 Oct 2026 · 16:46:31

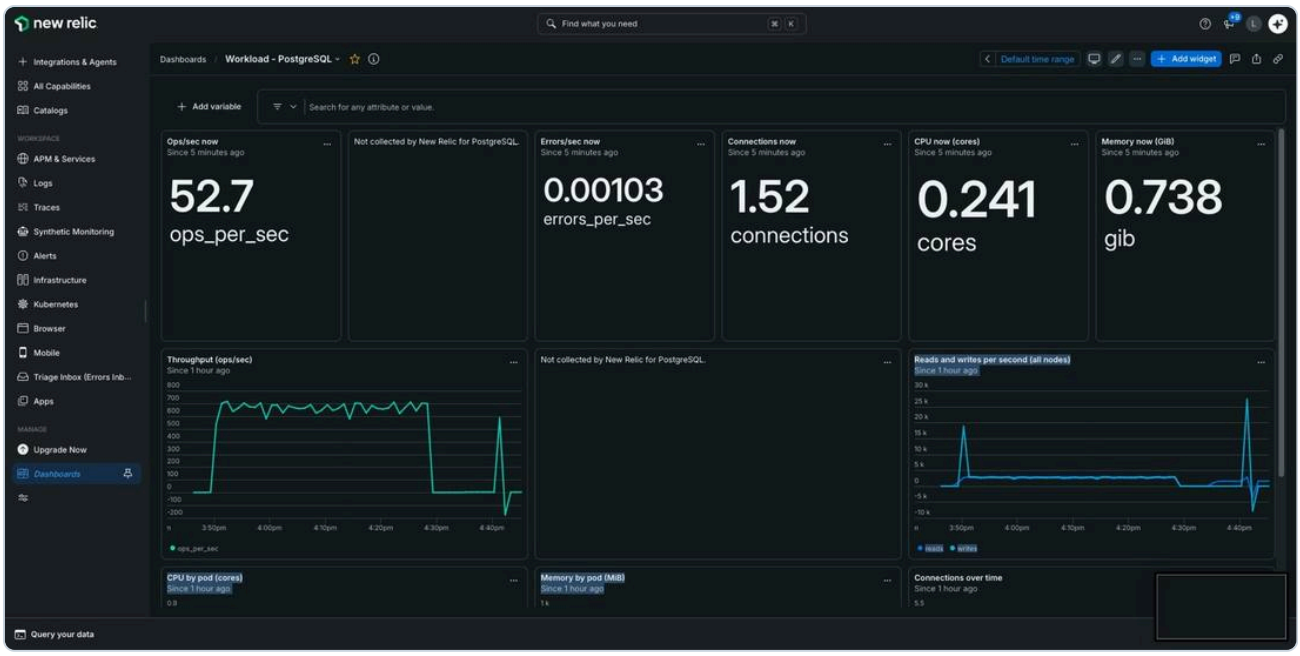


Figure 8.29 27 Oct 2026 · 16:46:33

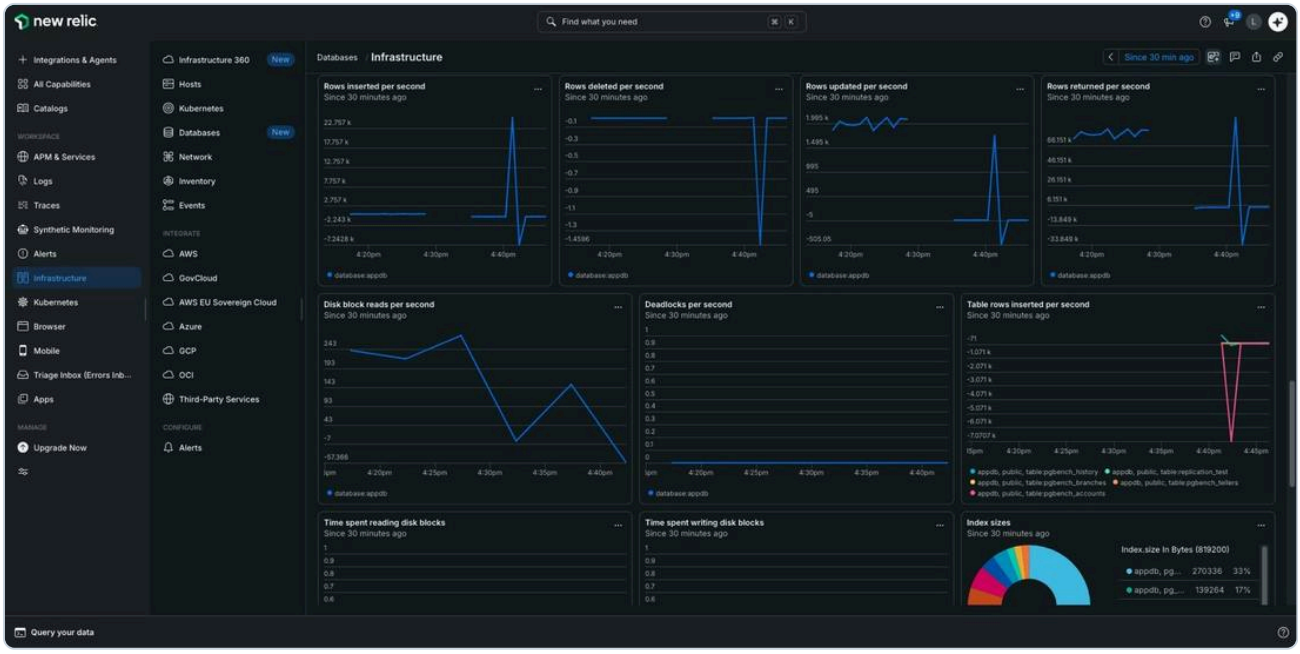


Figure 8.30 27 Oct 2026 · 16:47:48

8.32 Unregister PostgreSQL from New Relic

Unregister | New Relic | PostgreSQL | Completed

PostgreSQL is removed from New Relic; the database itself keeps running.

Started 27 Oct 2026, 16:48 · took 8s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/unregister.sh
```

✔ Kubernetes API reachable

✔ PostgreSQL integration configuration absent

- ✔ New Relic PostgreSQL telemetry secret absent
- ✔ newrelic-postgres-secret absent
- ✔ New Relic agents refreshed
- ✔ postgres-config.yml absent

8.33 Destroy PostgreSQL

Destroy | PostgreSQL | Completed

PostgreSQL is removed and its storage released.

Started 27 Oct 2026, 16:49 · took 48s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh postgres
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ PostgreSQL pods stopped (10s)
- ✔ CloudNativePG operator removed

8.34 Deploy MongoDB

Deploy database | MongoDB | Completed

RunShore deploys MongoDB on Microsoft Azure · AKS and waits until every node is ready.

Started 27 Oct 2026, 16:50 · took 4m 56s

COMMAND

```
platform/cloud/azure/orchestrator/deploy.sh mongodb
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ Bound PVCs: 3
- ✔ MongoDB platform validation complete

8.35 Failover test · MongoDB

Failover | MongoDB | Completed

The primary MongoDB node is stopped on purpose. RunShore measures how quickly another node

takes over, and checks that no data was lost.

Started 27 Oct 2026, 16:56 · took 48s

COMMAND

```
SCOPE=azure db-platform/mongodb/failover/kubernetes/run_failover.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ PSMDB cluster is ready.
- ✔ Replica set: rs0
- ✔ Ready members: 3/3
- ✔ HA validation marker survived failover.
- ✔ MongoDB replica set recovered to 3/3 ready members.

8.36 Workload · MongoDB

Workload | MongoDB | Completed

A load generator runs against MongoDB, so the monitoring shows real traffic. It stops by itself after a set time.

Started 27 Oct 2026, 16:57 · took 8s

COMMAND

```
bash db-platform/mongodb/workload/kubernetes/start.sh
```

- ✔ MongoDB YCSB workload Pod is Running.

8.37 Point-in-time recovery test · MongoDB

Assessment | MongoDB | Completed

RunShore restores MongoDB to a chosen moment from its backups and checks the restored data is correct.

Started 27 Oct 2026, 16:58 · took 1m 28s

COMMAND

```
SCOPE=azure db-platform/mongodb/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Cluster: mongodb-psmdb-db
- ✔ PITR archiving enabled
- ✔ Baseline seeded
- ✔ Post-marker orders inserted
- ✔ Archived oplog covers the target (2026-10-27T16:59:02 - 2026-10-27T16:59:23) after 10s

8.38 Register MongoDB with New Relic

Register | New Relic | MongoDB | Completed

MongoDB is connected to New Relic, so its metrics and dashboards appear there.

Started 27 Oct 2026, 17:11 · took 1m 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/register.sh
```

- ✔ Kubernetes API reachable
- ✔ New Relic deployment available
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ MongoDB namespace: mongodb
- ✔ Exporter reachable inside the cluster: http://mongodb-newrelic-exporter.mongodb.svc.cluster.local:9216/metri (mongodb_up 1)
- ✔ New Relic scraped MongoDB with no errors since 2026-10-27T17:12:09Z

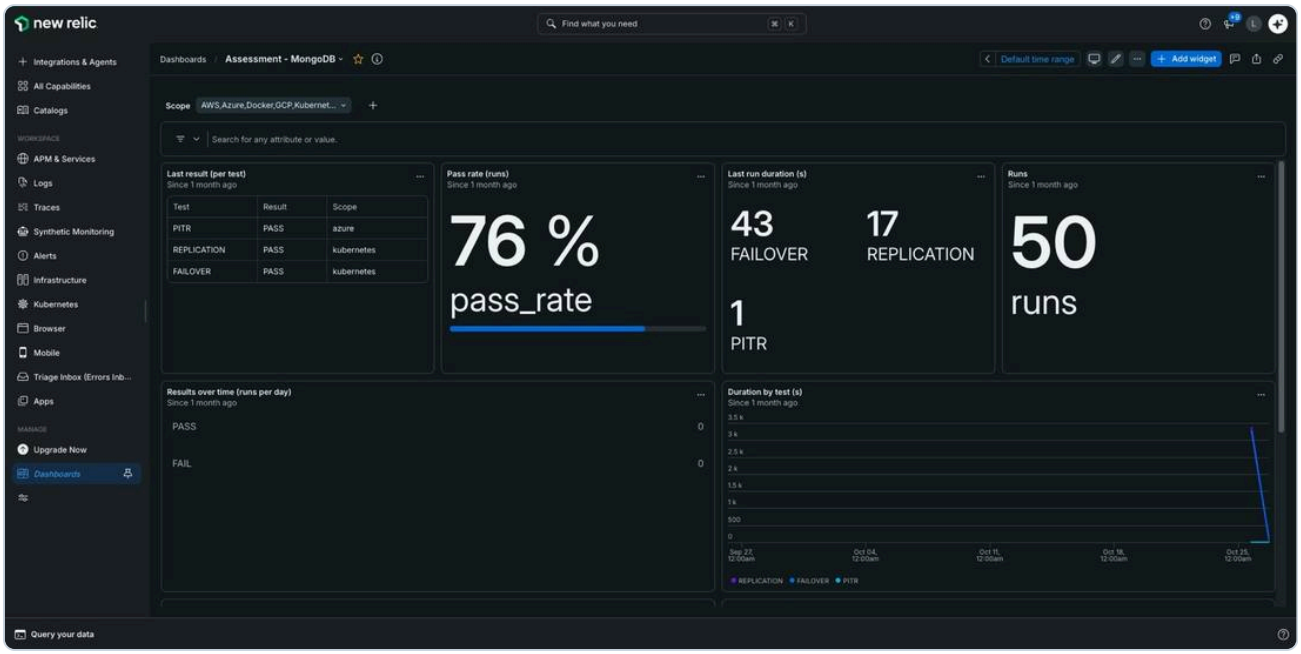


Figure 8.31 27 Oct 2026 · 18:17:25

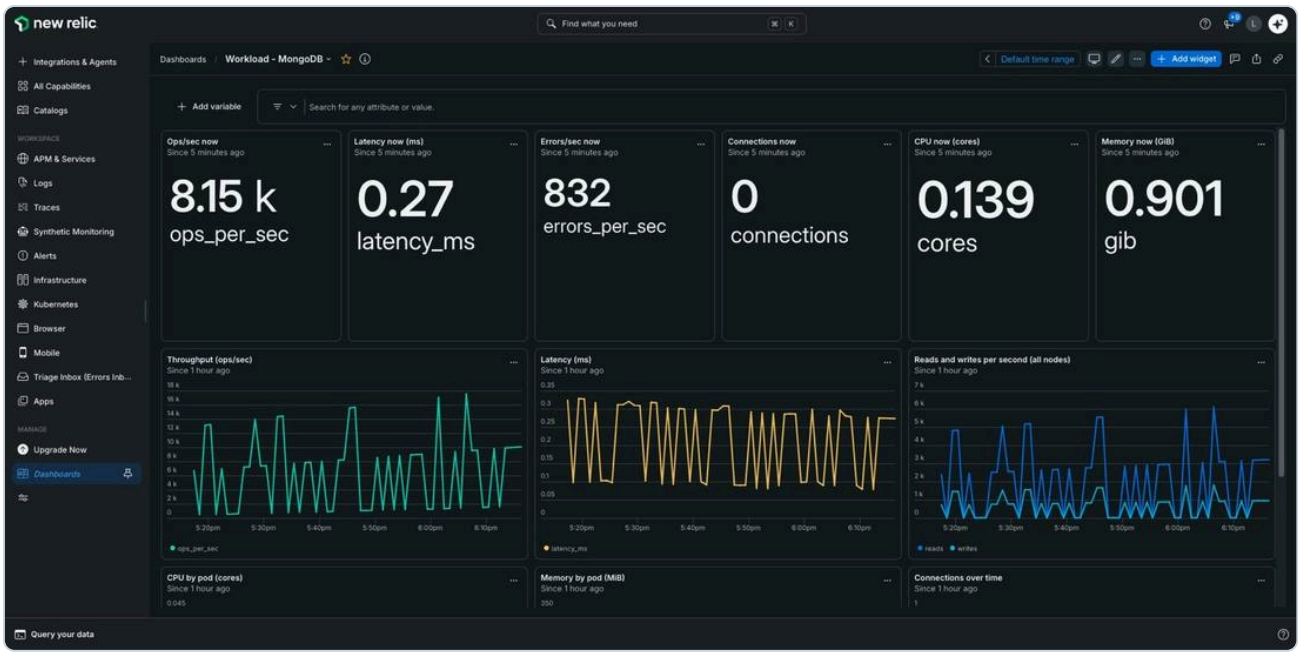


Figure 8.32 27 Oct 2026 · 18:17:43

8.39 Unregister MongoDB from New Relic

Unregister | New Relic | MongoDB | Completed

MongoDB is removed from New Relic; the database itself keeps running.

Started 27 Oct 2026, 18:18 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/unregister.sh
```

- ✔ Kubernetes API reachable
- ✔ MongoDB New Relic registration state removed
- ✔ Removed Deployment/mongodb-newrelic-exporter (namespace mongodb)
- ✔ Removed Service/mongodb-newrelic-exporter (namespace mongodb)
- ✔ New Relic kubelet agents refreshed
- ✔ mongodb-config.yml absent

8.40 Destroy MongoDB

Destroy | MongoDB | Completed

MongoDB is removed and its storage released.

Started 27 Oct 2026, 18:19 · took 56s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh mongodb
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ Kubernetes API reachable
- ✔ Azure/AKS context validated
- ✔ Tooling and Kubernetes API available
- ✔ MongoDB namespace absent

8.41 Register MySQL with New Relic

Register | New Relic | MySQL | Completed

MySQL is connected to New Relic, so its metrics and dashboards appear there.

Started 27 Oct 2026, 18:26 · took 20s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/register.sh
```

- ✔ Kubernetes, PXC and New Relic available
- ✔ PXC cluster ready
- ✔ MySQL endpoint discovered
- ✔ PXC administrative credential available
- ✔ PXC HAProxy endpoint present
- ✔ MySQL telemetry secret present

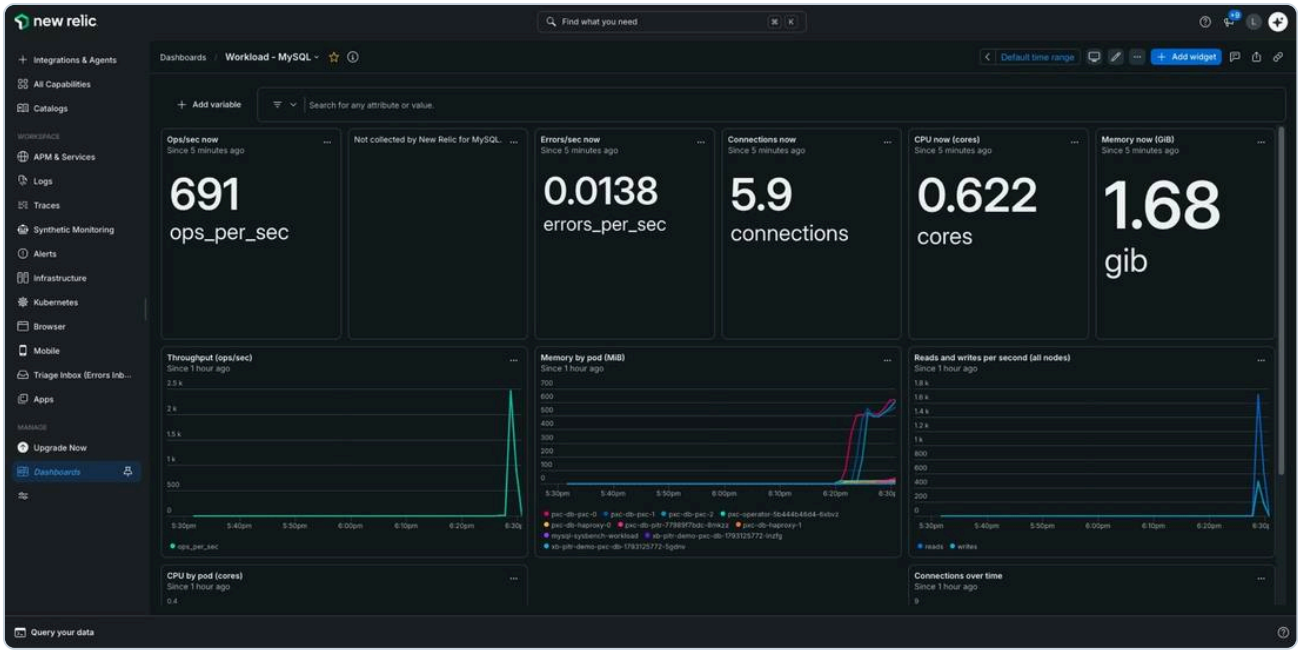


Figure 8.33 27 Oct 2026 · 18:31:58

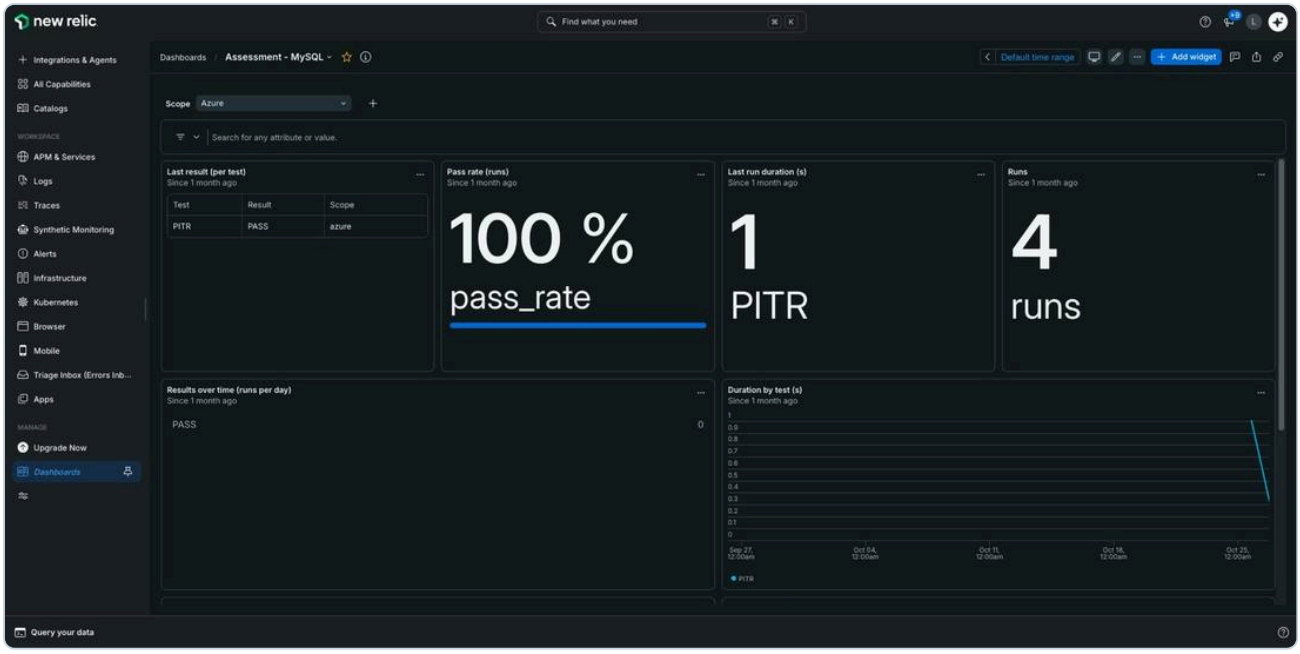


Figure 8.34 27 Oct 2026 · 18:33:56

8.42 Unregister MySQL from New Relic

Unregister	New Relic	MySQL	Completed

MySQL is removed from New Relic; the database itself keeps running.

Started 27 Oct 2026, 18:34 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/unregister.sh
```

- ✔ Kubernetes and New Relic available
- ✔ MySQL registration state cleared

- ✔ Removed mysql-config.yml
- ✔ Removed newrelic-mysql-secret
- ✔ mysql-config.yml absent
- ✔ newrelic-mysql-secret absent

8.43 Failover test · MySQL

Failover | MySQL | Completed

The primary MySQL node is stopped on purpose. RunShore measures how quickly another node takes over, and checks that no data was lost.

Started 27 Oct 2026, 18:45 · took 1m 16s

COMMAND

```
SCOPE=azure db-platform/mysql/failover/kubernetes/run_failover.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ PXC administrative credential available
- ✔ All three PXC members are Kubernetes-Ready and Primary/Synced/Ready
- ✔ Assessment data verified on pxc-db-pxc-2
- ✔ Final three-member cluster is Kubernetes-Ready and Primary/Synced/Ready

8.44 Destroy MySQL

Destroy | MySQL | Completed

MySQL is removed and its storage released.

Started 27 Oct 2026, 19:12 · took 9m 13s

COMMAND

```
platform/cloud/azure/orchestrator/destroy.sh kubernetes
```

- ✔ Terraform state: Azure storage account dbplattfdd7c99f20d82dc (resource group dbplat-tfstate-rg)
- ✔ kubectl connected to database-assessment-platform-aks
- ✔ no LoadBalancer services
- ✔ database objects deleted
- ✔ Resource group already absent: database-assessment-platform-rg
- ✔ removal requested -- the verify step waits until Azure has finished

09 Google GKE

Google Cloud · GKE

The same workflow on Google Cloud, using a GKE cluster built with Terraform, ending with a verified destroy.

SCOPE	DATABASES	OBSERVED WITH	EVIDENCE	CAPTURED
Google Cloud · GKE	MongoDB, MySQL, PostgreSQL	New Relic, PMM, Splunk	37 screens · 44 logs	27 Oct–28 Oct 2026

9.1 Build the Google Cloud Kubernetes cluster

Deploy database | Completed

RunShore creates the Google Cloud Kubernetes cluster with Terraform and connects to it.

Started 27 Oct 2026, 19:39 · took 10m 07s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh kubernetes
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ GCP europe-west1 CPUs: 200 vCPU free (limit 200, in use 0) -- cluster needs 12
- ✔ GCP all-regions CPUs: 32 vCPU free (limit 32, in use 0) -- cluster needs 12
- ✔ CPU quota sufficient
- ✔ GCP project: database-assessment-platform
- ✔ Kubernetes API reachable

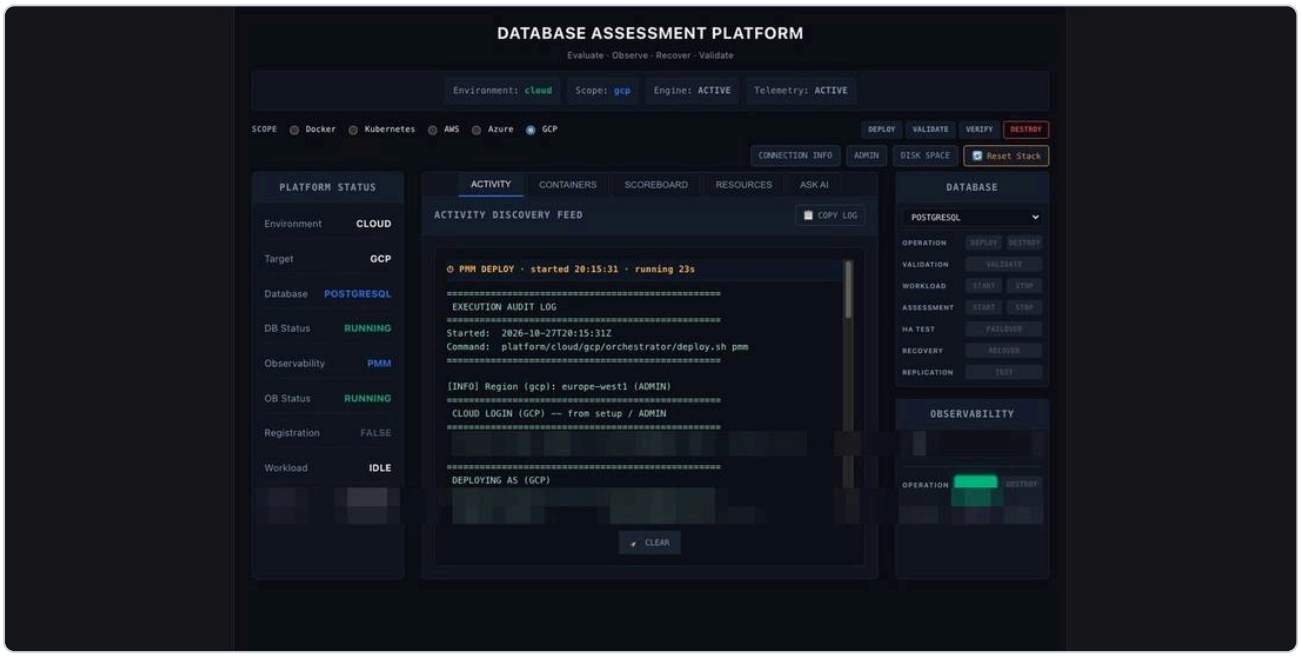


Figure 9.1 27 Oct 2026 · 20:15:55

9.2 Install PMM

Deploy tool	PMM	Completed

PMM is installed, ready to monitor the databases in this cluster.

Started 27 Oct 2026, 20:15 · took 4m 04s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh pmm
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ PMM Client pod pmm-client-6755cf5b44-5rkv9 is Ready
- ✔ PMM Client registered and connected to PMM Server

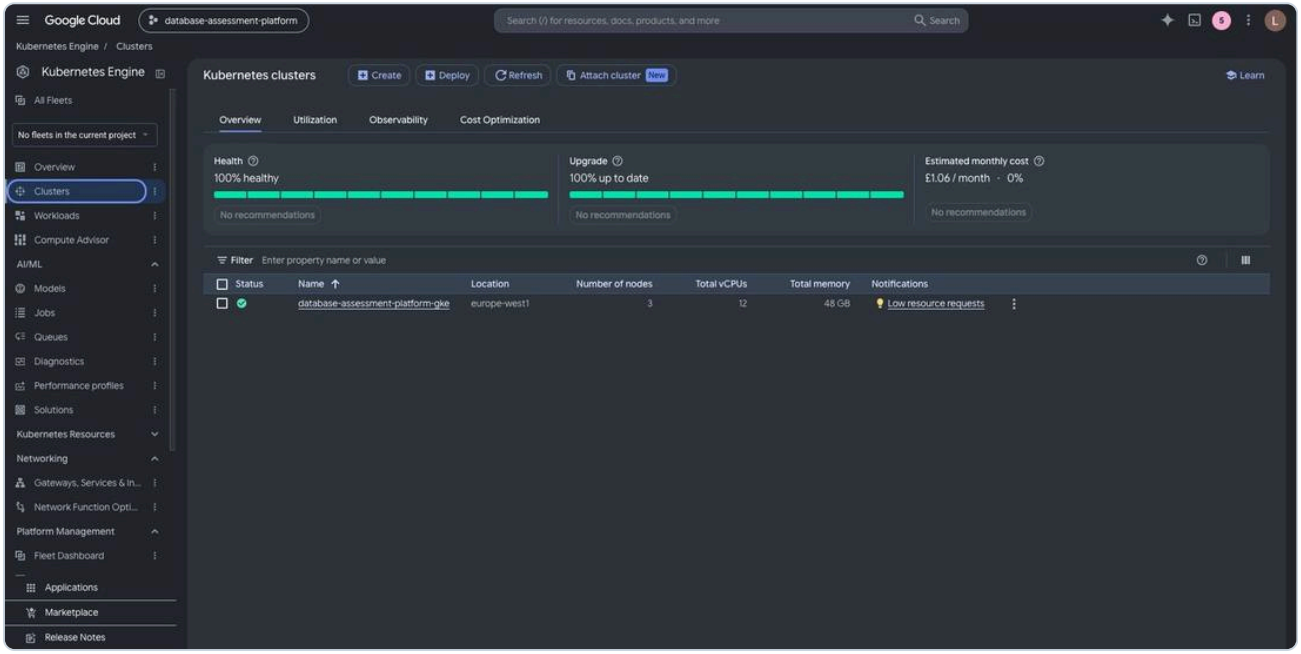


Figure 9.2 27 Oct 2026 · 20:26:50

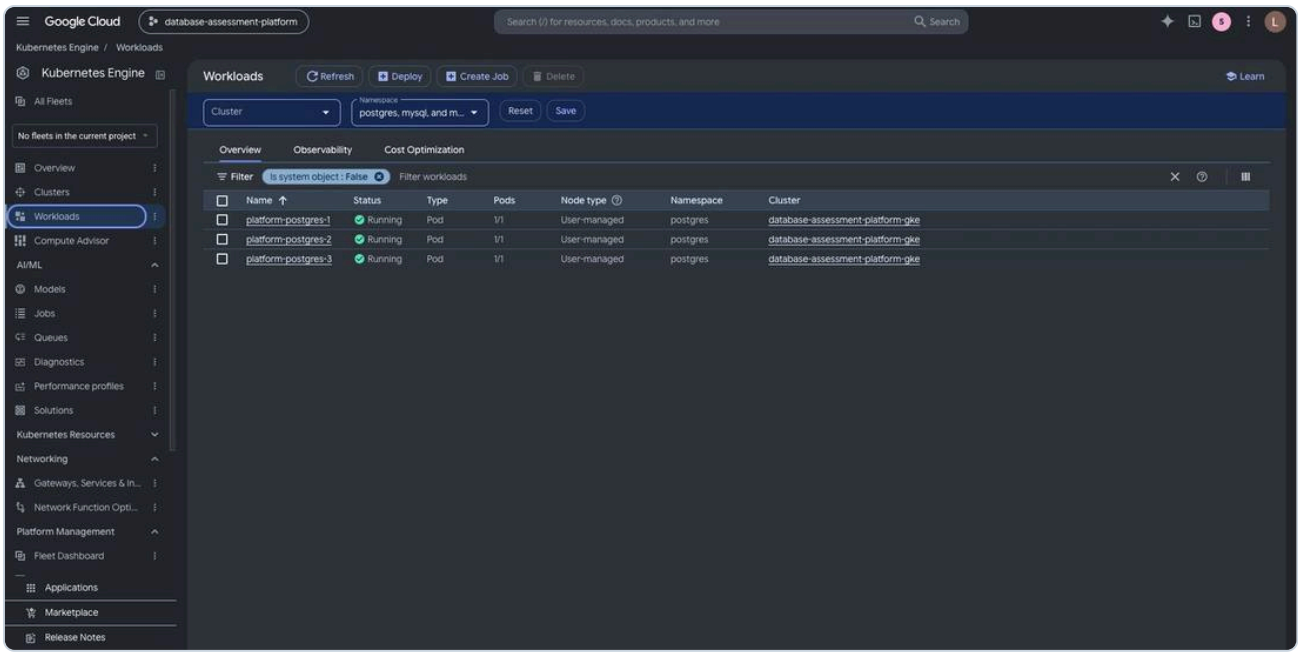


Figure 9.3 27 Oct 2026 · 20:28:08

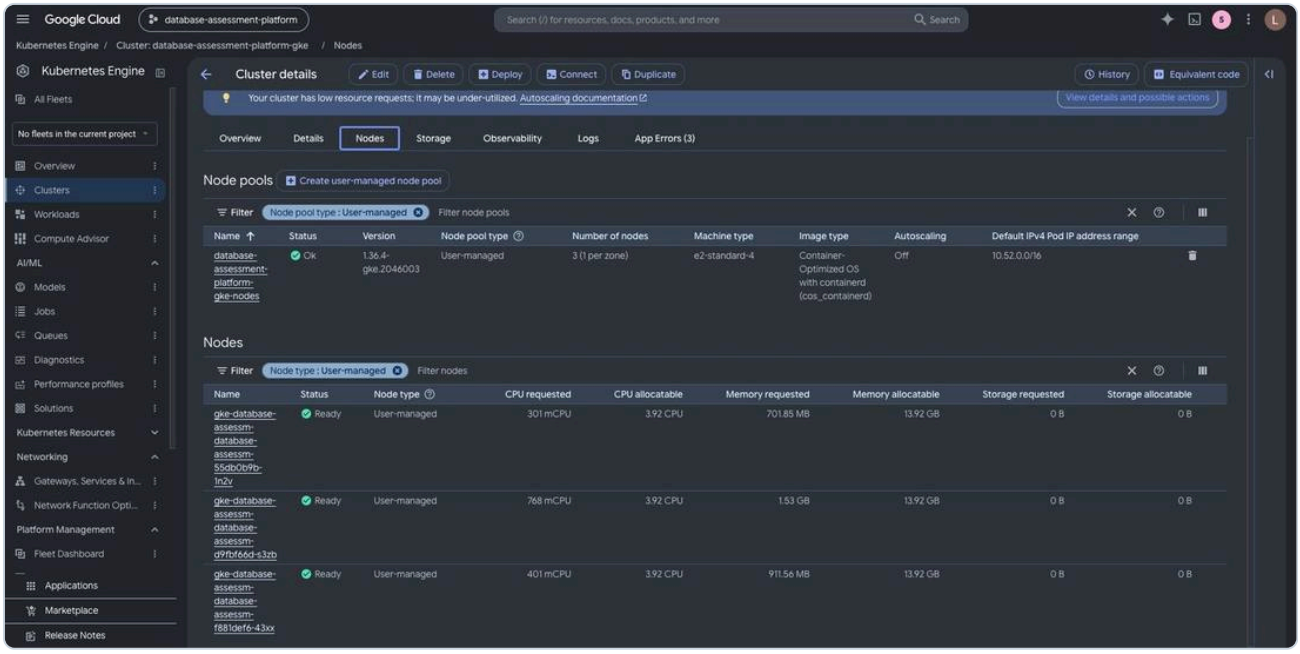


Figure 9.4 27 Oct 2026 · 20:29:35



Figure 9.5 27 Oct 2026 · 20:34:43

9.3 Deploy PostgreSQL

Deploy database | **PostgreSQL** | **Completed**

RunShore deploys PostgreSQL on Google Cloud · GKE and waits until every node is ready.

Started 27 Oct 2026, 21:11 · took 3m 04s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh postgres
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ WAL archive: <backup location>/platform-postgres-20261027211220/ (rendered: runtime/rendered/postgres/cluster.yaml)
- ✔ PostgreSQL platform deployed

9.4 Failover test · PostgreSQL

Failover | **PostgreSQL** | **Completed**

The primary PostgreSQL node is stopped on purpose. RunShore measures how quickly another node

takes over, and checks that no data was lost.

Started 27 Oct 2026, 21:20 · took 44s

COMMAND

```
SCOPE=gcp db-platform/postgres/failover/kubernetes/run_failover.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.
- ✔ Final CNPG phase: Cluster in healthy state
- ✔ Final ready instances: 3

9.5 Register PostgreSQL with PMM

Register | PMM | PostgreSQL | Completed

PostgreSQL is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 21:31 · took 24s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/register_cnpg_service.sh
```

- ✔ Credentials loaded
- ✔ pmm-0
- ✔ PostgreSQL successfully registered into PMM

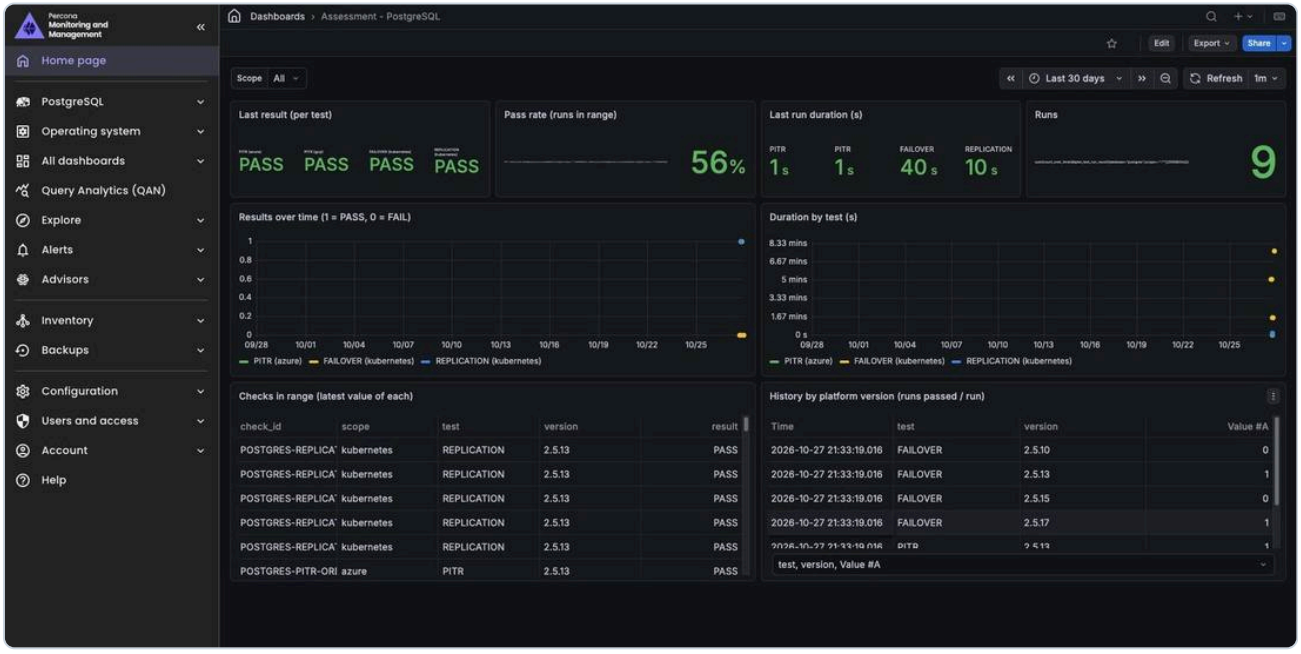


Figure 9.6 27 Oct 2026 · 21:33:33

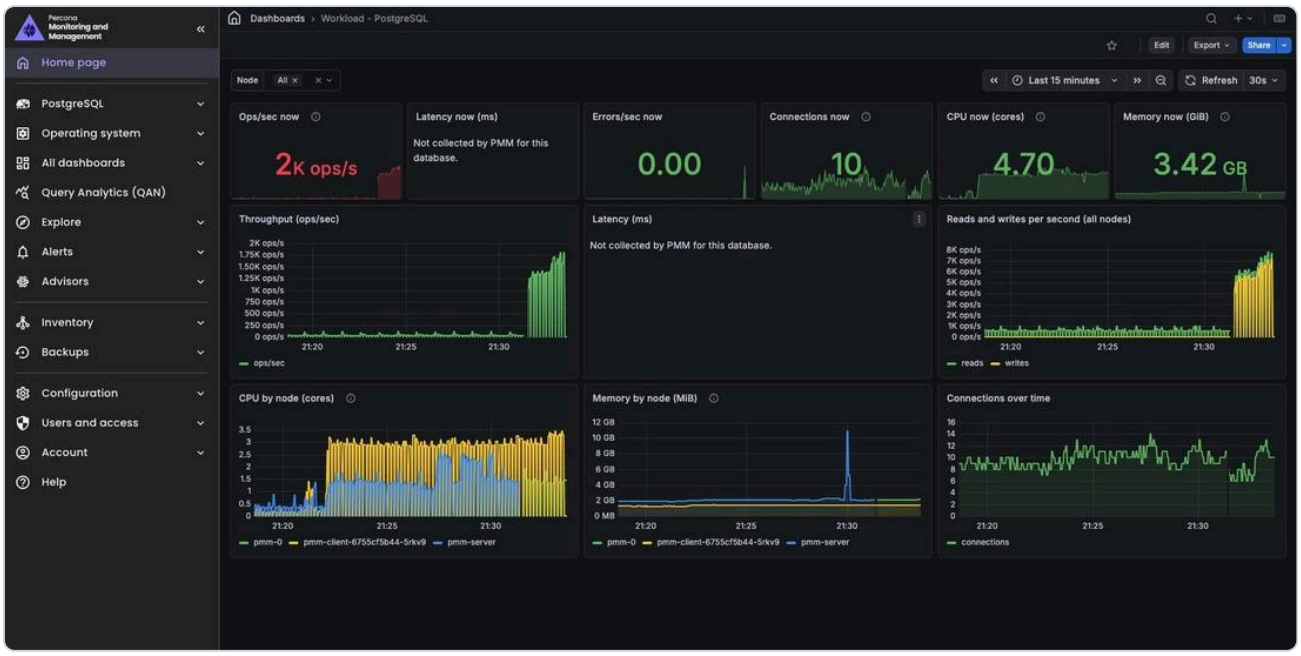


Figure 9.7 27 Oct 2026 · 21:34:04

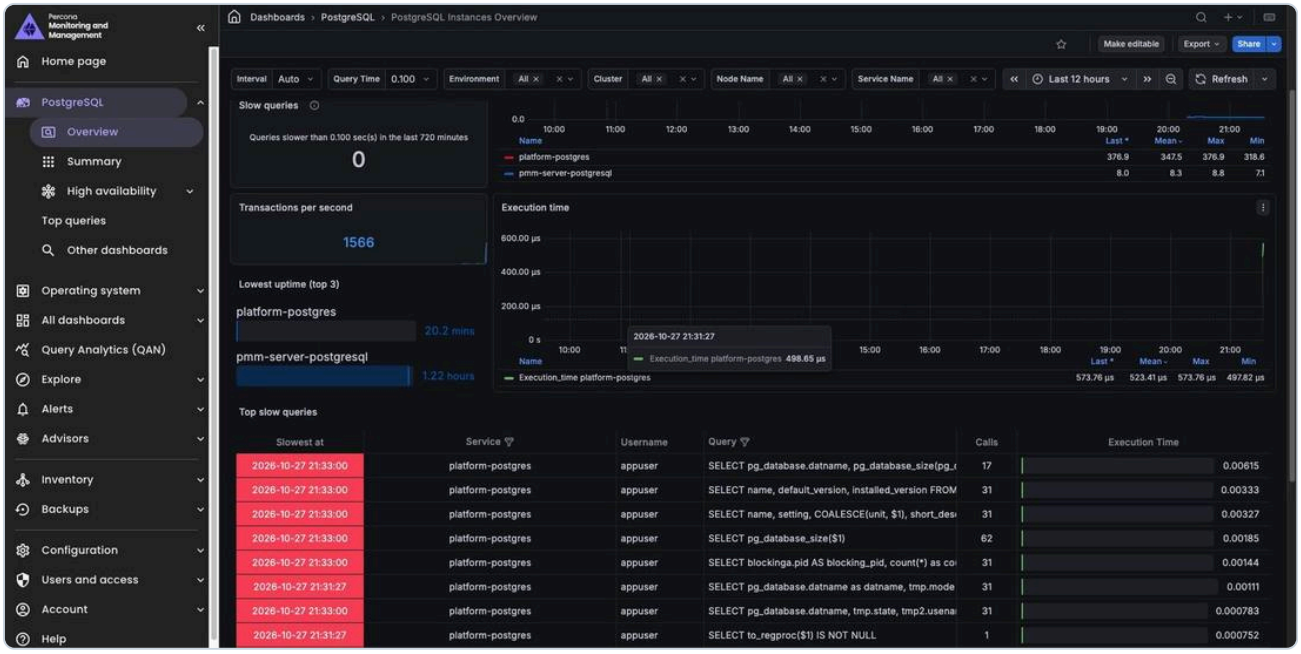


Figure 9.8 27 Oct 2026 · 21:34:25

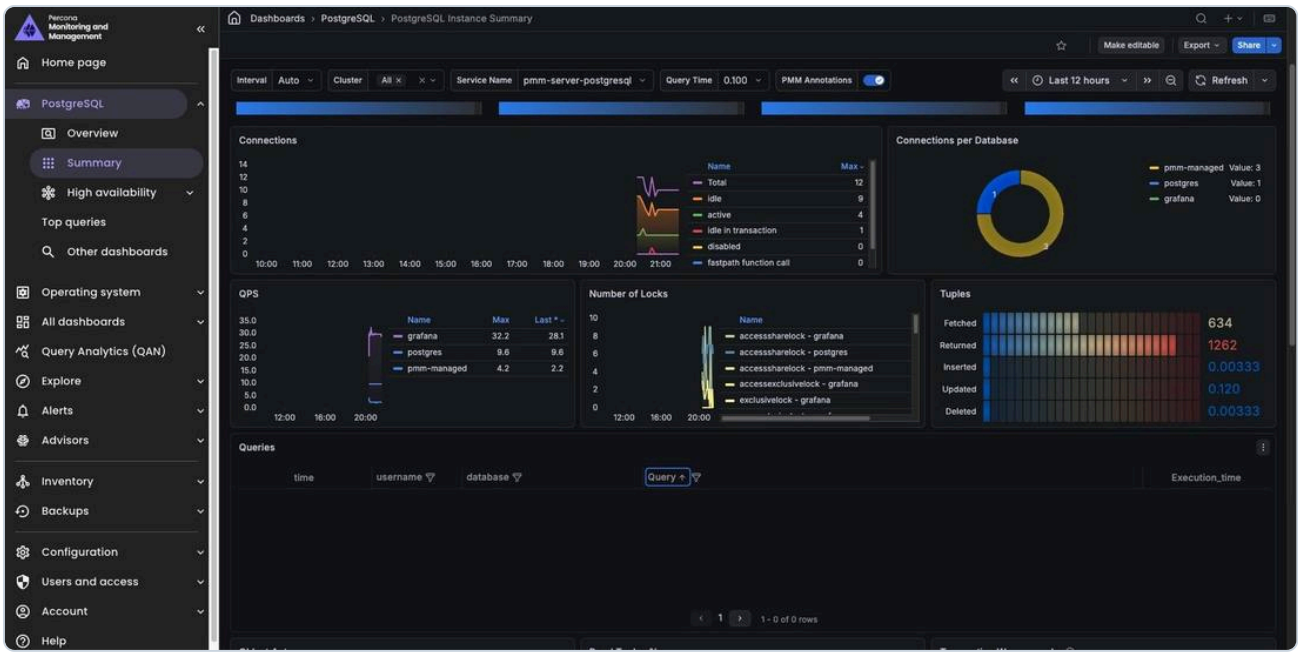


Figure 9.9 27 Oct 2026 · 21:35:18

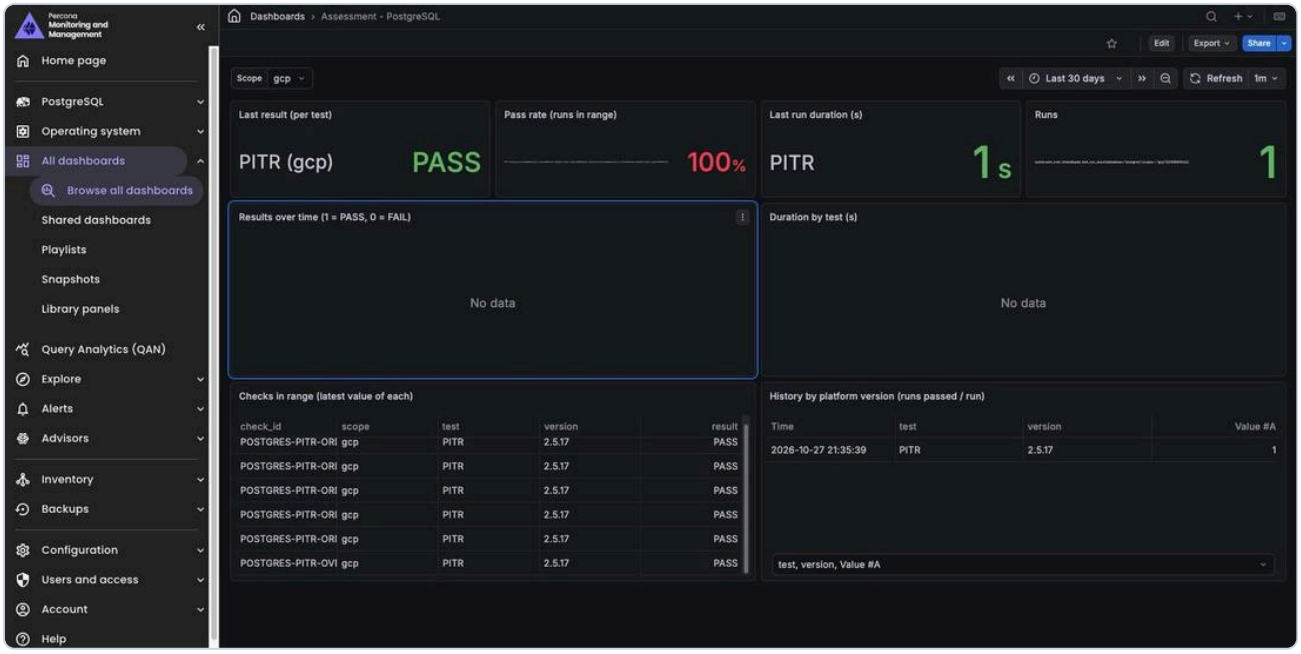


Figure 9.10 27 Oct 2026 · 21:36:00

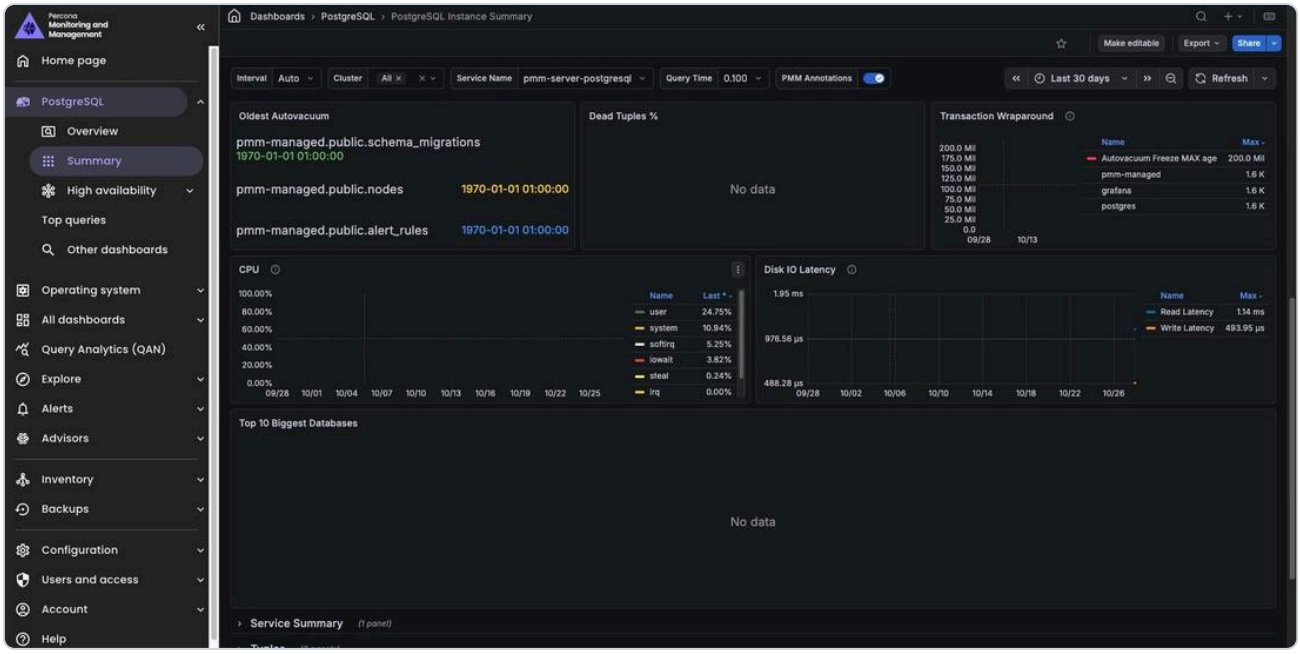


Figure 9.11 27 Oct 2026 · 21:36:20

9.6 Unregister PostgreSQL from PMM

Unregister | PMM | PostgreSQL | Completed

PostgreSQL is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 21:38 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/postgres/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

9.7 Destroy PostgreSQL

Destroy | PostgreSQL | Completed

PostgreSQL is removed and its storage released.

Started 27 Oct 2026, 21:54 · took 1m 29s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh postgres
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable

- ✔ GCP/GKE context validated
- ✔ PostgreSQL namespace removed
- ✔ PostgreSQL GCP service account deleted.

9.8 Deploy MongoDB

Deploy database	MongoDB	Completed

RunShore deploys MongoDB on Google Cloud · GKE and waits until every node is ready.

Started 27 Oct 2026, 21:57 · took 3m 33s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh mongodb
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ Bound PVCs: 3
- ✔ MongoDB platform validation complete

9.9 Register MongoDB with PMM

Register	PMM	MongoDB	Completed

MongoDB is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 22:02 · took 20s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/register_mongodb_replicaset.sh
```

- ✔ MongoDB namespace : mongodb
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ Replica set : rs0
- ✔ Running MongoDB members: 3
- ✔ Cleanup complete
- ✔ MongoDB service present in PMM inventory

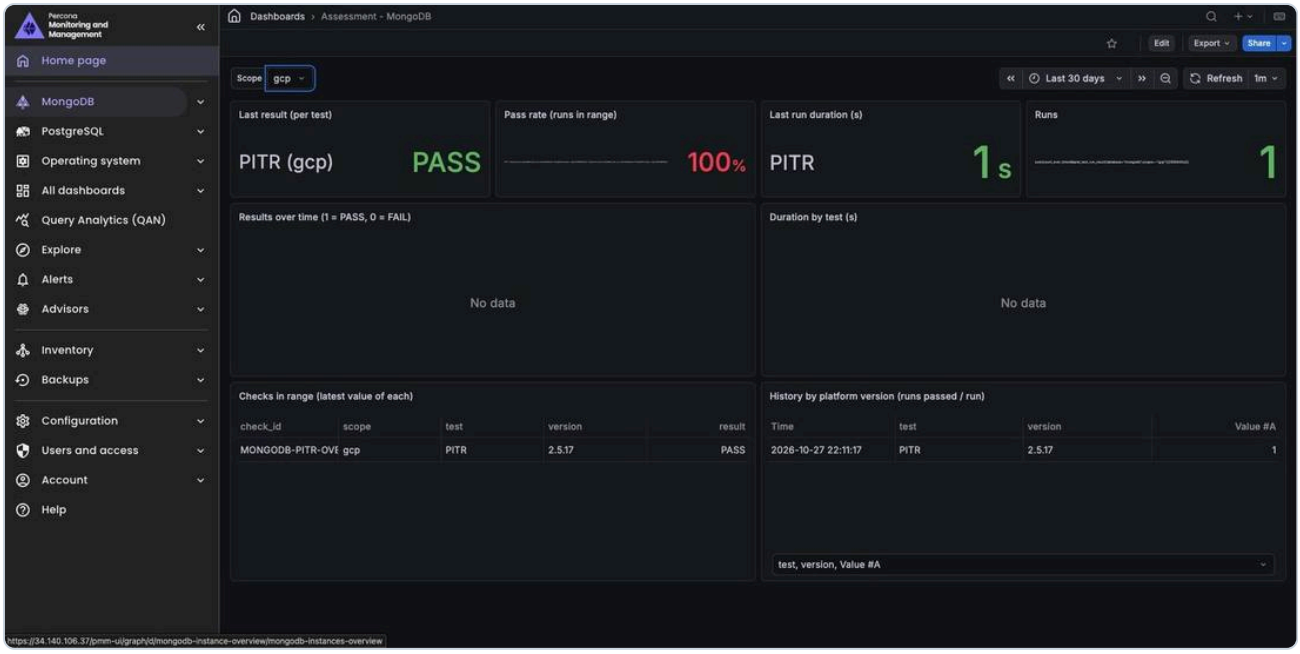


Figure 9.12 27 Oct 2026 · 22:11:30

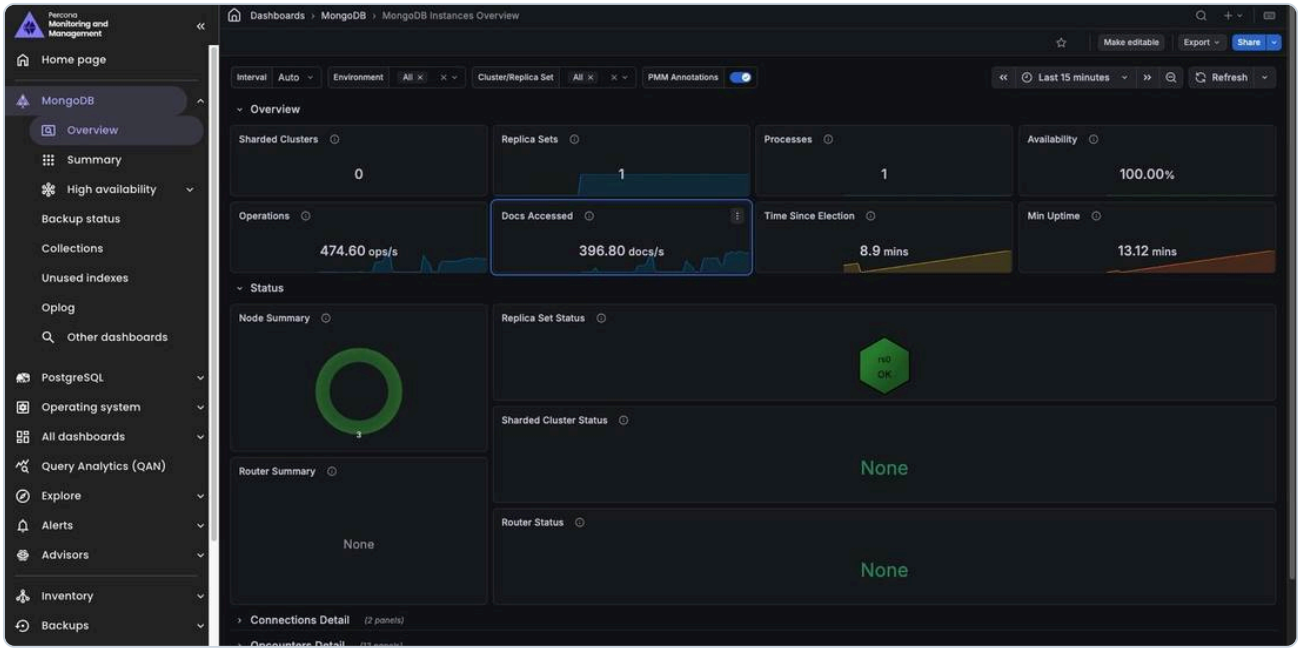


Figure 9.13 27 Oct 2026 · 22:12:46

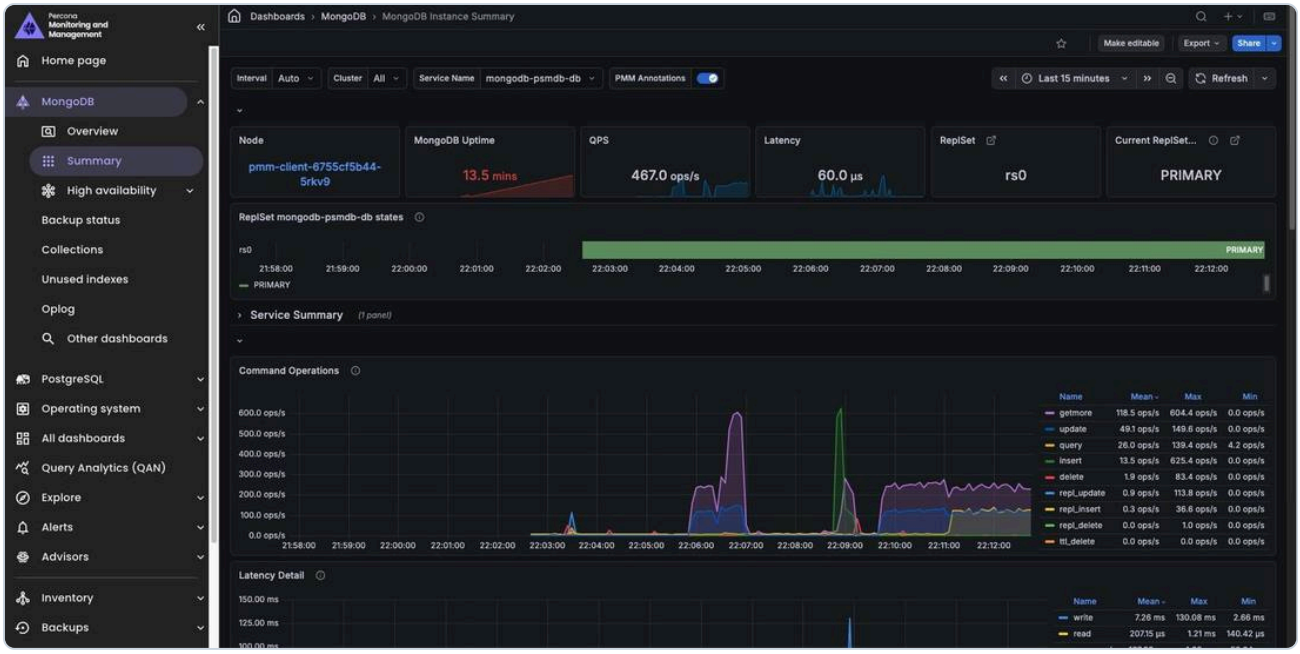


Figure 9.14 27 Oct 2026 · 22:12:55

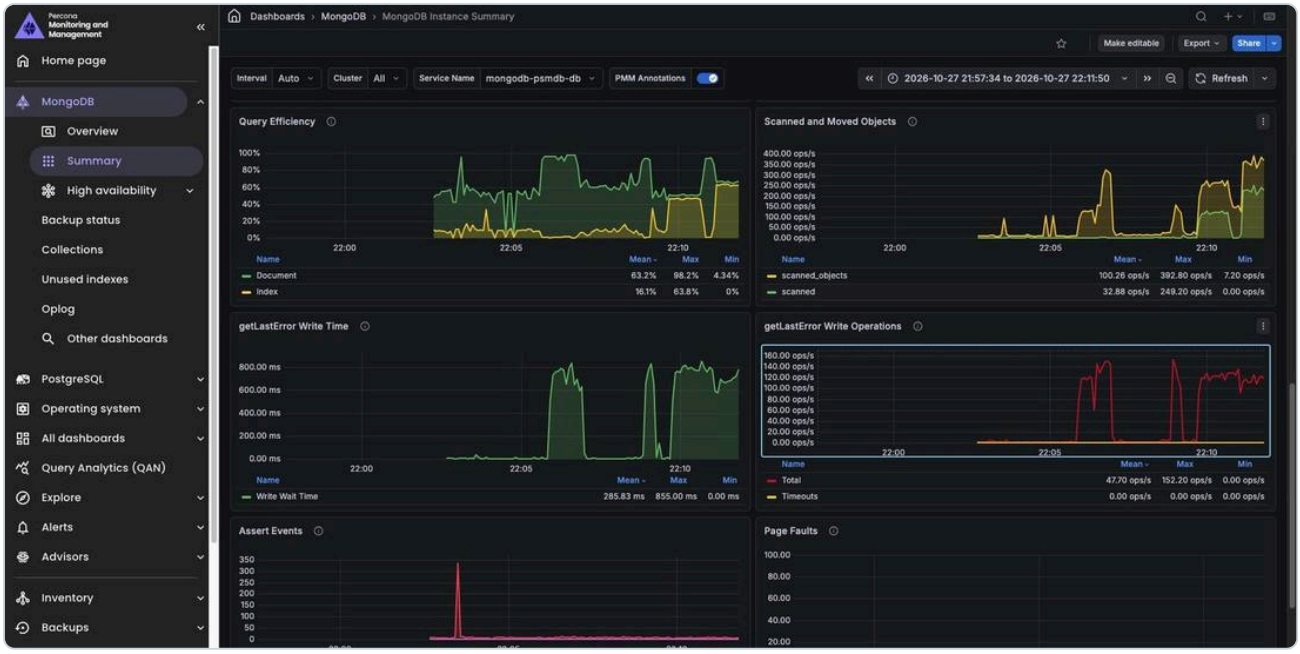


Figure 9.15 27 Oct 2026 · 22:13:47

9.10 Unregister MongoDB from PMM

Unregister	PMM	MongoDB	Completed

MongoDB is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 22:14 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/mongodb/unregister.sh
```

✔ pmm-0

- ✔ nothing left in PMM inventory -- Registration shows FALSE
- ✔ MongoDB service absent from PMM: mongodb-psmdb-db

9.11 Deploy MySQL

Deploy database	MySQL	Completed

RunShore deploys MySQL on Google Cloud · GKE and waits until every node is ready.

Started 27 Oct 2026, 22:19 · took 5m 57s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh mysql
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ PXC cluster health converged
- ✔ PXC deployment complete

9.12 Register MySQL with PMM

Register	PMM	MySQL	Completed

MySQL is connected to PMM, so its metrics and dashboards appear there.

Started 27 Oct 2026, 22:29 · took 20s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/register_pxc_service.sh
```

- ✔ pmm-0
- ✔ Credentials loaded
- ✔ PXC successfully registered into PMM

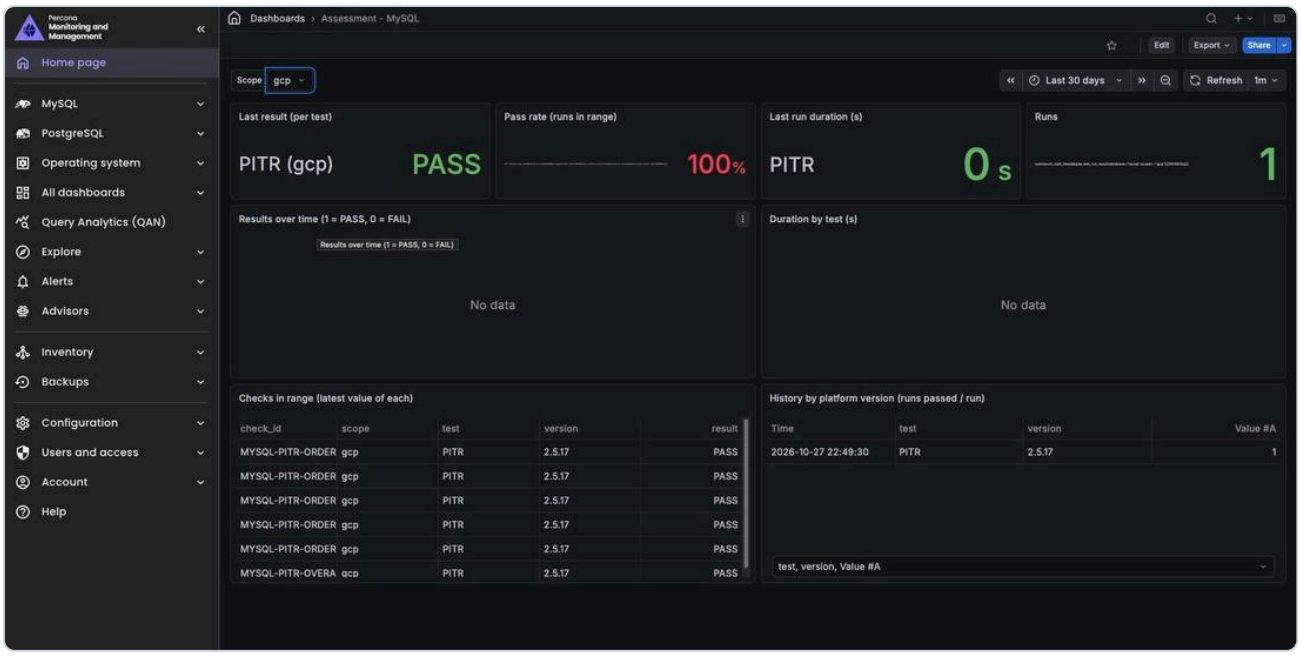


Figure 9.16 27 Oct 2026 · 22:49:51

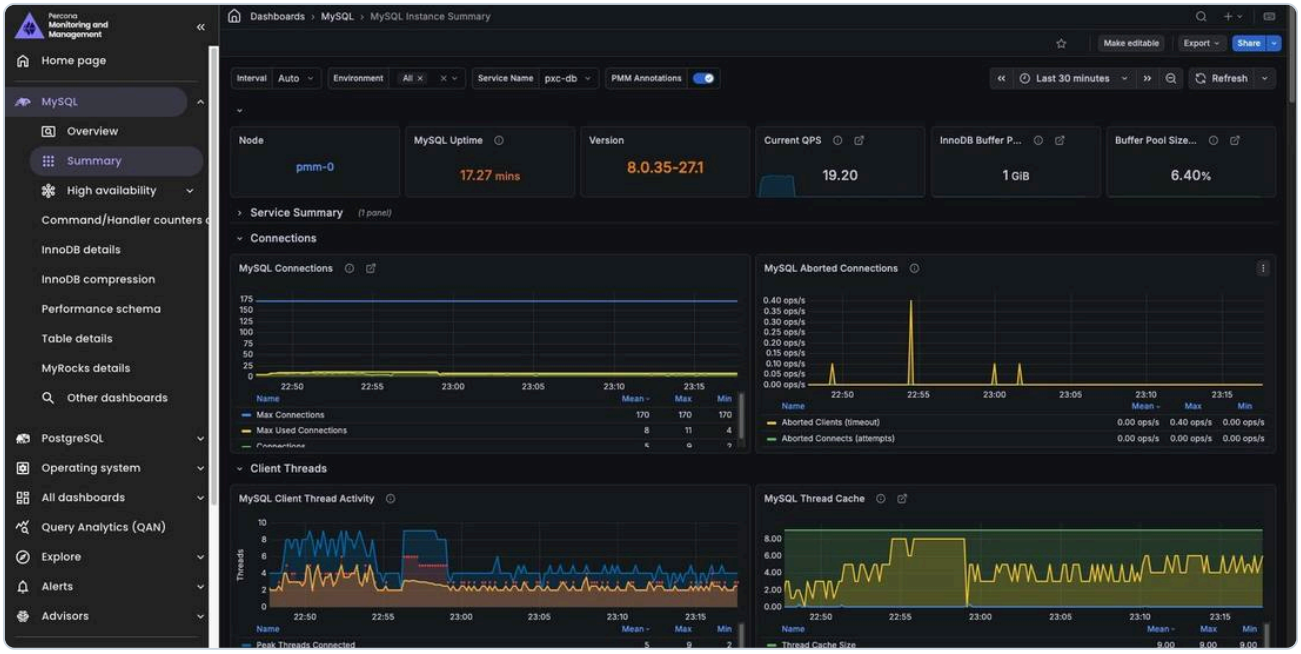


Figure 9.17 27 Oct 2026 · 23:17:53

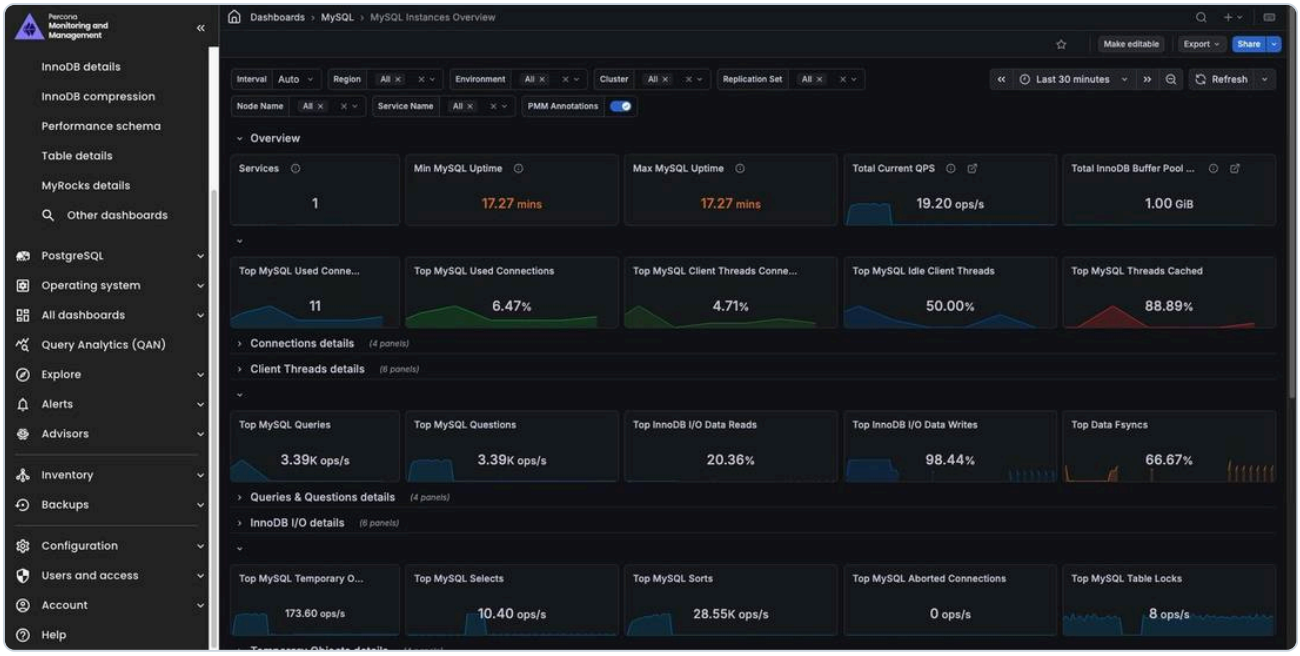


Figure 9.18 27 Oct 2026 · 23:18:06

9.13 Unregister MySQL from PMM

Unregister | PMM | MySQL | Completed

MySQL is removed from PMM; the database itself keeps running.

Started 27 Oct 2026, 23:18 · took 12s

COMMAND

```
bash observability/pmm/registration/kubernetes/mysql/unregister.sh
```

- ✔ pmm-0
- ✔ nothing left in PMM inventory -- Registration shows FALSE

9.14 Remove PMM

Unregister | PMM | Completed

PMM is removed from the cluster once no database is registered with it.

Started 27 Oct 2026, 23:19 · took 1m 05s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh pmm
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable

- ✔ GCP/GKE context validated
- ✔ PMM Client deployment removed
- ✔ PMM namespace absent

9.15 Install New Relic

Deploy tool | New Relic | Completed

New Relic is installed, ready to monitor the databases in this cluster.

Started 27 Oct 2026, 23:21 · took 1m 16s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh newrelic
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ New Relic MySQL registration state recorded
- ✔ MySQL collection converged

9.16 Register MySQL with New Relic

Register | New Relic | MySQL | Completed

MySQL is connected to New Relic, so its metrics and dashboards appear there.

Started 27 Oct 2026, 23:23 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/register.sh
```

- ✔ Kubernetes, PXC and New Relic available
- ✔ PXC cluster ready
- ✔ MySQL endpoint discovered
- ✔ PXC administrative credential available
- ✔ PXC HAProxy endpoint present
- ✔ MySQL telemetry secret present

9.17 Failover test · MySQL

Failover | MySQL | Completed

The primary MySQL node is stopped on purpose. RunShore measures how quickly another node

takes over, and checks that no data was lost.

Started 27 Oct 2026, 23:24 · took 1m 28s

COMMAND

```
SCOPE=gcp db-platform/mysql/failover/kubernetes/run_failover.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ PXC administrative credential available
- ✔ All three PXC members are Kubernetes-Ready and Primary/Synced/Ready
- ✔ Assessment data verified on pxc-db-pxc-2
- ✔ Final three-member cluster is Kubernetes-Ready and Primary/Synced/Ready

9.18 Replication test · MySQL

Replication | MySQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 27 Oct 2026, 23:26 · took 1m 37s

COMMAND

```
SCOPE=gcp db-platform/mysql/replication/kubernetes/run_replication.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.

9.19 Point-in-time recovery test · MySQL

Assessment | MySQL | Completed with warnings

RunShore restores MySQL to a chosen moment from its backups and checks the restored data is correct.

Started 27 Oct 2026, 23:28 · took 7m 40s

COMMAND

```
SCOPE=gcp db-platform/mysql/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ No unfinished restore and the cluster has 3 nodes -- nothing to do
- ✔ mysql: clean start done. The database and monitoring tools were kept.
- ✔ Live root credential resolved from Kubernetes Secret pxc-db-secrets
- ✔ Prerequisites available
- ✔ Restore completed: pitr-demo-restore-1793143867

✔ PXC cluster ready

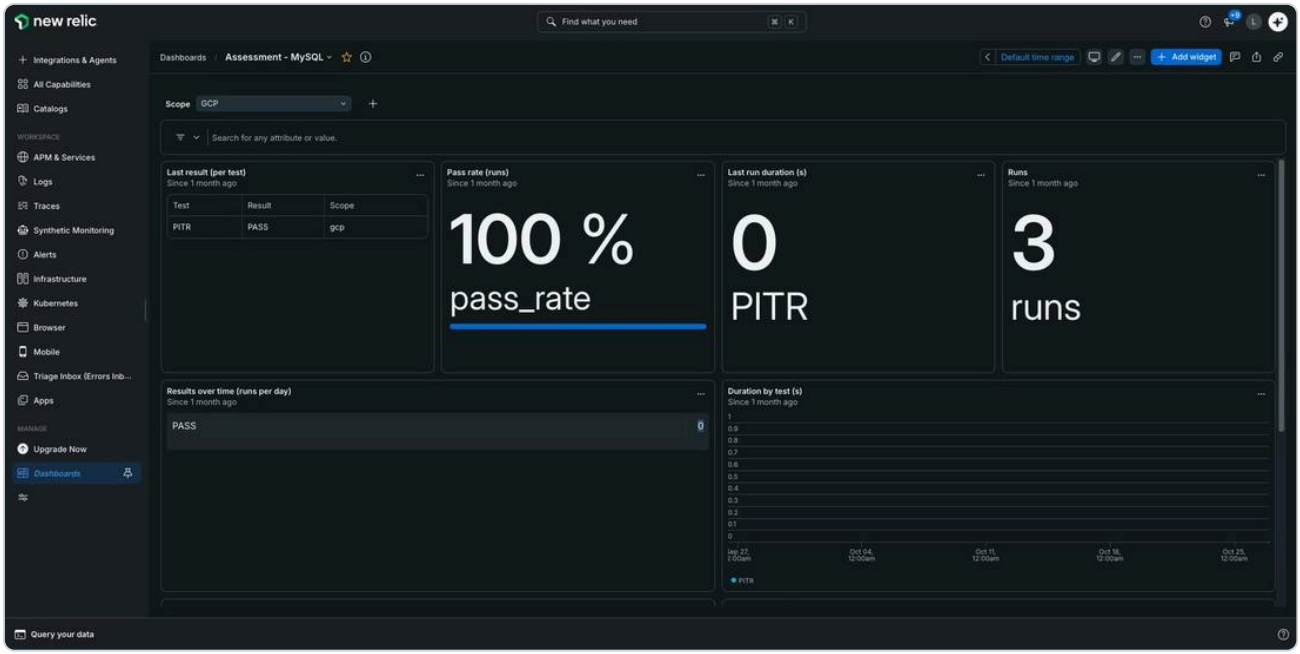


Figure 9.19 27 Oct 2026 · 23:38:36

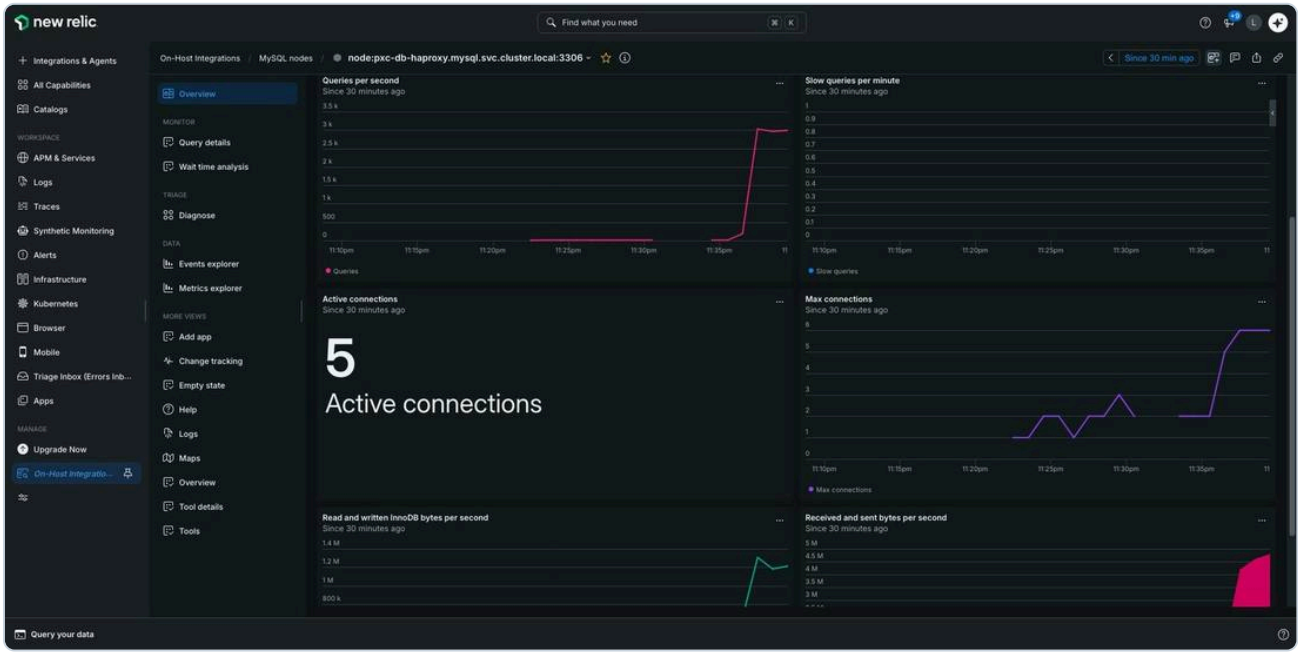


Figure 9.20 27 Oct 2026 · 23:41:21

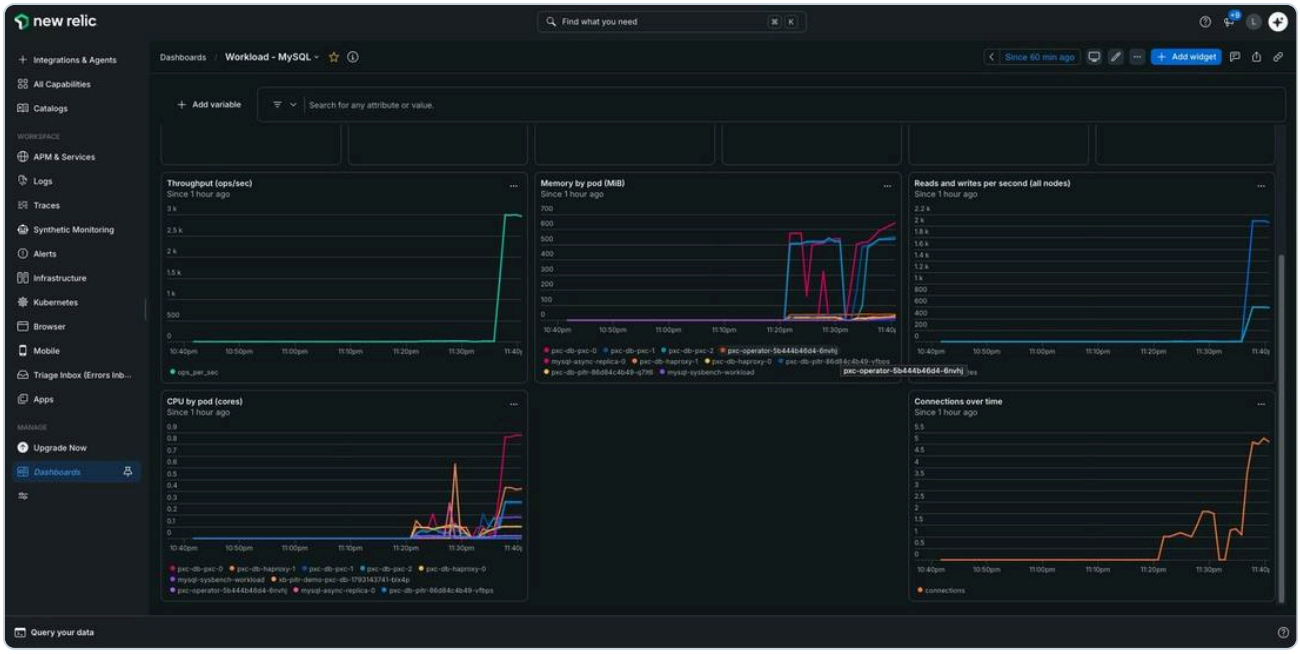


Figure 9.21 27 Oct 2026 · 23:41:58

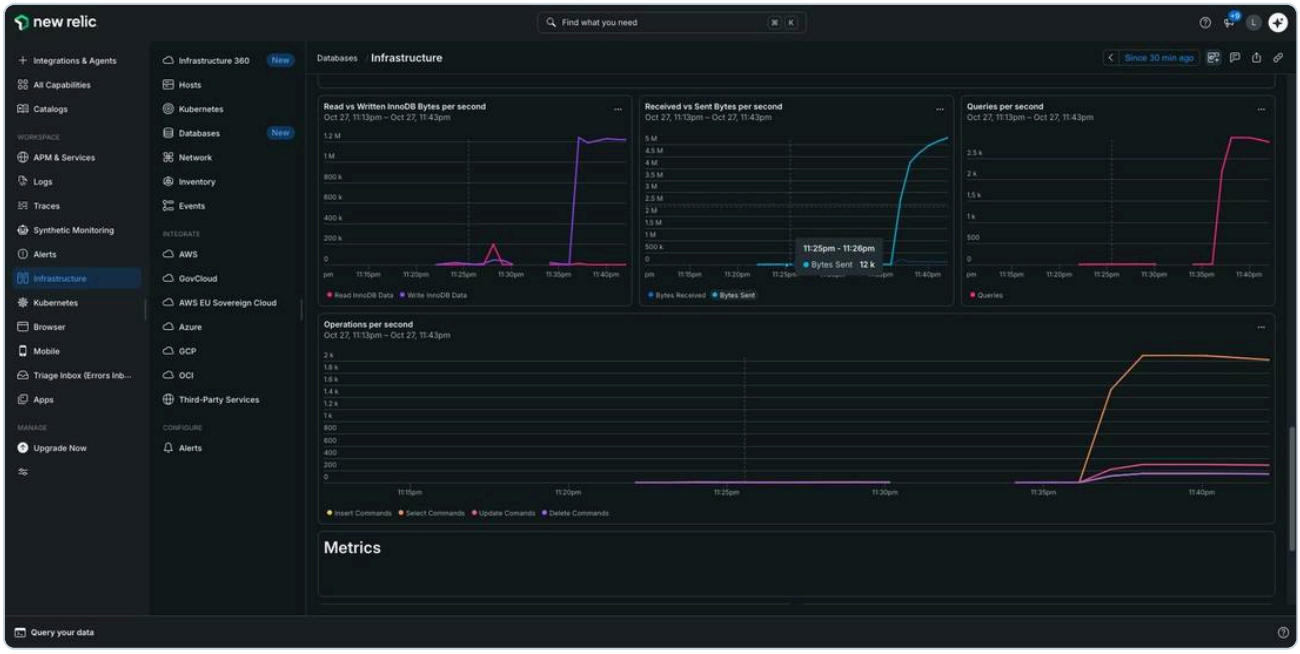


Figure 9.22 27 Oct 2026 · 23:43:15

9.20 Unregister MySQL from New Relic

Unregister	New Relic	MySQL	Completed

MySQL is removed from New Relic; the database itself keeps running.

Started 27 Oct 2026, 23:43 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mysql/unregister.sh
```

- ✔ Kubernetes and New Relic available
- ✔ MySQL registration state cleared

- ✔ Removed mysql-config.yml
- ✔ Removed newrelic-mysql-secret
- ✔ mysql-config.yml absent
- ✔ newrelic-mysql-secret absent

9.21 Destroy MySQL

Destroy | MySQL | Completed

MySQL is removed and its storage released.

Started 27 Oct 2026, 23:44 · took 1m 08s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh mysql
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ All namespace Pods terminated
- ✔ Namespace mysql fully removed

9.22 Register MongoDB with New Relic

Register | New Relic | MongoDB | Completed

MongoDB is connected to New Relic, so its metrics and dashboards appear there.

Started 27 Oct 2026, 23:49 · took 1m 36s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/register.sh
```

- ✔ Kubernetes API reachable
- ✔ New Relic deployment available
- ✔ MongoDB cluster : mongodb-psmdb-db
- ✔ MongoDB namespace: mongodb
- ✔ Exporter reachable inside the cluster: http://mongodb-newrelic-exporter.mongodb.svc.cluster.local:9216/metri (mongodb_up 1)
- ✔ New Relic scraped MongoDB with no errors since 2026-10-27T23:50:39Z

9.23 Failover test · MongoDB

Failover | MongoDB | Completed

The primary MongoDB node is stopped on purpose. RunShore measures how quickly another node

takes over, and checks that no data was lost.

Started 27 Oct 2026, 23:51 · took 48s

COMMAND

```
SCOPE=gcp db-platform/mongodb/failover/kubernetes/run_failover.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ PSMDB cluster is ready.
- ✔ Replica set: rs0
- ✔ Ready members: 3/3
- ✔ HA validation marker survived failover.
- ✔ MongoDB replica set recovered to 3/3 ready members.

9.24 Replication test · MongoDB

Replication | MongoDB | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 27 Oct 2026, 23:53 · took 32s

COMMAND

```
SCOPE=gcp db-platform/mongodb/replication/kubernetes/run_replication.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Three-member replica-set topology verified.
- ✔ PRIMARY is writable.
- ✔ SECONDARY is in secondary state.
- ✔ Replicated data matches.
- ✔ MONGODB REPLICATION: PASS

9.25 Point-in-time recovery test · MongoDB

Assessment | MongoDB | Completed

RunShore restores MongoDB to a chosen moment from its backups and checks the restored data is correct.

Started 27 Oct 2026, 23:54 · took 2m 33s

COMMAND

```
SCOPE=gcp db-platform/mongodb/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ mongodb: clean start done. The database and monitoring tools were kept.
- ✔ Cluster: mongodb-psmdb-db
- ✔ PITR archiving enabled
- ✔ Baseline seeded
- ✔ Post-marker orders inserted

- ✔ Archived oplog covers the target (2026-10-27T23:54:30 - 2026-10-27T23:55:56) after 70s

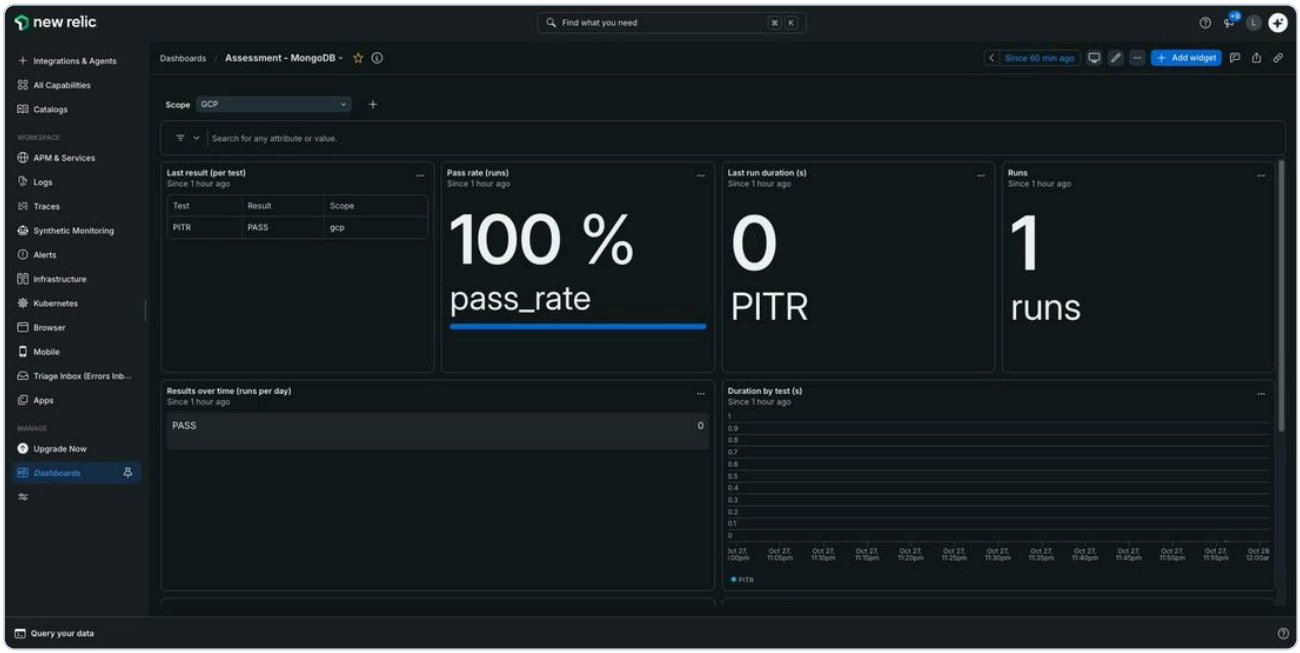


Figure 9.23 28 Oct 2026 · 00:02:43

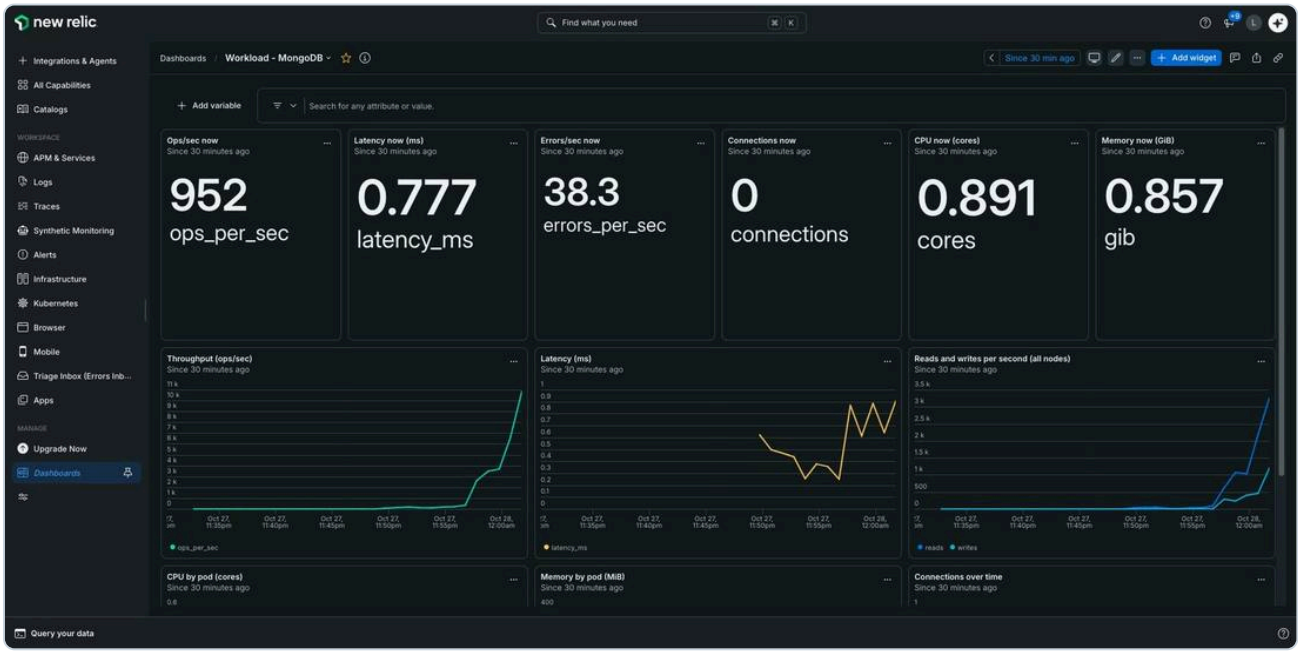


Figure 9.24 28 Oct 2026 · 00:02:53

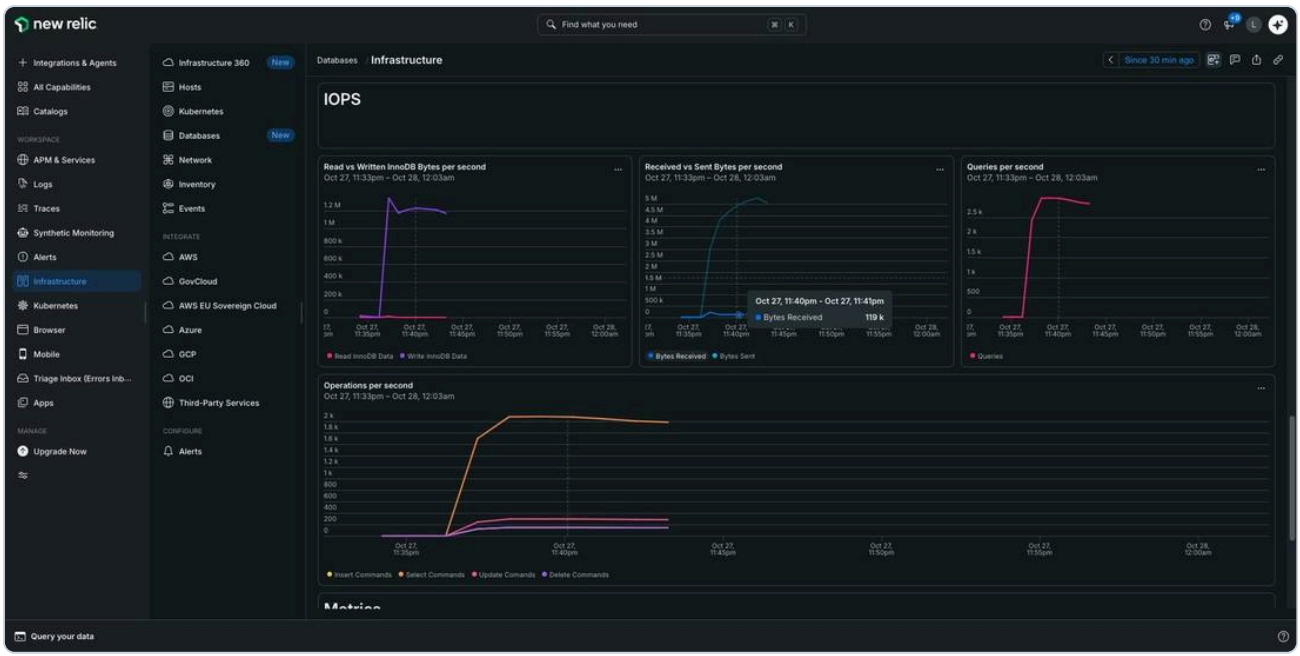


Figure 9.25 28 Oct 2026 · 00:03:24



Figure 9.26 28 Oct 2026 · 00:05:06

9.26 Unregister MongoDB from New Relic

Unregister	New Relic	MongoDB	Completed

MongoDB is removed from New Relic; the database itself keeps running.

Started 28 Oct 2026, 00:19 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/mongodb/unregister.sh
```

✔ Kubernetes API reachable

- ✔ MongoDB New Relic registration state removed
- ✔ Removed Deployment/mongodb-newrelic-exporter (namespace mongodb)
- ✔ Removed Service/mongodb-newrelic-exporter (namespace mongodb)
- ✔ New Relic kubelet agents refreshed
- ✔ mongodb-config.yml absent

9.27 Destroy MongoDB

Destroy | MongoDB | Completed

MongoDB is removed and its storage released.

Started 28 Oct 2026, 00:22 · took 1m 01s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh mongodb
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ Tooling and Kubernetes API available
- ✔ MongoDB namespace absent

9.28 Register PostgreSQL with New Relic

Register | New Relic | PostgreSQL | Completed

PostgreSQL is connected to New Relic, so its metrics and dashboards appear there.

Started 28 Oct 2026, 00:27 · took 16s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/register.sh
```

- ✔ Kubernetes, PostgreSQL and New Relic available
- ✔ Prerequisites available
- ✔ Monitoring secret available: postgres-monitoring
- ✔ Monitoring credentials loaded
- ✔ New Relic agents refreshed
- ✔ nri-postgresql configuration present

9.29 Replication test · PostgreSQL

Replication | PostgreSQL | Completed

RunShore writes on the primary and checks that the change reaches every replica.

Started 28 Oct 2026, 00:28 · took 16s

COMMAND

```
SCOPE=gcp db-platform/postgres/replication/kubernetes/run_replication.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped
- ✔ postgres: clean start done. The database and monitoring tools were kept.

9.30 Workload · PostgreSQL

Workload | PostgreSQL | Completed

A load generator runs against PostgreSQL, so the monitoring shows real traffic. It stops by itself after a set time.

Started 28 Oct 2026, 00:29 · took 8s

COMMAND

```
bash db-platform/postgres/workload/kubernetes/start.sh
```

- ✔ pgbench schema initialized.
- ✔ pgbench runtime permissions ensured for appuser.
- ✔ pgbench schema validated.

9.31 Point-in-time recovery test · PostgreSQL

Assessment | PostgreSQL | Completed with warnings

RunShore restores PostgreSQL to a chosen moment from its backups and checks the restored data is correct.

Started 28 Oct 2026, 00:30 · took 7m 53s

COMMAND

```
SCOPE=gcp db-platform/postgres/pitr/kubernetes/run_pitr_demo.sh
```

- ✔ POSTGRESQL REPLICATION TEST STOPPED
- ✔ Native CNPG replication was not modified.
- ✔ PostgreSQL Kubernetes workload stopped

✔ postgres: clean start done. The database and monitoring tools were kept.

! Could not resolve hostPath for volume pvc-32a029eb-2c15-4dc1-b398-6d3dcc66a419 — skipping permission fix.

✔ PITR clone ready

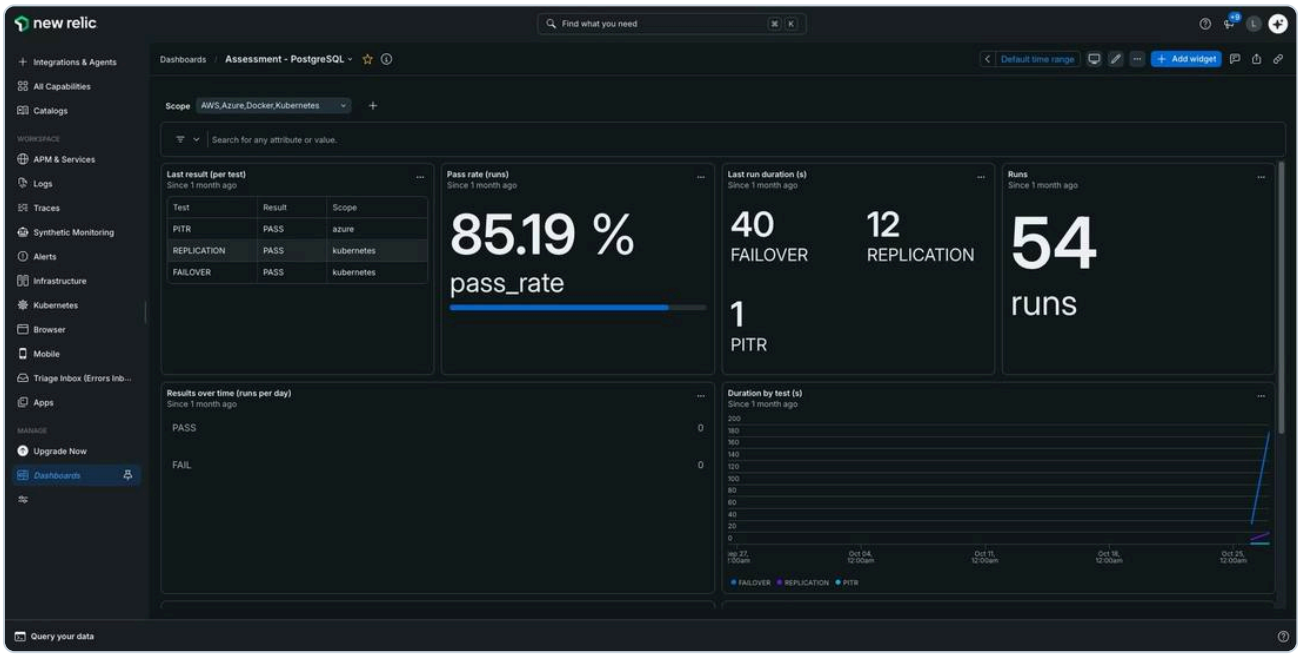


Figure 9.27 28 Oct 2026 · 00:50:03

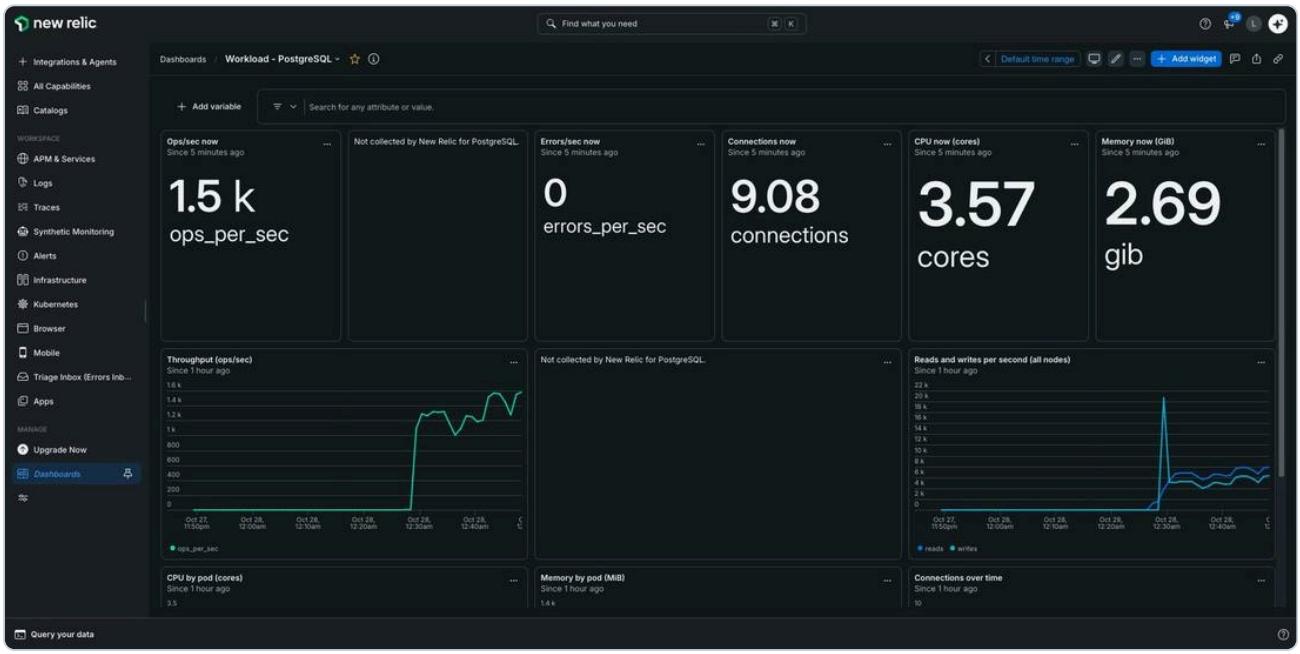


Figure 9.28 28 Oct 2026 · 00:50:17

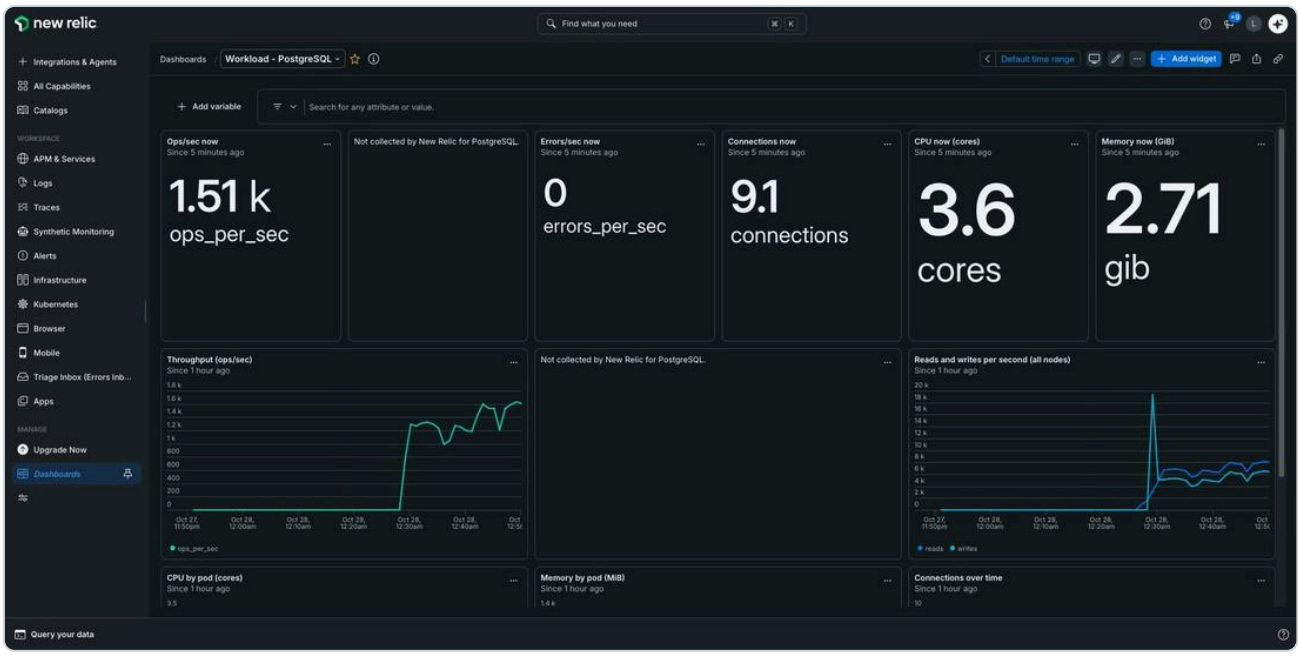


Figure 9.29 28 Oct 2026 · 00:51:23

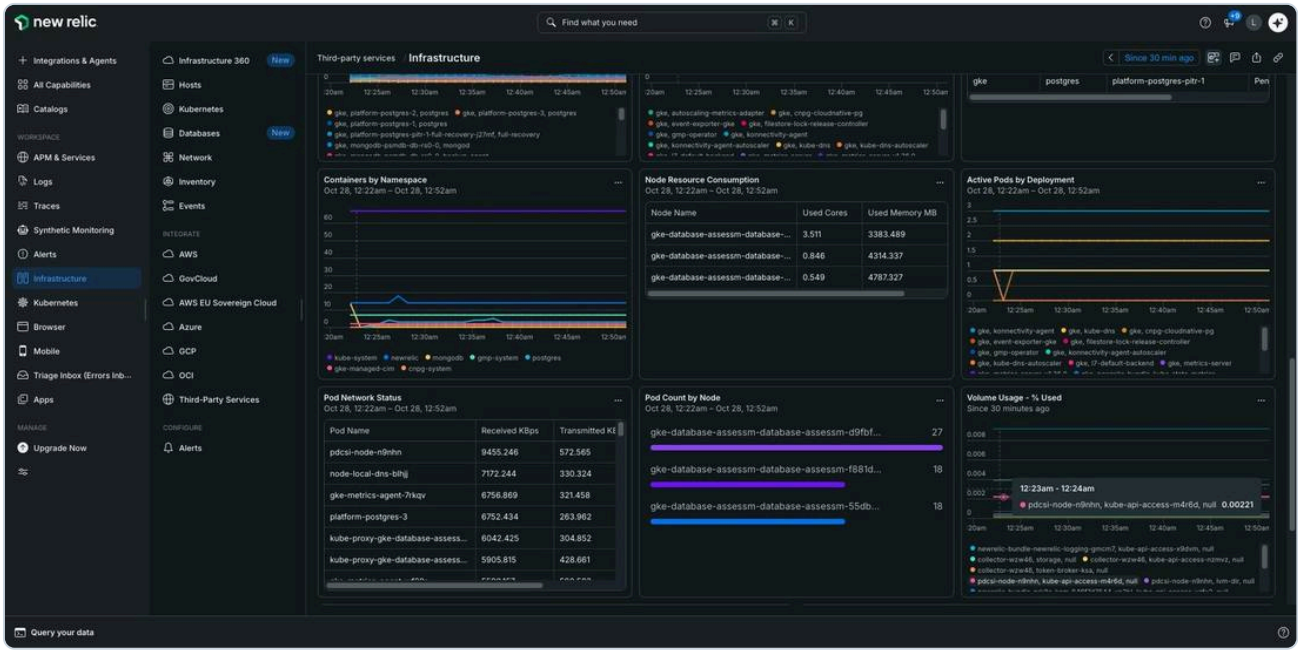


Figure 9.30 28 Oct 2026 · 00:52:32

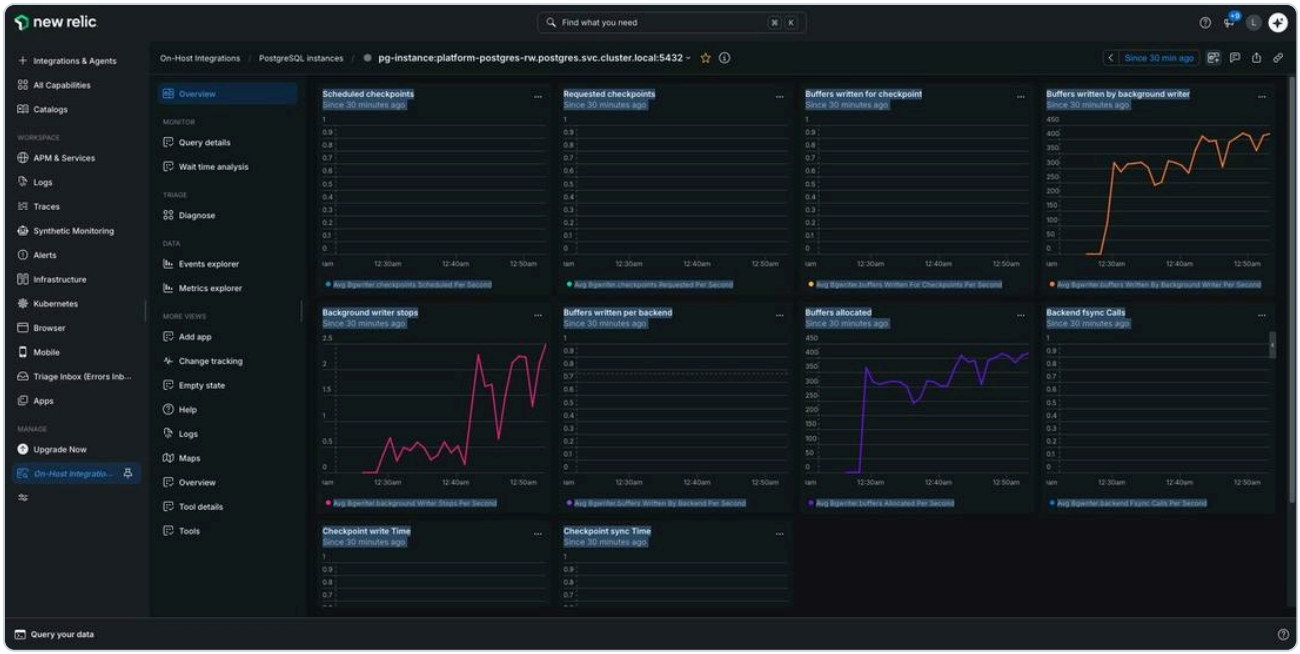


Figure 9.31 28 Oct 2026 · 00:54:27

9.32 Unregister PostgreSQL from New Relic

Unregister | New Relic | PostgreSQL | Completed

PostgreSQL is removed from New Relic; the database itself keeps running.

Started 28 Oct 2026, 00:55 · took 12s

COMMAND

```
bash observability/newrelic/registration/kubernetes/postgres/unregister.sh
```

- ✔ Kubernetes API reachable
- ✔ PostgreSQL integration configuration absent
- ✔ New Relic PostgreSQL telemetry secret absent
- ✔ New Relic agents refreshed
- ✔ postgres-config.yml absent
- ✔ newrelic-postgres-secret absent

9.33 Remove New Relic

Unregister | New Relic | Completed

New Relic is removed from the cluster once no database is registered with it.

Started 28 Oct 2026, 00:56 · took 36s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh newrelic
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke

- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ Helm release removed
- ✔ Namespace removed

9.34 Install Splunk

Deploy tool	Splunk	Completed with warnings

Splunk is installed, ready to monitor the databases in this cluster.

Started 28 Oct 2026, 00:58 · took 6m 53s

COMMAND

```
platform/cloud/gcp/orchestrator/deploy.sh splunk
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ No pods in a failed state
- ✔ Splunk accepted a HEC test event

9.35 Register PostgreSQL with Splunk

Register	Splunk	PostgreSQL	Completed

PostgreSQL is connected to Splunk, so its metrics and dashboards appear there.

Started 28 Oct 2026, 01:06 · took 16s

COMMAND

```
bash observability/splunk/registration/kubernetes/postgres/register.sh
```

- ✔ Resolved Splunk admin credentials from secret: splunk-s1-standalone-secret-v1
- ✔ Created HEC token 'postgres_hec_token'
- ✔ HEC token resolved
- ✔ No known pipeline errors detected
- ✔ Registration event confirmed in Splunk (index=main sourcetype="postgresql:postgres")
- ✔ Postgres logs successfully registered into Splunk

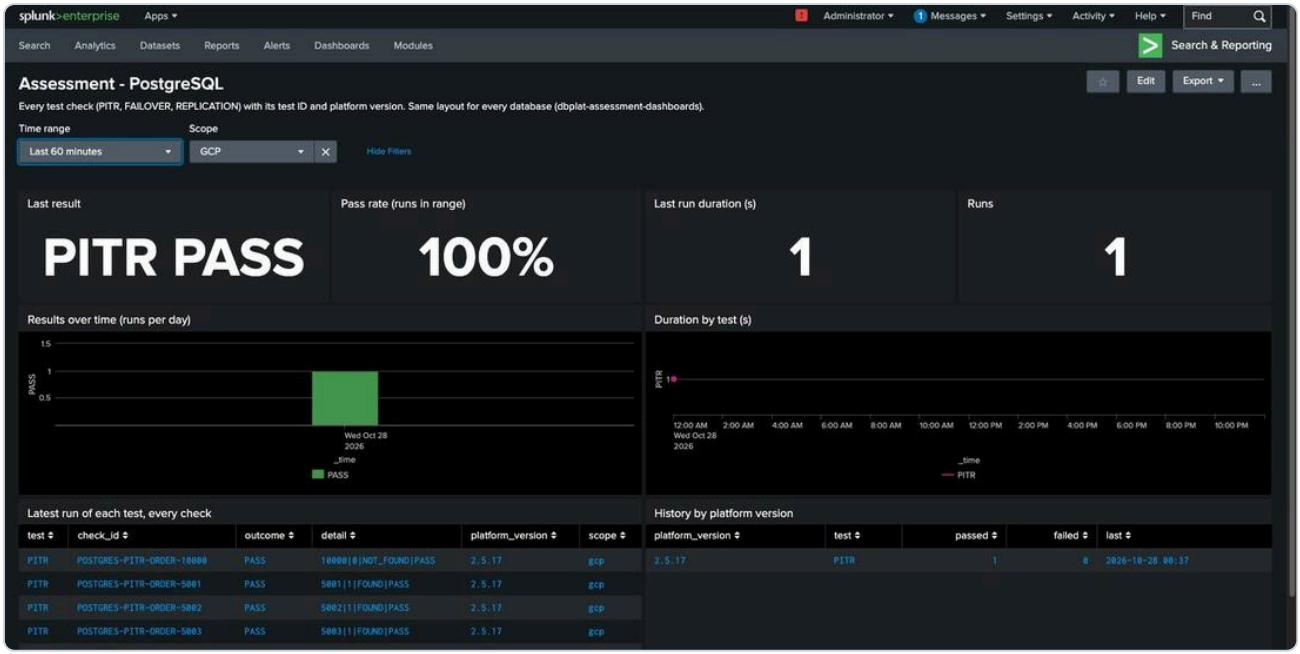


Figure 9.32 28 Oct 2026 · 01:12:08

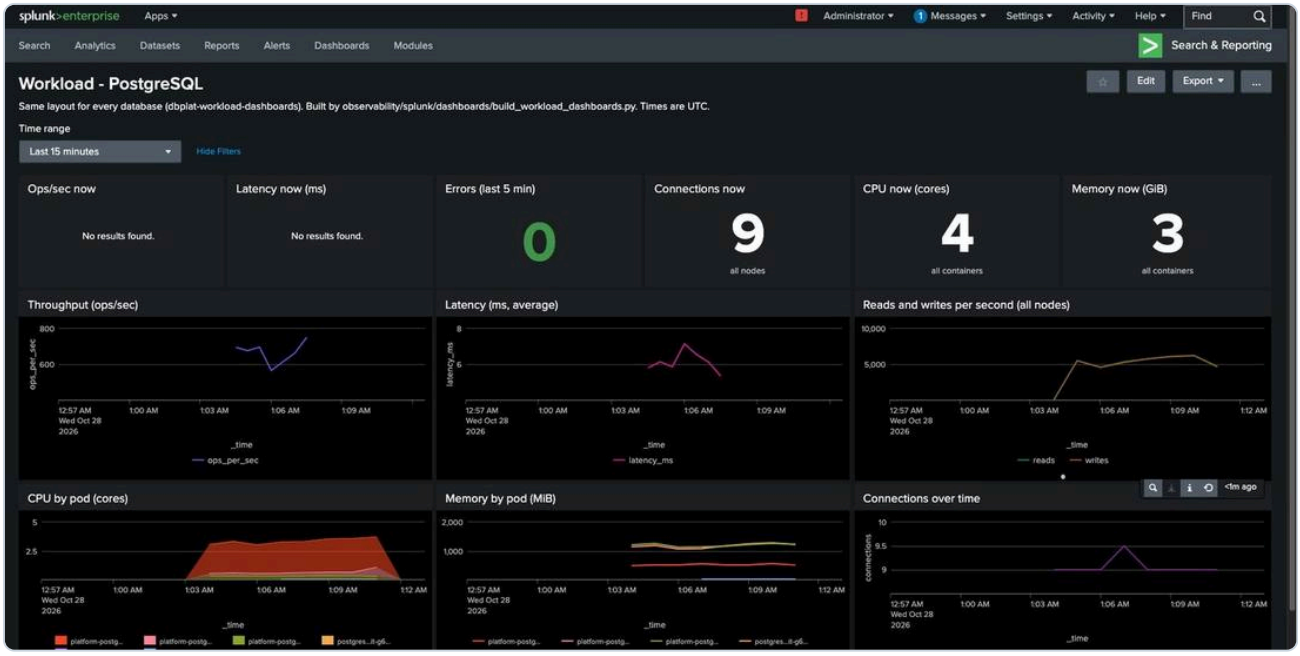


Figure 9.33 28 Oct 2026 · 01:12:20

9.36 Register MongoDB with Splunk

Register	Splunk	MongoDB	Completed

MongoDB is connected to Splunk, so its metrics and dashboards appear there.

Started 28 Oct 2026, 01:19 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/register.sh
```

✔ PSMDB: mongodb-psmdb-db

- ✔ Splunk HEC service: splunk-s1-standalone-service
- ✔ Backup written:
- ✔ MongoDB Splunk HEC secret converged
- ✔ MongoDB Splunk HEC secret validated
- ✔ MongoDB logs successfully registered into Splunk

9.37 Workload · MongoDB

Workload | MongoDB | Completed

A load generator runs against MongoDB, so the monitoring shows real traffic. It stops by itself after a set time.

Started 28 Oct 2026, 01:20 · took 12s

COMMAND

```
bash db-platform/mongodb/workload/kubernetes/start.sh
```

- ✔ MongoDB YCSB workload Pod is Running.

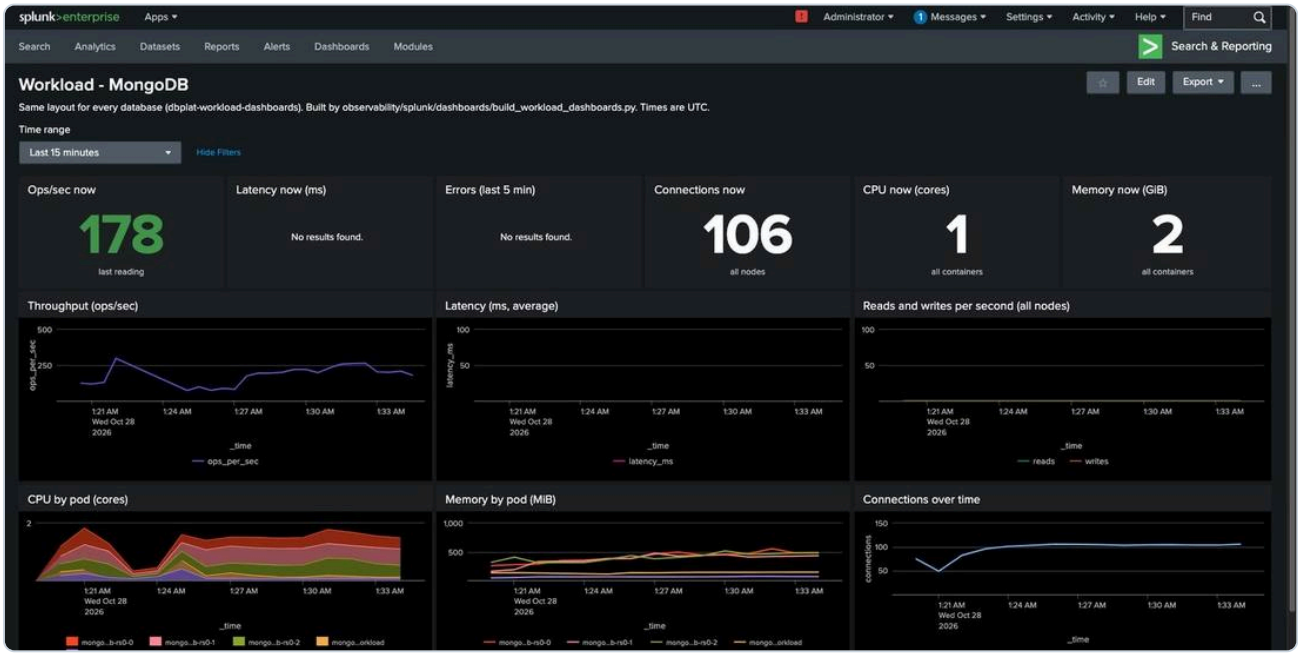


Figure 9.34 28 Oct 2026 · 01:35:09

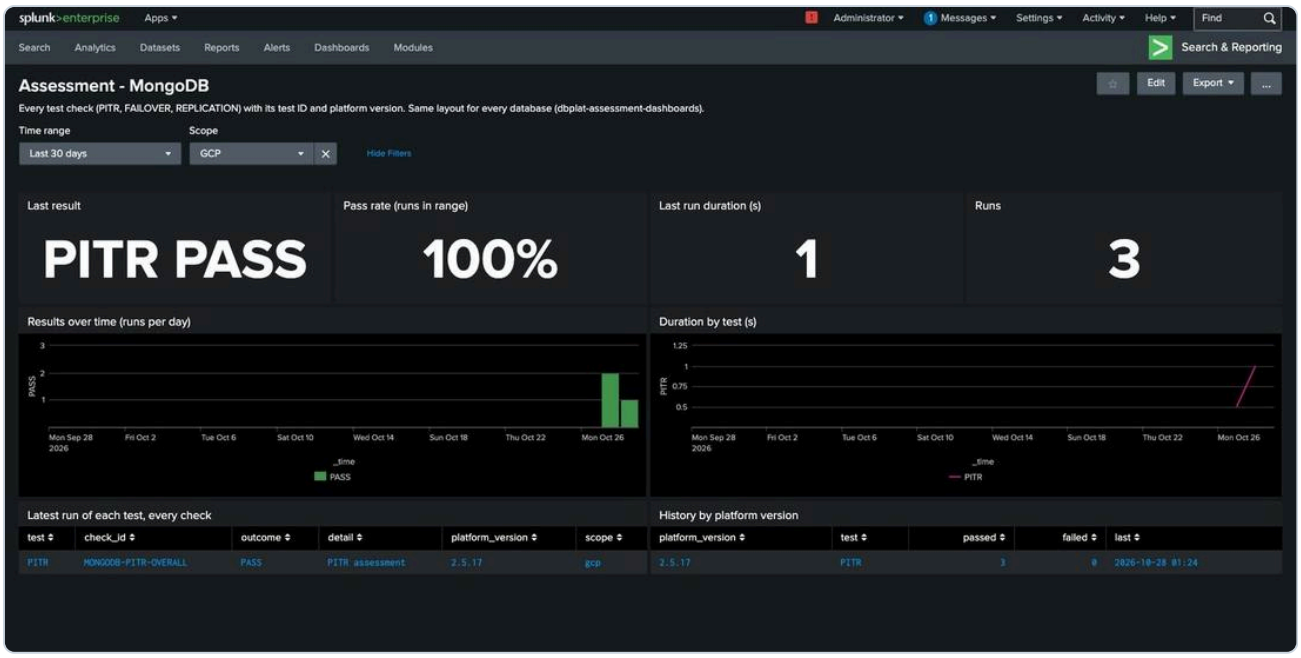


Figure 9.35 28 Oct 2026 · 01:35:28

9.38 Unregister MongoDB from Splunk

Unregister	Splunk	MongoDB	Completed

MongoDB is removed from Splunk; the database itself keeps running.

Started 28 Oct 2026, 01:35 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mongodb/unregister.sh
```

- ✔ Splunk logcollector configuration cleared
- ✔ Confirmed: no Splunk configuration or envFrom remains on the CR
- ✔ All pods have rolled — Splunk forwarding should now be fully stopped
- ✔ MongoDB Splunk log forwarding removed
- ✔ Registration state cleared

9.39 Register MySQL with Splunk

Register	Splunk	MySQL	Completed

MySQL is connected to Splunk, so its metrics and dashboards appear there.

Started 28 Oct 2026, 01:44 · took 3m 33s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/register.sh
```

- ✔ PXC: pxc-db

- ✔ Splunk HEC service: splunk-s1-standalone-service
- ✔ Splunk runtime Secret: splunk-splunk-secret
- ✔ mysql/mysql-splunk-hec converged
- ✔ Percona logrotate container present
- ✔ MySQL → Splunk OTel registration complete

9.40 Workload · MySQL

Workload | MySQL | Completed

A load generator runs against MySQL, so the monitoring shows real traffic. It stops by itself after a set time.

Started 28 Oct 2026, 01:48 · took 24s

COMMAND

```
bash db-platform/mysql/workload/start.sh
```

- ✔ MySQL assessment database available: assessmentdb

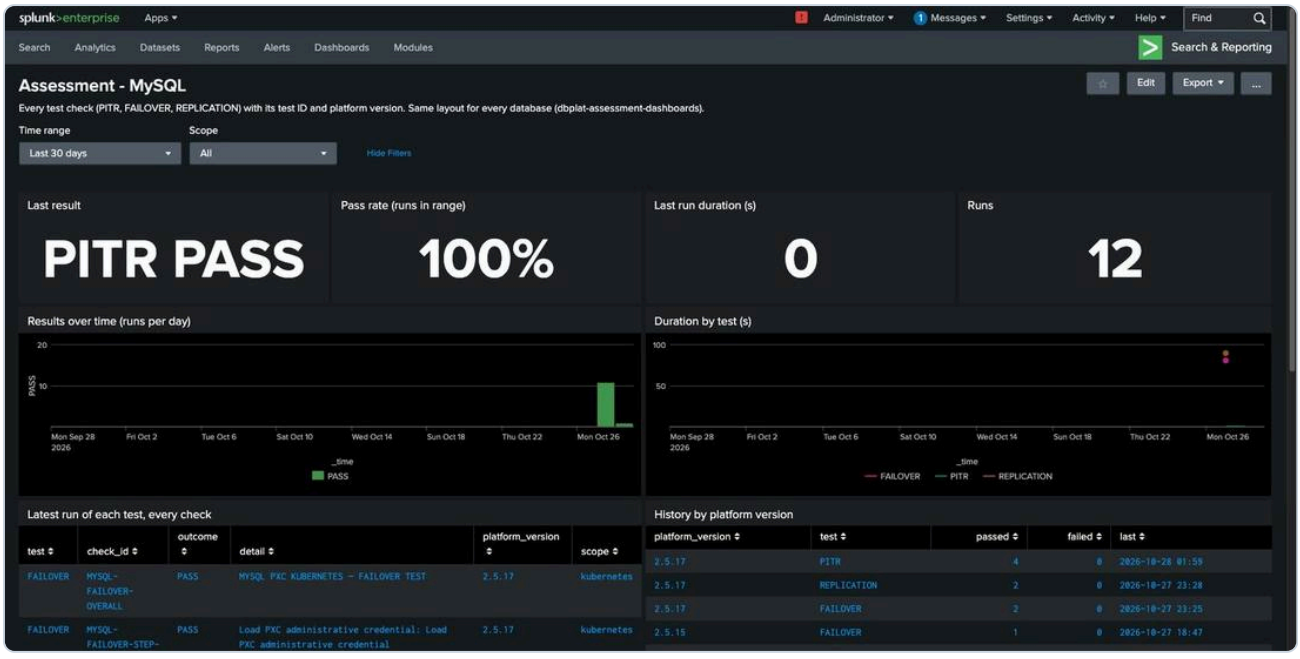


Figure 9.36 28 Oct 2026 · 02:07:55

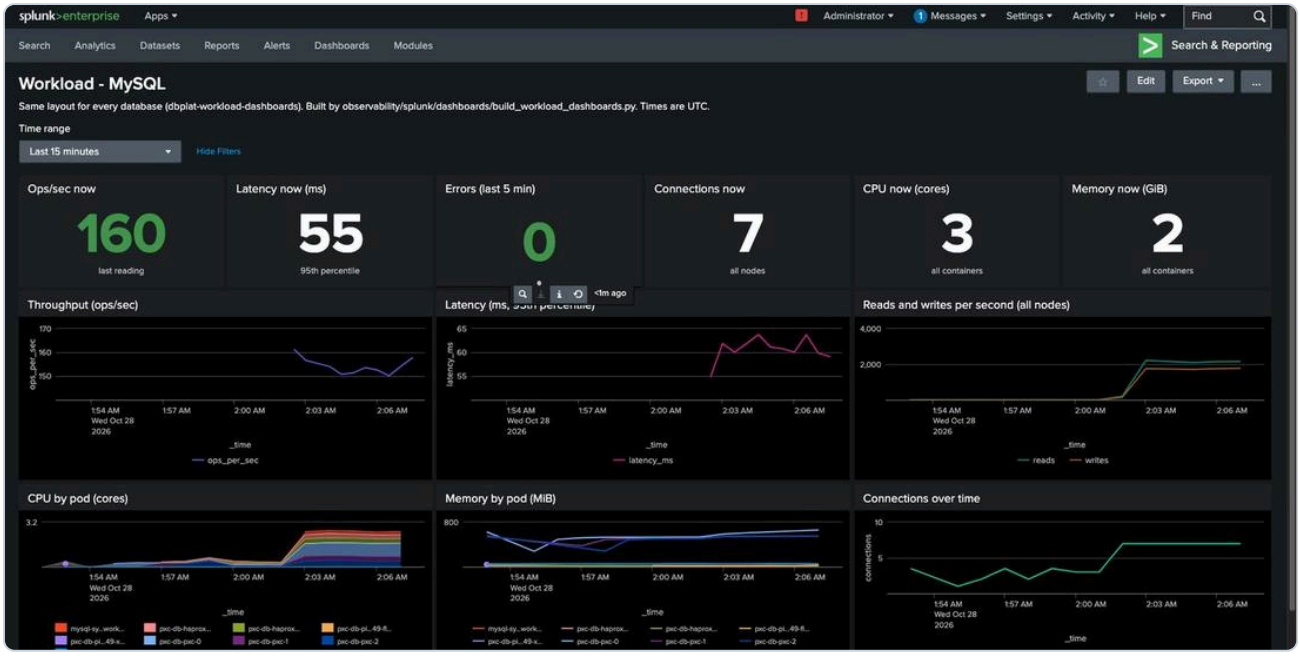


Figure 9.37 28 Oct 2026 · 02:08:01

9.41 Unregister MySQL from Splunk

Unregister	Splunk	MySQL	Completed

MySQL is removed from Splunk; the database itself keeps running.

Started 28 Oct 2026, 02:08 · took 4s

COMMAND

```
bash observability/splunk/registration/kubernetes/mysql/unregister.sh
```

- ✔ Percona LogCollector configuration disabled
- ✔ MySQL-local Splunk HEC Secret removed
- ✔ MySQL → Splunk OTel registration removed
- ✔ Registration state cleared

9.42 Unregister PostgreSQL from Splunk

Unregister	Splunk	PostgreSQL	Completed

PostgreSQL is removed from Splunk; the database itself keeps running.

Started 28 Oct 2026, 02:15 · took 5s

COMMAND

```
bash observability/splunk/registration/kubernetes/postgres/unregister.sh
```

- ✔ Registration state cleared
- ✔ Postgres Splunk log forwarding removed

9.43 Remove Splunk

Unregister | Splunk | Completed

Splunk is removed from the cluster once no database is registered with it.

Started 28 Oct 2026, 02:17 · took 1m 32s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh splunk
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ Kubernetes API reachable
- ✔ GCP/GKE context validated
- ✔ Splunk resources removed
- ✔ Splunk Web access removed

9.44 Destroy the Google Cloud environment

Destroy | Completed

Everything RunShore created on Google Cloud is removed, then checked: nothing that can still be billed is left behind.

Started 28 Oct 2026, 02:19 · took 7m 55s

COMMAND

```
platform/cloud/gcp/orchestrator/destroy.sh kubernetes
```

- ✔ Terraform state: GCS bucket dbplat-tfstate-database-assessment-platform
- ✔ kubectl connected to database-assessment-platform-gke
- ✔ no LoadBalancer services
- ✔ no platform namespaces
- ✔ GCP project database-assessment-platform is clean: nothing left that bills while idle
- ✔ sweep complete

What this shows

- The same RunShore workflow deploys, monitors, tests and destroys every database on local Kubernetes and on all three clouds.
- Every action leaves a log; each step above links to it.
- Recovery is tested, not assumed: failover, replication and point-in-time recovery runs are recorded with their results.
- Cloud environments are destroyed at the end of each chapter, and the destroy is verified.

RunShore · Database assessment and observability platform